Nutritious School Meals and Educational Outcomes

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Abstract

This paper evaluates the causal impact of improved school meal quality, holding caloric content constant, on academic performance among public school students in Chile. Exploiting the staggered rollout of a nationwide policy implemented in 2015 that raised the nutritional standards of school-provided meals, we employ a difference-in-differences strategy using national, student-level administrative records covering seven years. The intention-to-treat estimates indicate that exposure to healthier meals leads to a statistically significant increase of 0.039 SD in standardized Math and Reading test scores. Turning to mechanisms, we find larger effects among students more likely to receive the meal, proxied by imputed program eligibility, full-day schooling, and rural location, and among students in the poorest income tercile, whose diet outside school is likely to be of lower quality. Consistent with a change in diet, we document a positive spillover onto food choices, with an increase in the consumption of healthy snacks at school and in a household healthy dietary index, and no offsetting change in unhealthy items, which rules out crowding out of healthy food at home. The results are not driven by caloric content, which is unchanged by design, nor by body mass, school attendance, parental investment and expectations, student behavior, or teacher-reported classroom climate, all of which show null effects. A back-of-the-envelope calculation suggests that improving meal quality is cost-effective, at approximately $121.9 per student per year for each 0.1 SD in test scores.

Keywords: School meals, nutrition, academic performance, educational outcomes, public policy, cost-effectiveness, child development, Chile

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1 Introduction

Disparities in human capital accumulation are a key determinant of cross-country differences in economic development (Barro 1991). In response, governments and development agencies in low- and middle-income countries have implemented a range of costly interventions aimed at increasing educational participation and improving the underlying determinants of human capital formation. Among the most prevalent of these initiatives are school meal programs (SMPs), which are now implemented in 161 countries, so that approximately one in every two school-age children worldwide receives meals through a government-supported program (WFP 2020). Their primary objectives are to reduce the private cost of schooling, mitigate the adverse effects of poverty, and enhance child health, each of which plays a critical role in human capital formation (Alderman et al. 2018).

A growing body of empirical evidence documents the positive effects of these programs. School meals increase student attendance in settings with historically low participation rates (Adelman et al. 2008), and improve child health outcomes regardless of baseline school participation (Aurino et al. 2020, Wang et al. 2021, Bommer et al. 2020). Given this evidence on the extensive margin, policymakers have increasingly shifted their focus from the quantity to the quality of the calories provided. Improving the nutritional content of school meals, particularly through improved micronutrient intake, can enhance cognitive performance and reduce susceptibility to infections even in the absence of changes in total caloric consumption (Adelman et al. 2008), and several countries have made meal quality a central policy objective (USDA 2012, WFP 2017, GCNF 2019). This makes it important to identify rigorously the causal effect of food quality on cognitive development.

This paper asks whether, and to what extent, the quality of calories influences later student performance. To address this question, we exploit variation in the timing at which Chilean schools were required to improve the nutritional composition of the meals provided by their vendors. Beginning in 2015, the agency in charge of the federal resources allocated to school meals (Junaeb) implemented a nationwide policy mandating enhancements in nutritional quality. For the purpose of this study, a nutritious meal follows the Chilean national dietary guidelines (Minsal 2013), which favor fresh, minimally processed foods of high nutritional value (fresh fruits and vegetables, fish, legumes, and water) over energy-dense, low-nutritional-value items high in added sugars, sodium, and saturated fats (such as sugary foods and beverages, fried foods, and processed meats).

The timing of these improvements was determined by a pre-existing national contracting schedule in place since the late 1990s, when Junaeb replaced its previous discretionary award criteria with a sequential combinatorial auction designed to minimize costs and avoid market concentration (Catalán et al. 2009). Under this system, meal vendors are assigned to schools through three-year contracts, the country is divided into three groups of municipalities, and each year one group receives a new contract with either a new or a continuing vendor. Following the introduction of the nutritional guidelines in 2015, Group 1 schools were affected immediately, Group 2 in 2016,

and Group 3 in 2017. Crucially for identification, the grouping of municipalities responds to these procurement objectives, not to test scores, school characteristics, or other municipal conditions, so these factors did not dictate the timing of adoption of the healthier menu.

To recover an intention-to-treat (ITT) effect, we use a difference-in-differences (DiD) strategy based on Callaway and Sant'Anna 2021, given the staggered nature of adoption. We focus on public schools, which enroll 40% of national students, the majority from more disadvantaged backgrounds, and where virtually all schools receive meals through the SMP. Our main outcome comes from individual administrative records of national standardized test scores, a seven-year repeated cross-section covering all Chilean 4th graders attending public schools (approximately 64,000 students and 4,245 schools per year). To study mechanisms we complement these data with the SIMCE student, teacher, and parent questionnaires, the Junaeb anthropometrics' *Mapa Nutricional*, and the Encuesta Longitudinal de la Primera Infancia (ELPI) longitudinal survey.

The main results indicate that the introduction of a healthier school menu increases standardized test scores by approximately 0.039 SD on average, with gains of 0.033 SD in Reading and 0.038 SD in Math relative to the previous, less nutritious menu. These magnitudes are comparable to those of educational interventions such as adding one hour to the school week (Lavy 2016) or extending the school year by five days (Hansen 2011). From a cost-benefit perspective, improved meal quality is particularly attractive: raising a student's test score by 0.1 SD through better meals costs approximately $121.9 per year, against $297 under North Carolina's weekend feeding program[1].

We next provide supporting evidence that these gains can be attributed to the adoption of healthier meals. Event-study specifications show no differential pre-trends, and estimates remain robust across alternative estimators, including De Chaisemartin and d'Haultfoeuille 2020 and traditional TWFE, as well as alternative clustering levels and spatial trends. A falsification test using contract rotations under unchanged menus confirms that vendor turnover alone generates no effect. Finally, a shorter sixth-grade panel yields consistent, albeit more muted, gains, in line with greater nutritional sensitivity at younger ages, all reported in Section 3

We then use a set of complementary tests to narrow the range of plausible mechanisms, since our data do not allow us to test the nutritional-biological channel directly. The pattern of effects is consistent with exposure to the healthier meals: gains are concentrated among a proxy for eligible students (0.045 SD against 0.002 SD for non-eligible students), students in full-day schools (0.043 SD against 0.016 SD), and students in rural schools (0.071 SD against 0.032 SD). They are also consistent with baseline dietary quality, since the effect declines from the first to the second income tercile (0.066 SD and 0.036 SD) even though coverage is likely universal in both. Consistent with a change in diet, we find an increase of 1.7 percentage point in the consumption of healthy snacks at

[1] All values are inflation-adjusted to 2026 U.S. dollars.

school and an increase of about 0.15 SD in a household Healthy Dietary Index, with no change in sugary drinks, fast food, candy, or an Unhealthy Dietary Index, which rules out crowding out of healthy food at home. Taken together, these results suggest that the effect on test scores combines two channels: exposure to nutritious foods at school, and a shift in students' preferences that changes what they eat at home. In contrast, the remaining candidate channels are not supported by the data: the caloric content of the meals did not change, body mass shows null effects across three datasets and three empirical strategies, attendance shows a precisely estimated zero, and parental investment and expectations, classroom climate, and peer victimization show no effects.

The contributions of this paper are threefold. First, we contribute to the literature on the education production function (Hanushek 2002, Hanushek and Luque 2003, Das et al. 2013), in which school meals are an important input. A broad set of studies shows that the provision of meals improves student outcomes (Dotter 2013, Imberman and Kugler 2014, Frisvold 2015, Corcoran et al. 2016, Kurtz et al. 2020, Schwartz and Rothbart 2020, Ruffini 2021), in some cases with long-term gains in educational, health, and economic outcomes (Lundborg et al. 2022). While this work has focused on the extensive margin, whether or not meals are provided, we shift the focus to the intensive margin, that is, the nutritional quality of a meal that students were already receiving. This dimension of the education production function determines the return to a large share of SMP spending, yet it remains underexplored.

Second, we add to a growing body of work on the role of food quality in shaping educational outcomes. Medical research documents positive associations between specific nutrients and cognitive development (K. F. Bell et al. 2015, Goodwill and Szoeke 2017, Francis and Stevenson 2013), though usually in small-scale studies with limited external validity. In the economics literature, a handful of studies examine nutrient supplementation, such as iron and folic acid (Berry et al. 2020) or double-fortified salt (Krämer et al. 2021), and only Belot and James 2011 and Anderson et al. 2018 examine broader improvements in school food quality. We improve upon these by exploiting a staggered rollout driven by a pre-existing meal vendor assignment schedule that is plausibly exogenous to school characteristics and outcomes, while the universal application of the policy and the uniform nutritional standards imposed across vendors strengthen the external validity of our results.

Third, we narrow the set of channels through which better school food affects learning and identify spillovers. Existing work establishes that improved meal quality raises test scores but cannot separate why, since the interventions studied change several inputs at once and the available outcomes rarely allow competing channels to be tested. Using a policy that altered only the nutritional composition of the meals, together with data on diet at school and at home, anthropometrics, attendance, parental investment, and classroom behavior, we show that the effect increases with the probability of receiving the meal and is larger where baseline diet is likely worse. Whether institutional food

provision reinforces or displaces healthy eating outside school is a central question in the economics of education literature (Afridi 2010; Adamson et al. 2013; Valizadeh and Ng 2020; Marcus and Yewell 2022), and the sign of the effect is theoretically ambiguous. We find an increase in the consumption of healthy snacks at school and of healthy foods at home, that is, reinforcement rather than displacement. This evidence aligns with theories of preference formation in which children need repeated exposure to a food before accepting it (Wardle et al. 2003; Savage et al. 2007), and in which this acceptance is what changes household purchases (Monalisa et al. 2023). By contrast, calories, body mass, attendance, and behavioral channels are not supported by the data. While we cannot identify the biological mechanism directly, the evidence narrows the range of explanations that remain plausible.

More broadly, our findings speak to the set of interventions known to improve outcomes during late childhood, the stage between early childhood and adolescence at which a large proportion of children can be reached through the education system (Lundborg et al. 2022). The evidence at this stage comes mainly from schooling inputs, such as class size reductions and performance-based teacher pay (Gilraine et al. 2018, Biasi 2021), and from health policies targeting school-age children outside the classroom, such as hookworm eradication, mandatory vaccination laws, and expanded access to health insurance (Bleakley 2007, Luca 2016, Arenberg et al. 2024, Bütikofer and Salvanes 2020). School meals combine both features, since they are delivered through the school system and act on a health-related margin, and our results show that improving their nutritional quality raises achievement at a cost per unit of test score gain well below that of the schooling inputs studied in this literature.

This paper proceeds in Section 2 by reviewing the literature on SMPs and the one that links healthy food and cognition, describing the Chilean School Feeding Program, and presenting the identification strategy. Section 3 describes the data, the empirical strategy, and the internal validity of the estimates. Section 4 presents the main results. Section 5 tests which mechanisms could explain them. Section 6 presents a discussion and a cost-benefit analysis, and Section 7 summarizes the findings and concludes.

2 Background

2.1 Linking Nutritious School Meals and Cognitive Development

The SMPs seek to improve children's outcomes through two broad mechanisms. First, by reducing the cost of schooling for families, they make it more attractive and financially feasible for parents to send their children to school and invest in their education. Second, beyond this cost-reduction effect, SMPs serve as a direct input in the education production function, with the potential to improve student attainment directly. The empirical literature examining the effect of SMPs on academic achievement has yielded mixed results. Most studies report positive impacts (Hinrichs 2010, Dotter

2013, Imberman and Kugler 2014, Frisvold 2015, Chakraborty and Jayaraman 2019, Schwartz and Rothbart 2020, Kurtz et al. 2020, Y. Kim 2021, Ruffini 2021), but research has also found null effects (McEwan 2013, Leos-Urbel et al. 2013, Corcoran et al. 2016, Cuadros-Meñaca et al. 2022).

These two mechanisms operate through three most common channels identified in the literature: increased enrollment and attendance, an outcome of the cost-reduction effect, and caloric intake and improvements in the nutritional quality of food, both instances of the direct-input effect. This paper focuses on the third channel, nutritional quality. The remainder of this subsection discusses the evidence linking dietary composition to student achievement.

Much of the current understanding regarding the effects of diet quality on cognitive outcomes originates from the medical literature, primarily through observational studies and clinical trials. Table A.5 summarizes this evidence for school aged children by food or compound, including the proposed channel, and the reported effect sizes. On the one hand, a number of nutrients have been found to positively influence cognitive and behavioral outcomes. For example, deficiencies in vitamins B and D, or in omega-3 fatty acids, are associated with impulsive behavior and symptoms of ADHD (Patrick and Ames 2015, Kaplan et al. 2004). Similarly, antioxidants found in fruits and vegetables help prevent neuronal oxidative damage (J. H. Kang et al. 2005), while long-chain omega-3 polyunsaturated fatty acids are linked to improve learning and memory (Stonehouse 2014). Iron deficiency impairs dopaminergic neurotransmission and hippocampal function, and iron supplementation has been shown to improve intelligence, attention, and memory test scores, with larger effects among children who are anemic at baseline (Gutema et al. 2023, Krämer et al. 2021, Low et al. 2013). Additional nutrients, such as dietary fiber, protein, water, and vitamins B and D, have also been shown to support cognitive functioning (Goodwill and Szoeke 2017, Naiman A Khan et al. 2015, Mahoney et al. 2005, Khan et al. 2015).

On the other hand, poor dietary components also have adverse effects on cognition. Evidence on glycemic index is mixed: some studies report better verbal memory following low-glycemic-index meals, while others find faster processing speed and vigilance after high-glycemic-index meals, with the direction of the effect depending on the cognitive domain measured (Micha et al. 2011, Ingwersen et al. 2007, Jansen et al. 2020). Refined sugars have also been linked to worse cognitive performance: higher consumption of sugar-sweetened beverages is associated with slower reaction times and greater odds of impaired performance on working-memory and executive-function tasks in adolescents (Yan et al. 2022, Ren et al. 2022, Zhang et al. 2022).

Connecting these magnitudes to the effects of a school meals program requires some care. The medical literature reviewed above examines the effects of specific, isolated nutrients under controlled conditions, whereas a program-level nutritional reform bundles multiple dietary changes into a single intervention delivered in a real-world school setting. Sample sizes, intervention spans, testing

methods, and the outcomes used to measure cognitive performance also vary considerably across these studies, which limits how precisely any single estimate can be benchmarked against this evidence. For these reasons, the effects of the SMP reform we study should be expected to be more modest than the effects documented in clinical trials of individual nutrients.

Nutritious meals may also influence student achievement through broader health-related mechanisms, which are likely to materialize over longer time horizons. First, healthier meals can strengthen students' overall health, making them less susceptible to illness. Healthier students tend to miss fewer school days, which increases instructional time and learning opportunities[2]. Second, in contexts where malnutrition due to excess is more prevalent, improved nutrition may reduce childhood obesity, which in turn has been linked to better cognitive outcomes. For instance, Liang et al. 2014 document a negative relationship between obesity and cognitive development. One potential mechanism for this association is described by Stanek et al. 2011, who show that, even in otherwise healthy individuals, obesity is associated with reduced white matter integrity. Disruptions in white matter pathways can impair neural transmission speed, executive functioning, and working memory, all of which are essential components of the learning process.

2.2 The Chilean School Meal Program

The Chilean SMP is a nationally mandated, government-administered initiative operated by the National Board of School Aid and Scholarships (Junaeb, for its Spanish acronym, Junta Nacional de Auxilio Escolar y Becas). Initially created to combat high rates of child undernutrition, the program has evolved to respond to shifts in dietary needs, particularly the growing concern over the quality of students' energy and nutrient intake. Today, the SMP primarily provides breakfast and lunch, with a small share of students also receiving an evening snack.[3]

Meals are provided daily for roughly 200 days per year, meeting between 40 and 60 percent of a student's recommended daily caloric intake. The meals are designed to resemble home-cooked dishes, and, with very few exceptions, avoid ultra-processed foods which tend to be high in salt, sugar, or fat. See Figure A.1 for examples of Chilean school meals.

In terms of food satisfaction, Junaeb conducts annual qualitative evaluations of student satisfaction with the meals. Students rate various aspects of the service, including taste, variety, temperature, and serving size. On a scale from 1 to 7, more than 65% of students assign a rating of 5 or higher. The level of satisfaction have not change over time, despite modifications on the menu (Junaeb 2011; Consulting 2012; ClioDinamica 2013; ClioDinamica 2014; ClioDinamica 2015; Junaeb 2016)

[2]See Jomaa et al. 2011 for a review of SMP health effects in developing countries, and Hinrichs 2010, Gundersen and Kreider 2009 for evidence from developed countries.

[3]Junaeb. "Programa De Alimentación Escolar (PAE)." Accessed November 28, 2021. https://www.junaeb.cl/programa-de-alimentacion-escolar.

The program targets all students enrolled in public or charter schools (i.e., private subsidized schools), which together serve over 90 percent of Chilean students. Eligibility is based on socioeconomic status. There is no direct administrative flag for SMP student eligibility. However, the Longitudinal Early Childhood Survey (ELPI), a nationally representative survey of Chilean children born between 2006 and 2016, shows that the take-up rate is almost 70% in urban settings and 95% in rural areas. (See Section 5.1 for more information).

2.2.1 The SMP reform

In 2015, Junaeb modified the nutritional requirement of the SMP to improve its nutritional content, aiming to address rising childhood obesity rates. While undernutrition had been largely eradicated by the late 1980s (Vio et al. 2008), recent years have seen a rapid increase in childhood overweight and obesity. Current estimates indicate that 24 percent of children are obese and 51 percent are overweight. These rates vary substantially by socioeconomic status: children from the lowest income quintile are 40 percent more likely to be obese than their peers in the highest quintile. Additionally, obesity rates are 5 percentage point higher in rural areas than in urban areas (Junaeb 2016).

The 2015 nutritional reform covered all meal types (breakfast, lunch, and the evening snack) and aimed to improve dietary quality without altering total caloric content. Specifically, Table 1 details these modifications, showing an increase in the presence of foods rich in nutrients associated with improved cognitive outcomes (e.g., fruits, vegetables, fish, meat, dried fruits, and water) and a reduction in components found to have negative cognitive effects, such as refined carbohydrates and added sugars. Moreover, portion sizes are adjusted: side dishes were reduced (except when the side dish is salad), while the share of protein was increased. Table A.6 reorganizes these compositional changes by nutrient category. The reform increased the frequency and portion size of foods rich in antioxidants (salads, vegetables, and fruit), vitamins, micronutrients, and omega-3 fatty acids (avocado, fish, and produce), and iron and other micronutrients from animal-source foods (meat, chicken, and fish), and made water a mandatory component of the tray, while reducing the frequency of foods high in refined carbohydrates and added sugars (white bread, packaged cereal, sweetened spreads, and oat cookies). As discussed in Section 2.1 and summarized in Table A.5, these are precisely the nutrient categories the medical literature links to cognitive performance in children: antioxidant intake to working memory and processing speed, micronutrient adequacy to verbal learning and memory, iron intake to intelligence and attention, and reduced added-sugar intake to academic achievement and executive function.

It is also relevant to understand children's baseline eating habits and how they compare to recommended dietary guidelines. In Table 2[4] we show eating patterns of families with primary school

[4]We measure food consumption using Chile's Household Budget Survey *Encuesta de Presupuestos Familiares*

Table 1: Change of nutritional requirements for 4th grade students meals

	Before 2015 auction	After 2015 auction	Change
Calories			
Breakfast	250	no change	–
Lunch	450	no change	–
Evening snack	250	200	↓
Total sucrose intake	23 grams	15 grams	↓
Breakfast ***(in times per month)***			
White bread	12	8	↓
Jam, honey or dulce de leche	4	2	↓
Margarine	3	2	↓
Egg	4	3	↓
Partly skimmed milk	20 to 22	16	↓
Breakfast cereal	2	eliminated	↓
Large oat cookie	2	eliminated	↓
Yogurt	2	4 to 8	↑
Whole wheat bread	4	6	↑
Avocado	2	3	↑
Cheese	2	3	↑
Granola	–	incorporated	↑
Oats	2	4	↑
Dried fruit	–	incorporated	↑
Lunch ***(in times per month)***			
Jelly with fruit	4	2	↓
Potato, rice and pasta	250, 70 and 70 g.	220, 60 and 60 g.	↓
Legumes side dish	1	eliminated	↓
Salads	4	6 (w/ larger portions)	↑
Salad + protein (tuna, egg, chicken)	4	5	↑
Meats (beef, chicken, pork)	7	8 (w/ larger portions)	↑
Fish	4	4 (w/ larger portions)	↑
Vegetables side dish	5	6	↑
Fresh fruit	8	10	↑
Legumes main dish (w/ egg, beef, cheese)	5	5	–
Egg	4	4	–
Dairy-based desserts	6	6	–
Yogurt	2	2	–
Canned fruit	2	2	–
Water	Available upon request	Mandatory in tray	
Evening Snack			
Dried fruit	-	incorporated	↑
Large oat cookie or similar	60 grams max	40 grams max	↓

Note: This table illustrates changes in nutritional requirements after the 2015 auction. Modifications decreased the grams of daily sucrose intake, the number of processed foods on the menu and increased fresh fruits and vegetables. Schools 2015 started receiving this menu in 2015, Schools 2016 in 2016 and Schools 2017 in 2017. Source: Prepared by the author based on information of auctions 85-16-LP12 and 85-10-LP14.

(EPF), a nationally representative household budget survey fielded by the Instituto Nacional de Estadísticas. For each household we collapse all recorded purchases of a given product (identified by its CCIF code) into a single monthly quantity, and classify each product into a food group (meats, fish, dairy, vegetables, fruits, legumes, bread, snacks, sugary drinks, and others) based on the CCIF classification. Quantities are summed within each food group and household, converted to grams, divided by household size, and divided by 4.33 to obtain a weekly per capita quantity for each food group. To express this quantity in servings, we divide the weekly per capita grams by a representative serving size for each food group, drawn from Chile's clinical portion-size references and the *Guías Alimentarias para Chile* (INTA/MINSAL), and round to the nearest whole serving. The table reports population-weighted averages (using the survey's expansion factor) of these servings. The sample includes households with at least one child currently

aged children attending public schools. Several things are worth noting. First, consumption rises across income terciles for nearly every food type, both healthier (fish, fruits, legumes) and less healthy (snacks, sugary drinks, treats), with the exception of bread, whose consumption falls as income rises.[5] Second, with the exception of meat, which exceeds the recommended level in every tercile, and bread, which exceeds the recommended level by at least four times, servings of every other food category fall short of Junaeb's recommended weekly servings across all three terciles. This indicates that, since families are not currently meeting the recommended weekly servings of these healthier food groups, there is room for the SMP's healthier menu to affect student outcomes.

During our study period, a second national policy acted on the school food environment: Law 20.606, the Law on the Nutritional Composition of Foods and their Advertising. The law operates through three instruments: mandatory front-of-package "high in" warning labels on products that exceed set limits for calories, sodium, sugars, or saturated fats; a ban on advertising such products to children under fourteen; and a prohibition on their sale within pre-school, primary, and secondary schools (Minsal 2012). Its restrictions were phased in over product nutritional thresholds (see Table A.13 in the Appendix, for the exact nutrient thresholds by stage), in June 2016, June 2018, and June 2019 (Minsal 2015). Because the law was implemented nationwide, entered into force only a year and a half after the SMP reform began for the first treated cohort, and since it was gradually implemented, we suggest this Law did not affect our estimates. We discuss the law and its implications for our identification strategy in more detail in the Internal Validity Section 3.4.5.

2.3 Identification Strategy

The identification strategy leverages the staggered adoption of healthier school meals, determined by a pre-existing schedule of public procurement auctions. This auction schedule, in place since the late 1990s, governs the assignment of meal providers to schools via three-year contracts.

Until the late 1990s, Junaeb awarded contracts in a non-standardized fashion, based on subjective, and quite rudimentary, criteria (Catalán et al. 2009). Consequently, vendors lacked incentives to

enrolled in a public educational establishment at the primary level; each such child contributes one observation, so households with more than one primary-school-age child are represented multiple times, and the reported averages reflect the food environment experienced by children rather than by households. Households are split into terciles of per capita disposable household income, excluding imputed rent, computed using the survey's expansion factor, among households with a primary-school child; T1 denotes the lowest income tercile and T3 the highest. The EPF 2011-2012 would have provided a better description of baseline differences, but that wave did not record purchased quantities, only spending. Table A.1 in Appendix reports differences in spending by tercile. The gradient is the same as the one documented here, though spending is the product of quantity and unit value, so it does not separate the amount of food households buy from the prices they pay for it. The 2012 gradient therefore cannot be read as a difference in quantities alone.

[5]The number of weekly servings of bread is unusually high for two reasons. First, families underestimate how much bread constitutes a serving: 96% do not know that a serving of marraqueta (the most commonly eaten bread in Chile) is one piece, and instead assume a serving is two or four pieces (Hermosilla 2024). Second, bread is used as the main component of both breakfast and the evening meal: Chileans typically do not eat a hot dinner, and instead have bread-based sandwiches.

Table 2: Dietary diversity, by Household Income Tercile - EPF Survey

	Number servings					Number servings			
	T1	T2	T3	Junaeb		T1	T2	T3	Junaeb
Meats[a]	5.65	6.76	7.96	3	Dairy	1.76	1.98	2.57	21
	(5.97)	(5.49)	(7.46)			(2.88)	(2.42)	(2.96)	
Fish	0.84	1.21	1.21	2	Bread	37.63	34.59	32.34	7
	(1.60)	(2.13)	(1.98)			(26.83)	(21.61)	(22.16)	
Fruits	2.90	3.30	3.52	14	Snacks	0.41	0.64	1.00	n.a.
	(4.11)	(3.89)	(3.87)			(1.21)	(1.29)	(2.30)	
Vegetables	9.70	10.57	11.69	21	Sugary drinks	3.96	5.42	6.41	n.a.
	(7.91)	(11.56)	(11.18)			(4.08)	(4.91)	(6.38)	
Legumes	0.52	0.55	0.76	2	Treats	7.65	8.75	13.17	n.a.
	(1.30)	(1.37)	(1.56)			(9.28)	(9.75)	(13.81)	

Notes: Author's calculations using the VIII Household Budget Survey *Encuesta de Presupuestos Familiares* (INE, 2016-2017). Sample: children enrolled in public primary education (one observation per child). Servings are weekly per capita quantities, converted from purchased grams using standard Chilean portion sizes and rounded to the nearest whole serving. Means are weighted by the survey's expansion factor, standard deviations (in parentheses) are weighted the same way. Households are split into terciles of per capita disposable household income (excluding imputed rent, computed using the survey's expansion factor) among households with a primary-school child attending public schools. T1 is the lowest tercile and T3 the highest. The Junaeb column reports the recommended weekly servings from Junaeb's "Guía de Alimentación del Escolar 6-10 años," converted from the guide's daily/weekly frequencies. [a]Junaeb's guideline (3 servings/week) is for lean meat or egg jointly, a narrower category than total meat purchases reported here. The EPF 2011-2012 would have provided a better description of baseline differences, but that wave did not record purchased quantities, only spending. Table A.1 in Appendix, reports differences in spending by tercile. The gradient is the same as the one documented here, though spending is the product of quantity and unit value, so it does not separate the amount of food households buy from the prices they pay for it. The 2012 gradient therefore cannot be read as a difference in quantities alone.

reduce costs, and Junaeb ended up paying a high price for low-quality products (Catalán et al. 2009). In 1997, Junaeb's board asked a team of academics from the University of Chile's Department of Industrial Engineering to design and implement a new mechanism for auctioning its school meals service, awarding contracts through a single-round, sealed-bid, first-price sequential combinatorial auction (combinatorial auctions are multi-unit auctions in which suppliers bid for packages of items). The new mechanism, aimed at minimizing costs while avoiding market concentration, saved the government an estimated US$40 million a year, the equivalent of the cost of feeding 300,000 children (Catalán et al. 2009).

Under this system, the country is divided into three groups of municipalities (see Figure 1), with each group comprising municipalities that may be geographically proximate or distant[6]. Each year, one of these three groups receives a new contract, either with an existing or a new vendor, on a rotating basis: Group 1 in the first year, Group 2 in the second, and Group 3 in the third. Because contracts are renewed every three years, the cycle restarts in the fourth year.

[6]For example, in the group of schools treated in 2015, the municipalities of Loncoche and Torres del Paine are approximately 1,200 kilometers (730 miles) apart, with no other municipality from the same treatment group located between them. However, within this group, Villarrica is adjacent to Loncoche, and Natales is adjacent to Torres del Paine.

The SMP auction process follows a two-stage procedure. In the first stage, the agency defines technical requirements related to nutritional standards, minimum operational quality, and infrastructure. In the second stage, firms who meet these criteria compete on prices. This phase also imposes market share restrictions and awards contracts through multiple sequential auctions to promote diversification and enhance competition (See Epstein et al. 2004; Catalán et al. 2009; Olivares et al. 2009 for a detailed explanation of the design of combinatorial auctions for procurement which created the groups for the school meals auction).

This institutional structure enables us to exploit quasi-random variation in treatment timing. Specifically, we combine the predetermined contract schedule with the rollout of nutritional improvements that begins in 2015. In that year, only schools assigned to new contracts received the updated, healthier meals, while the remaining schools continue under the previous contract terms. In 2016 and 2017, Groups 2 and 3, respectively, were brought under the new menu (see Figure 2). This staggered implementation generates clearly defined treatment and control groups during the initial two years of the reform, facilitating a credible empirical strategy. It is not possible to analyze the effect in the third year, since all three groups are treated, and no control group remains. Critical to the identification strategy, we argue that student performance, individual characteristics, and school or municipal attributes did not determine the rollout of healthier meals across schools, as it followed the preexisting process of meal vendor assignment which was designed primarily to enhance market competition among vendors, as described above in this section. Moreover, the general population, and teachers and parents in particular, are not aware of how this assignment process operates.

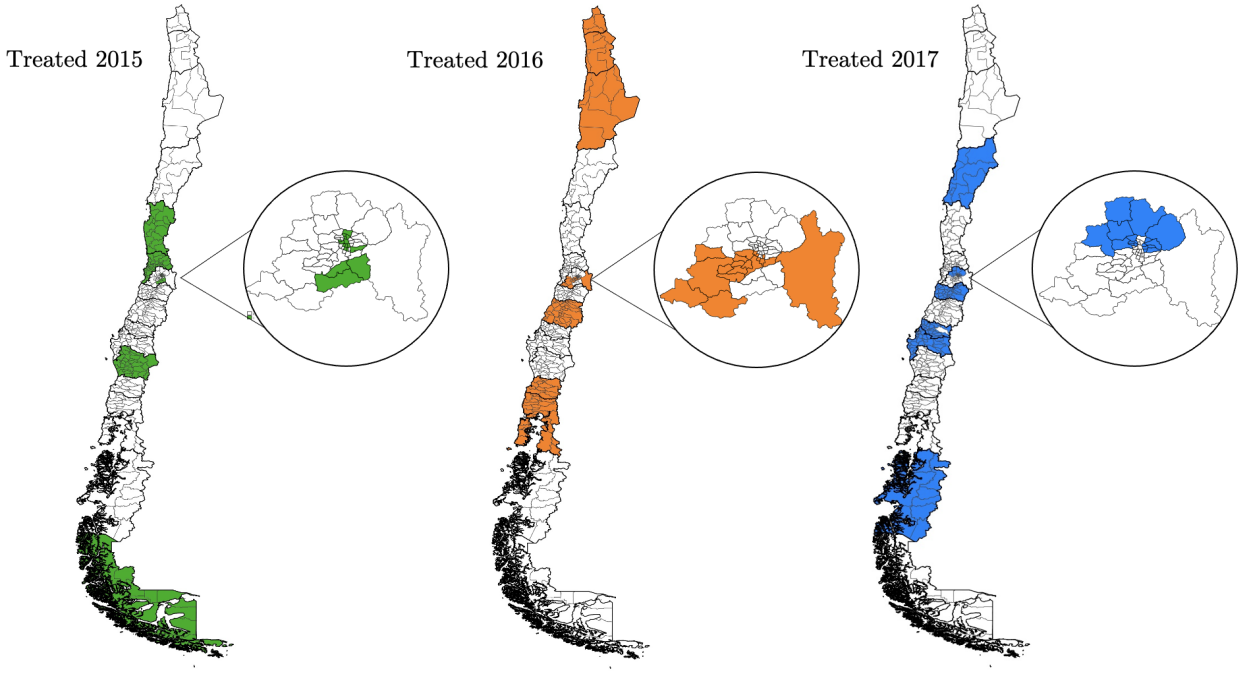
3 Data and Empirical Strategy

3.1 Data

To analyze the effects of changes in the nutritional composition of school meals on all students in 4th grade attending public schools, we use data from three sources: the Ministry of Education, the Quality of Education Agency, and the Chilean public procurement market. These databases contain individual standardized test results, student and school characteristics, schools assigned to each group, and the nutritional requirements for meal vendors each year. The data range from 2010 to 2016, and the sample consists of approximately 64,000 students and 4,245 schools per year. We describe each type of information in detail below.

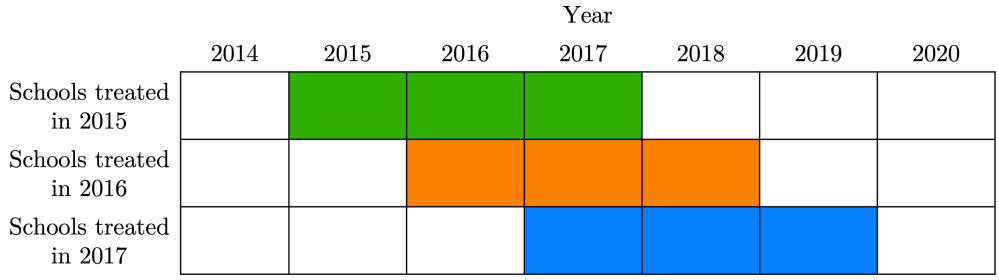
Individual standardized test results. The Education Quality Measurement System (SIMCE, for the acronym in Spanish of Sistema de Medición de Calidad de la Educación) is a centralized battery of tests managed by the Quality of Education Agency, used to measure learning outcomes based on the national curriculum. Depending on various scheduling factors, cohorts of students are tested in 2nd, 4th, 6th, and 8th grades (primary and middle school), and in 10th or 11th grades

Figure 1: Rollout of the Healthier Meals



Note: Figure shows the geographic roll-out of the implementation of healthier meals across schools. The zoomed portion corresponds to the Metropolitan Region, where 70% of the population resides. This picture illustrates that adjacent and afar municipalities were treated at the same time.

Figure 2: Treated and Control groups



Note: Figure shows the temporal roll-out of the implementation of healthier meals across schools. Schools treated in 2015 received the healthier menu during years 2015, 2016 and 2017; Schools treated in 2016 received healthier meals until year 2018 and schools treated in 2017 received healthier meals until year 2019. Schools treated in 2015 where subject to a different contract in year 2018. This new contract followed the same healthy guidelines as the previous one, but it is not the exact same menu as the one implemented in 2015.

(secondary school). SIMCE assessments cover subjects including Mathematics, Reading and Writing (henceforth, Reading), Social Sciences, and Natural Sciences.

Starting in 2005, 4th-grade students have taken the test every year at the end of the school year.

Students in 10th (8th) grade are assessed in even (odd) years, while the Ministry of Education has not yet established fixed testing frequencies for grades 2, 6, and 11. Importantly, SIMCE consistently tests only Math and Reading annually, while Social Sciences and Natural Sciences are tested alternately[7]. Because the nutritional improvements were implemented in a staggered fashion, and test score data are only available annually for 4th-grade students in Math and Reading, the main empirical analysis in this paper focuses on 4th graders exclusively.

Control variables. Student-level data are sourced from both the SIMCE student survey and the Ministry of Education's enrollment records. The SIMCE survey provides information on household income and maternal education, while the enrollment data provide demographic variables such as gender, age, and school attendance. School-level variables, including whether a school is located in a rural or urban area and whether it operates under the full school day program, are also drawn from the enrollment database. In addition, student–teacher ratios are obtained from the Educators Database.

Table 3 reports pre-reform (2014) baseline characteristics across treatment cohorts. Because of the large sample size, two-sample t-tests reject equality of means across groups. However, evaluating standardized mean differences demonstrates that these statistical differences are economically negligible. Across virtually all key demographic and socioeconomic dimensions, including baseline test scores, maternal education, and household income, Cohen's d effect sizes remain well below the standard 0.20 threshold (Imbens and Rubin 2015) for baseline imbalance ($d \leq 0.163$ for test scores; $d \leq 0.088$ for income[8]).

Minor structural variations exist in rurality (15.2% in 2015 vs. 22.9% in 2016; $d = 0.196$) and student-to-teacher ratios ($d = 0.498$). To ensure these baseline differences do not confound our causal estimates, the main econometric strategy conditions on this rich set of time-varying covariates while leveraging the doubly robust Callaway and Sant'Anna 2021 estimator to account for staggered treatment adoption dynamics.

Changes in nutritional requirements of meals. Information regarding the timing of when a school contract starts and ends, and the minimal nutritional requirements, comes from the Chilean public procurement market. Each year, the Junaeb uploads the terms to the public procurement market to its website. This document defines technical specifications regarding nutritional requirements, minimum quality operating conditions, and infrastructure.

[7]For example, if 4th graders in year t took the Natural Sciences test, then 4th graders in year $t+1$ would take the Social Sciences test.

[8]We further examined the full pre-treatment income distributions rather than the means, and find that the difference in means is generated by the top 5 percent of households: baseline income percentiles coincide across the 2015, 2016 and 2017 cohorts up to the 95th percentile, and the distributions are otherwise virtually identical. Since those upper tail households lie above the eligibility threshold and never receive program meals, we don't think they are an issue in this study. Figure A.2 in the Appendix reports these distributions.

Table 3: Pre-treatment baseline descriptive statistics

	(1) Schools 2015	(2) Schools 2016	(3) Schools 2017	(1) vs. (2) p-val / [cohen-d]	(1) vs. (3) p-val / [cohen-d]	(2) vs. (3) p-val / [cohen-d]
Math Score	237.771	242.009	245.649	0.000	0.000	0.000
	(47.783)	(48.196)	(48.571)	[0.088]	[0.163]	[0.075]
Reading Score	248.163	251.966	254.360	0.000	0.000	0.000
	(48.911)	(49.090)	(48.587)	[0.078]	[0.127]	[0.049]
Gender	0.480	0.480	0.483	0.967	0.576	0.587
	(0.500)	(0.500)	(0.500)	[0.000]	[0.005]	[0.005]
Age	10.126	10.082	10.092	0.000	0.000	0.065
	(0.659)	(0.639)	(0.622)	[0.068]	[0.053]	[0.016]
Mother schooling	11.674	11.404	11.422	0.000	0.000	0.551
	(3.219)	(3.301)	(3.329)	[0.083]	[0.077]	[0.005]
Income	288,240.6	309,522.7	308,358.3	0.000	0.000	0.622
	(231,534.1)	(251,527.6)	(268,387.9)	[0.088]	[0.080]	[0.004]
Rural (=1)	0.152	0.229	0.210	0.000	0.000	0.000
	(0.359)	(0.420)	(0.407)	[0.196]	[0.149]	[0.048]
Full school day (=1)	0.804	0.752	0.727	0.000	0.000	0.000
	(0.397)	(0.432)	(0.446)	[0.127]	[0.183]	[0.057]
Students per teacher	16.264	19.031	17.055	0.000	0.000	0.000
	(4.814)	(6.110)	(5.656)	[0.498]	[0.150]	[0.335]
Observations	22,626	27,292	26,565			

Note: Pre-treatment baseline averages (year 2014) restricting to students with non-missing test scores with standard deviations in parentheses. Income is monthly income in Chilean Pesos (1 USD $\approx$ 1,000 CLP). Columns (4) through (6) report two-sample t-test p-values with corresponding Cohen's-d effect sizes enclosed in brackets underneath to show that statistical differences are economically negligible. See Figure A.2 in the Appendix for the income distribution of the three groups.

Longitudinal Early Childhood Survey. To complement our main administrative records, we incorporate microdata from the Longitudinal Early Childhood Survey (*Encuesta Longitudinal de la Primera Infancia*, ELPI). The ELPI is a nationally representative longitudinal panel that tracks the same cohort of Chilean children across waves (2012 and 2017 in this paper). Because the survey records the specific educational establishment each child attends in each wave, we are able to link these individual panel records directly to our administrative dataset, mapping the staggered adoption timing of the school meals program to each child.

This linked panel provides rich child- and household-level data unavailable in standard educational records. It contains detailed socio-demographic, employment, and income information for the household. At the child level, the survey records the frequency of consumption for specific food categories at home, capturing a detailed picture of both healthy and less healthy home dietary habits. Furthermore, the ELPI includes objective anthropometric measurements, specifically, weight and height collected in the 2012 and 2017 waves, alongside extensive modules on parenting practices and parental educational expectations. Exploiting the longitudinal structure of these waves allows us to observe household environments, dietary inputs, and developmental indicators within the exact same child over time, enabling estimations that net out time-invariant household characteristics.

3.2 Difference-in-Differences Specification

The identification strategy exploits the staggered adoption of the SMP reform, allowing us to compare the academic performance of students in early-adopting schools with those in schools not yet treated. Because all schools are treated by 2017, our analysis focuses on the transition window where treated and untreated schools coexist.[9]

To avoid the well-documented biases of two-way fixed effects (TWFE) under staggered adoption and heterogeneous treatment effects (Goodman-Bacon 2021; Sun and Abraham 2021), our primary specification relies on the doubly robust estimator developed by Callaway and Sant'Anna 2021 (CS)[10]. This approach estimates group-time average treatment effects, $ATT(g,t)$, comparing the evolution of outcomes for a cohort treated in year g to the evolution for not-yet-treated cohorts:

$$ATT(g,t) = \mathbb{E}[Y_t - Y_{g-1} \mid G_g = 1] - \mathbb{E}[Y_t - Y_{g-1} \mid D_t = 0] \tag{1}$$

Where G_g is an indicator for initial treatment in year g (2015, 2016, or 2017), and D_t indicates treatment status in year t. We aggregate these parameters into a single overall treatment effect δ:

$$\delta = \frac{1}{\kappa} \sum_{g=2015}^{2017} \sum_{t=2010}^{2016} w(g,t) ATT(g,t) \tag{2}$$

Where $w(g,t) = \mathbb{1}\{t \geq g\} P(G = g \mid G \leq 2016)$ and $\kappa = \sum_{g=2015}^{2017} \sum_{t=2010}^{2016} P(G = g \mid G \leq 2016)$.

The coefficient of interest, δ, captures an intention-to-treat (ITT) parameter. While virtually all public schools in Chile participate in the SMP, not all students within these schools receive the meals[11]. Therefore, we interpret δ as the average effect of exposure to the healthier SMP on student test scores for students attending treated schools, relative to those in schools not yet treated. Standard errors are clustered at the municipality level, as the assignment of meal vendor contracts is determined at this level[12].

[9] Schools treated in the final year of implementation serve as never-treated units for identification purposes, as all schools are exposed by 2017.

[10] While imputation-based approaches (Borusyak et al. 2024) are alternative solutions, we prefer Callaway and Sant'Anna 2021 as its identification rests strictly on local comparisons between each treated cohort and units not yet treated in the same period. For the same reason, our robustness checks utilize De Chaisemartin and d'Haultfoeuille 2020.

[11] The SIMCE dataset lacks a direct administrative flag for PAE eligibility. Section 5.1 explores effect of the program by different eligibility scenarios Also, see Section A.1 for eligibility requirements

[12] We cluster standard errors at the municipality level. Treatment timing is assigned at the procurement-group level, but the level of assignment and the appropriate level of clustering need not coincide: following Abadie et al. 2017, the relevant criterion is the level at which residuals in the outcome equation are correlated. Public school administration in Chile was decentralized to municipalities under DFL N° 1-3063 (Ministerio del Interior 1980), which made municipalities the legal employer of teaching staff and placed hiring, the allocation of staff and resources across schools, infrastructure, and the administration of the per-student subsidy under local government control, while regional governments hold no formal authority over public school management (Došek and Eaton 2025; Brosio and Jiménez 2013). Even for geographically broad shocks such as natural disasters (such as wildfires and earthquakes), the emergency response operates below the regional level, since decrees under Ley N° 16.282 (Congreso Nacional de Chile 1977) define their scope by province and municipality and the Ficha Básica de Emergencia targets affected

The identification strategy relies on the key assumption that, in the absence of the improvements in nutritional requirements, students in treated and not-yet-treated schools would have followed parallel trends in test scores. Although this assumption cannot be directly tested, we provide supportive evidence through graphical analysis of pre-treatment trends and an event-study specification that controls for covariates. Finally, for baseline comparison, we also report standard TWFE estimates[13].

3.3 Event Study Specification

To evaluate the parallel trends assumption and the dynamic evolution of the treatment effect, we estimate an event-study specification. Using the CS framework, the dynamic effects are aggregated as:

$$\delta_{es}(e) = \sum_{g=2015}^{2017} \sum_{t=2010}^{2016} \mathbb{1}\{t - g = e\} P(G = g \mid t - g = e) ATT(g, t) \tag{4}$$

Where $\delta_{es}(e)$ captures the average effect e periods relative to the onset of treatment[14].
The coefficients $\delta_{es}(e)$ measure the difference in outcomes between treated and not-yet-treated schools k years from reform implementation, normalized to the period strictly prior to treatment ($k = -1$).

This event-study specification is particularly valuable in our context, as it enables us to examine the dynamics of the treatment effect over time. In addition, it allows us to visually assess the plausibility of the parallel trends assumption, which is central to the identification strategy. Specifically, if the test scores of treated and untreated schools would have evolved similarly in the absence of the intervention, then the coefficients δ_{-6} through δ_{-2} should not be statistically different from zero. Significant deviations in these pre-treatment periods would suggest potential violations of the parallel trends assumption and raise concerns about the validity of the identification strategy.

households within municipalities (Campos and Gil 2025; Sehnbruch et al. 2017). Outside the Metropolitan Region, treatment cohort and region coincide, since each non-metropolitan region was assigned to a single procurement group, so municipal, cohort, and regional clustering cannot be separately distinguished on assignment grounds alone. Our choice therefore rests on the absence of an institutional mechanism through which region-level shocks to test scores would arise independently of municipal-level factors already captured by the specification, rather than on within-region variation in treatment timing. Table A.9 reports results under alternative clustering levels.

[13]The equation to estimate in TWFE is:

$$y_{ist} = \theta + \delta H_{ist} + \beta X_{it} + \alpha Z_{st} + \lambda_s + \gamma_t + \varepsilon_{ist} \tag{3}$$

Where y_{ist} is the standardized test score for student i in school s and year t. H_{ist} indicates exposure to the healthier meal. X_{it} and Z_{st} are student and school characteristics, respectively, while λ_s and γ_t are school and year fixed effects.

[14]We estimate the analogous TWFE event study as:

$$y_{ist} = \theta + \sum_{k=-6, k\neq -1}^{1} \delta_k D_{isk} + \beta X_{it} + \alpha Z_{st} + \lambda_s + \gamma_t + \varepsilon_{ist} \tag{5}$$

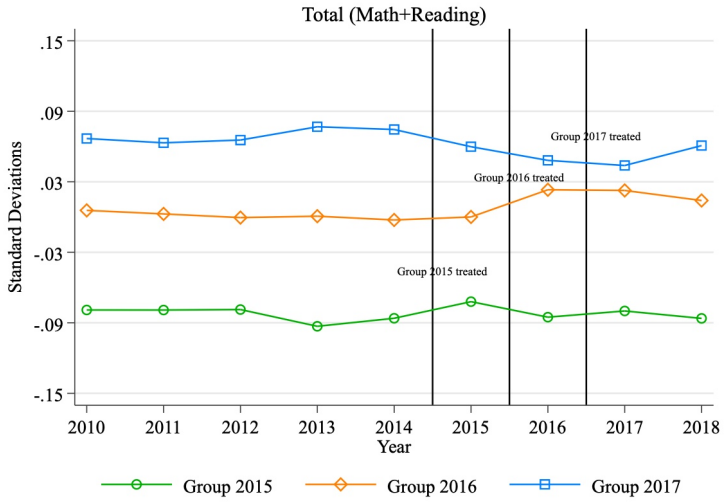
3.4 Internal Validity

3.4.1 Raw Test Scores

We conduct several tests to assess the validity of the parallel trends assumption. First, we examine the raw pre-treatment test score trends across treatment and control groups. Figure 3 presents the average standardized Math and Reading test scores for schools assigned to different treatment cohorts. Specifically, schools treated in 2015 represent the early adopters, those treated in 2016 the mid adopters, and schools treated in 2017 serve as the latest adopters, functioning as a pure control group in the initial periods. A key requirement for the validity of the DiD strategy is that the outcome trends across groups evolve similarly prior to treatment.

Figure 3 shows that, prior to 2015, the standardized test scores in Math and Reading followed similar trajectories across the three groups. Once the 2015 cohort begins receiving the healthier meals, we observe a relative increase in test scores for this group, while the 2016 and 2017 groups continue to exhibit similar trends. A comparable pattern emerges in 2016 following the treatment of the second cohort, with a visible rise in test scores for the newly treated schools. These trends provide visual evidence in support of the parallel trends assumption: prior to treatment, the performance trajectories of early-treated schools do not appear to differ from those of schools not yet treated.[15].

Figure 3: Raw Math+Reading Test Scores 4th grade: 2010-2018



Note: This figure shows raw test scores across the three groups to provide visual evidence supporting the parallel trends assumption. The green bottom line corresponds to the group treated in 2015, the orange middle line to the group treated in 2016, and the blue top line to the group treated in 2017. The timing of the treatment depends on groups of municipalities defined centrally in the late 1990s to promote competition and reduce school meal prices. See Section 2.2.1 for a detailed explanation of the bidding process.

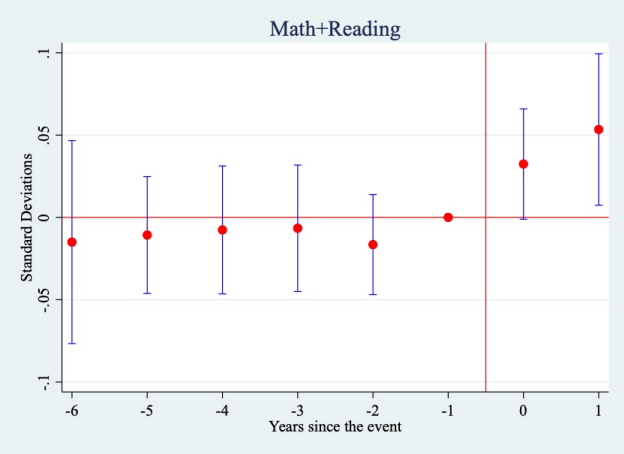
[15] Appendix Figure A.10 displays Math and Reading test scores separately, reinforcing the patterns observed in the combined scores.

3.4.2 Parallel Trends Assumption

Figure 4 presents the event-time study estimates for total test scores based on the method of Callaway and Sant'Anna 2021. The results support the parallel trends assumption, as all coefficients for the pre-treatment periods, from -6 to -2, are statistically indistinguishable from zero. Disaggregating by subject, Figure 5 displays the event-time estimates for Math and Reading scores separately. These results are consistent with the total score findings, as none of the pre-treatment coefficients in either subject are statistically different from zero, further supporting the validity of the parallel trends assumption.

To formally assess the joint significance of the pre-treatment coefficients, we test the null hypothesis that all pre-treatment effects are equal to zero. For combined scores, Math, and Reading, the null is not rejected, with p-values of 0.60, 0.43, and 0.70, respectively (see "Pretrends=0 (p-value)" in Panel A in the Table 5). These findings provide additional reassurance that there are no systematic differences in trends prior to treatment and lend credibility to a causal interpretation of the estimated effects.

Figure 4: Event Study for Math + Reading Test Scores

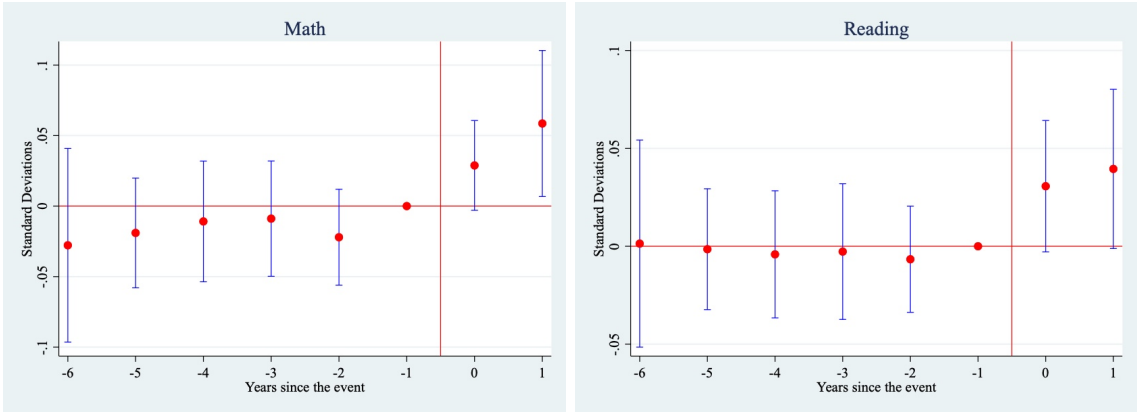


Note: The figure presents the event study using the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Observations are at the individual year lev el. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teachers ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

3.4.3 Selection on Observables

One potential threat to the identification strategy is that the estimated effect may be confounded by changes in the composition of school groups that coincide with the adoption of healthier meals. If such changes occurred, the estimated impact could reflect shifts in students' individual characteristics rather than the effect of improved nutritional quality. To address this concern, we examine whether

Figure 5: Event Study by Subject



Notes: The figure presents the event study using the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Observations are at the individual year level. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teachers ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

the timing of the introduction of the new menu is systematically correlated with student, family or school characteristics.

Table 4 reports the estimated correlations between the implementation of the healthier SMP and a set of student and school-level observables. The results indicate no economically meaningful correlation in any specification. While column 2 shows a statistically significant coefficient, the magnitude of the effect is negligible. Specifically, the likelihood of being a girl is 0.8 percentage point higher in treated schools, relative to a mean of 50 percent. This difference is too small to plausibly drive the estimated treatment effects.

Table 4: The correlation between student and school characteristics and the timing of the healthier SMP.

	(1) Mother Educ.	(2) Girl†	(3) Age	(4) Income	(5) Students/Teacher	(6) Rural†	(7) Full School Day†
Healthier	-0.0495	0.0084*	0.0023	-594.9	0.2165	-0.0083	-0.0162
	(0.0355)	(0.0048)	(0.0038)	(3,379.8)	(0.1452)	(0.0051)	(0.0083)
Mean Y	10.91	0.50	9.95	288,822.61	18.98	0.21	0.85
Observations	445,775	445,775	445,775	445,775	445,775	445,775	445,775

Note: Each column represents a separate regression using Callaway and Sant'Anna 2021 method. The dependent variable is a student characteristic. Observations are at the individual school/year level. All columns include school and year fixed effects. † means that the variable is a dummy. Standard errors are clustered at the municipal level and are in parentheses. *p < 0.05, ** p < 0.01, *** p < 0.001.

3.4.4 Selection into test taking

Test-day attendance might not be random with respect to academic performance. Cuesta et al. 2020 show that although overall school attendance in Chile is high (92 to 95 percent), it rises by roughly 3 percentage point for high-performing students and falls by roughly 2 percentage points for low-performing students on test days, a systematic pattern that distorts school-level scores by about 0.10 SD. If the nutritious meals intervention affects test-day presence, this compositional shift could confound our ITT estimates.

To assess this possibility directly, we replace the dependent variable with a binary indicator equal to 1 if the student took the test and 0 if the student was enrolled in the school but did not take it, and we re-estimate our main specification on this outcome. The estimated effect of the intervention on the probability of taking the test is positive, small and statistically indistinguishable from zero (Coefficient = 0.0098, $p = 0.209$), and a joint test of pre-trends finds no significant differences between treatment and control groups prior to the intervention ($p = 0.238$). Figure A.14 reports the corresponding event-study estimates, which are consistent with the parallel trends assumption. Because we find no evidence that the intervention causally affects test-day attendance, the composition of test-takers is balanced with respect to treatment, and our main ITT estimates are not confounded by selective attendance.

3.4.5 Law 20.606: the Law on the Nutritional Composition of Foods and their Advertising

Law 20.606 altered the same school food environment we study, through three instruments phased in between June 27, 2016 and June 27, 2019: front-of-package "high in" warning labels (Table A.13 in the Appendix reports the exact nutrient thresholds by stage), a ban on advertising labeled products to children under fourteen, and a prohibition on selling, promoting, or distributing labeled products within schools. Law 20.606 is, in principle, a plausible concurrent shock to the same food environment. We now explain why we suggest it does not confound our identification strategy.

Two features of our design address this. First, SIMCE is administered at the end of the school year, so the first post-treatment test for the 2015 cohort was taken roughly nine months before Law 20.606's first stage took effect by the end of June 2016. The estimated effect of 0.032 (s.e. 0.018) from that cohort's first year of exposure cannot mechanically reflect the law's warning labels, advertising restrictions, or school-sales ban, since none of these instruments existed at the time of testing. In addition, the 2016 cohort was exposed to Law 20.606 by the end of June, or to be more realistic, mid July since the first two weeks of July correspond to winter break. Since the SIMCE test is written in October, cohort 2016 would have had, roughly, 3 months of exposition to this intervention. Second, and more generally, Law 20.606's phase-in is defined over product nutritional thresholds, not over schools: the same thresholds apply nationwide and reach every school simultaneously at each stage. Because our main estimates follow the group-time approach of

Callaway and Sant'Anna 2021, each comparison is between a treated cohort and not-yet-treated schools observed over the same calendar period, so a nationwide change common to both groups enters both sides of the comparison and cancels out. For the law to confound our estimates, it would have to affect treated schools differently from not-yet-treated schools in the same period, which its national, product-based design rules out.

As an additional contextual point, the nutritional thresholds were more lenient in the first stage of implementation and tightened only in June 2018 and June 2019. Some products could therefore be sold without a warning seal in 2016 and require one from June 2018 onward[16]. This confirms that the phase-in operates on a product's nutrient content rather than on its identity, timing, or category, consistent with the point above that it generates no school-level variation. It also implies that the law's regulatory intensity was modest at the outset, since the first-stage cutoffs still permitted many such products to remain available in school kiosks, which likely limits the law's measured impact on student consumption in our earliest sample periods.

To provide empirical support for this identifying assumption, we leverage two questions from the SIMCE student questionnaire to test whether the implementation of Law 20.606 differentially affected treated and not-yet-treated schools. Recall that one of the policies introduced by Law 20.606 was a ban on school advertising of all foods carrying warning labels. Accordingly, the first question we use asks students whether they have been exposed to advertising in school promoting sugary drinks or fast food. We estimate a null average effect for 2015 and 2016 (0.005) and, as shown in the event study (Figure A.25), we observe no significant differential effect in 2016, following the law's introduction.

In addition, as described in Section 2.3, the reformulation of the SMP was not known to parents or teachers. This is not the case for Law 20.606, which was widely publicized. For that reason, our second question asks whether teachers encouraged students to bring healthy snacks. If we find an effect on this outcome, students might be eating healthier as a spillover of the law rather than exclusively as a result of the reformulation of the SMP. For this outcome, we find a small, statistically insignificant effect in the pre-treatment period in 2015 (0.007, see the event study in the same figure). While data limitations prevent us from observing this variable in 2016, the 2015 null result mitigates concerns about anticipatory behavioral shifts among school staff prior to the law's enforcement. Both event studies support the parallel trends assumption for these school environment proxies. Taken together, these results provide suggestive evidence that our main parameter isolates the causal effect of Junaeb's menu upgrade, net of a national food-environment policy that affected

[16] An example that illustrates the relative leniency of the first stage: Powerade Mountain Berry Blast, a widely sold U.S. sports drink, contains 80 calories, 240 mg sodium, and 21 g total sugars per 12 fl oz serving, equivalent to 22.5 kcal, 67.6 mg sodium, and 5.92 g sugar per 100mL. Energy and sodium remain under the threshold at every stage, but sugar falls just under the Stage 1 cutoff of 6 g/100mL, while exceeding the Stage 2 and Stage 3 cutoff of 5 g/100mL (see Table A.13).

treated and not-yet-treated schools alike.

4 Results

4.1 Main Results and Event Study Analysis

Table 5 reports the main results from Equation 2 for combined Math and Reading test scores, as well as for each subject separately. Panel A presents the estimates using the Callaway and Sant'Anna 2021 approach, while Panel B reports the corresponding two-way fixed effects (TWFE) estimates for comparison.

Column 1 in Panel A shows that the average effect of the healthier SMP on combined test scores is an increase of 0.039 SD. This effect is equivalent to adding one hour to the school week (Lavy 2016), or extending the school year by five days (Hansen 2011). The analogous TWFE estimate in Panel B, Column 1, indicates a smaller increase of 0.028 SD, suggesting that in this setting, TWFE tends to underestimate the treatment effect. Columns 2 and 3 in Panel A present results by subject. The healthier SMP increased Math and Reading scores by 0.038 and 0.033 SD, respectively. As with the combined scores, the TWFE estimates for each subject are slightly lower, although the differences relative to the CS estimates are not statistically significant[17].

Figure 4 displays the dynamic treatment effects. We observe a positive trajectory in combined test scores following the implementation of the policy, that is, the effect in the second period is larger than in the first. When disaggregated by subject in Figure 5, the results mirror the pattern observed in the combined scores. Although the differences between the first and second periods are not statistically significant, the increase in magnitude over time provides suggestive evidence of dynamic treatment effects[18].

To provide a more comprehensive picture of student performance, we also examine two additional administrative outcomes: teacher-assigned grades (GPA) and end-of-year grade retention. We find no statistically significant effects of the nutritional reform on either of these margins (Figure A.33 in the Appendix). The divergence between standardized test score gains and flat GPAs is a

[17] Standard errors are clustered at the municipality level in our preferred specification, justified on institutional grounds in Section 3.2. As an additional clustering errors at clustering at the regional level, we implement cluster jackknife inference following MacKinnon and M. D. Webb 2018 via M. Webb 2025 the only feasible correction given that wild cluster bootstrap is not currently available for the Callaway and Sant'Anna 2021 estimator. Jackknife standard errors are modestly larger than the standard cluster-robust estimates, and all three outcomes remain significant at the 10% level (0.061, 0.074, and 0.058 for Total, Math, and Reading, respectively), though not at 5%. We do not treat this exercise as conclusive regarding inference at the regional level, given the cluster jackknife inference known limitations relative to wild cluster bootstrap, but report it for completeness given the alternative clustering assumption discussed above.

[18] In the appendix, Figure A.11 presents the event-time studies for total, Math, and Reading test scores using the TWFE approach. Unlike the Callaway and Sant'Anna method, TWFE estimates suggest smaller impacts in the second period relative to the first. This discrepancy highlights that the Callaway and Sant'Anna method accounts for heterogeneous treatment effects across groups and over time.

Table 5: The effect of healthier SMP on test scores

	(1) Math+Reading	(2) Math	(3) Reading
Panel A: Callaway & Sant'Anna			
Healthier	0.0389**	0.0380**	0.0334**
	(0.0163)	(0.0164)	(0.0158)
Pretrends = 0 (p-value)	0.6041	0.4342	0.7019
Panel B: TWFE			
Healthier	0.0277**	0.0287**	0.0219*
	(0.0140)	(0.0146)	(0.0131)
Pretrends = 0 (p-value)	0.8115	0.746	0.8887
Observations	440,690	444,404	442,538

Notes: Each column represents a separate regression. The dependent variable is the test scores measured in standard deviations. Observations are at the individual year level. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teachers ratio. Panel A uses Callaway and Sant'Anna 2021 method. Panel B include year and school fixed effects and uses TWFE method. The null hypothesis of the pretrends test is the presence of parallel trends. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 340. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

common feature in the literature, as teachers frequently grade on a relative curve or adjust their grading standards to classroom averages rather than absolute knowledge. Similarly, grade retention represents a discrete and relatively extreme administrative outcome that may not be sufficiently sensitive to the continuous improvements induced by the dietary changes. This is particularly true in our context, as baseline retention rates for fourth-grade students are already structurally very low, leaving little margin to detect policy-induced reductions.

4.2 Robustness Checks

4.2.1 De Chaisemartin and d'Haultfoeuille estimator

In this section, we conduct robustness checks on our main results using the estimator proposed by De Chaisemartin and d'Haultfoeuille 2020 (CH), which like Callaway and Sant'Anna 2021 accounts for heterogeneous dynamic and group effects. The key distinction is that the parallel trends assumption in this approach is unconditional, meaning it does not require conditioning on observed covariates to hold. This feature allows us, for instance, to include linear geographical trends, thereby relaxing the identification assumption by focusing on deviations in test scores from long-term trends caused by the implementation of healthier meals.

Table A.7 presents the results for combined Math and Reading scores. Column 1 replicates our preferred CS estimate, while columns 2 to 6 report the CH estimates, including and excluding covariates and adding municipality-specific linear trends. The estimated effects range from 0.040 to 0.050 SDs, similar in magnitude to our preferred estimate in column 1, reinforcing the robustness of the main findings. Table A.8 reports the CH estimates by subject, and the conclusions remain consistent with those of the main analysis. Finally, we present event studies using the CH methodology in Figure A.12. These event studies employ placebo estimates rather than traditional pre-treatment estimates, using period -1 as the baseline.[19] The results support the same conclusions as those obtained from our primary empirical approach.

4.2.2 Placebos, Clustering and Pseudo-ATTs.

Placebo A remaining concern is that our estimates capture the effect of a change in meal vendor rather than the effect of the change in the nutritional composition of the menu. The staggered assignment of contracts operates in every three-year cycle, but nutritional requirements were revised only in the cycle beginning in 2015, so the preceding cycle provides a natural falsification test. We therefore re-estimate our main specification on the 2012-2014 cycle, in which Groups 1, 2, and 3 received new contracts in 2012, 2013, and 2014 respectively, under unchanged nutritional guidelines. We use the same estimator, the same outcomes, the same covariates, and standard errors clustered at the municipality level. Figure A.13 reports the resulting event-study estimates for the overall test score and, separately, for mathematics and reading. Post-contract coefficients are small and statistically indistinguishable from zero for all three outcomes, and pre-contract coefficients show no differential pre-trends. The absence of a measurable effect when contracts rotate without a change in menu composition supports interpreting our main estimates as the effect of improved nutritional quality, not of vendor turnover as such.

Levels of clustering and geographical trends Table A.9 in the appendix shows that the estimated effects are robust to alternative clustering levels, including clustering at the school and province levels. We additionally report inference clustered at the regional level, using the cluster jackknife of MacKinnon and M. D. Webb 2018[20]. Finally, Table A.10 shows that the TWFE estimates, underestimated compared to CS, are also robust to the inclusion of geographical linear trends.

[19] These event studies use placebo estimates in place of pre-treatment coefficients, employing period -1 as the reference period. The placebo method is described in detail in the following paragraph.

[20] Since the wild cluster bootstrap is not currently available for the Callaway and Sant'Anna 2021 we use M. Webb 2025 implementation. Jackknife standard errors are modestly larger than the cluster-robust ones, and all three outcomes remain significant at the 10% level (p-values of 0.061, 0.074, and 0.058 for Total, Math, and Reading, respectively), though not at the 5% level. We do not treat this exercise as conclusive regarding inference at the regional level, given the known limitations of the cluster jackknife relative to the wild cluster bootstrap, but report it for completeness given the alternative clustering assumptions discussed above.

Pseudo-ATTs We estimate pre-treatment placebos or pseudo-ATTs. The pre-trend coefficients from De Chaisemartin and d'Haultfoeuille 2020 differ from the standard pre-trend tests used in event studies, as the latter may be invalid when treatment effects are heterogeneous. These placebo ATT estimates correspond to what would have been estimated if the treatment had started t periods prior to its actual implementation. Consequently, the placebo coefficients provide a partial test of the parallel trends assumption by evaluating whether test scores for adopters and non-adopters followed similar trajectories prior to the implementation of the healthier meals. The results of these estimates are shown in Figure A.35, where none of the pre-treatment pseudo-ATTs or placebos are statistically significant. This suggests the absence of anticipation effects and reinforces the plausibility of the parallel trends assumption. Furthermore, Figure A.35 presents the estimates for Math and Reading separately. In both cases, the interpretation remains consistent with the results based on combined test scores.

4.2.3 Sixth graders

Finally, we use a sample of 6th graders instead of the 4th graders employed in the main estimates. If the improved meals positively affect test scores, we should observe a similar pattern in other grades. The caveat for this analysis is that we have fewer years available to estimate the effect (two years pre- and post-treatment) and to assess pre-treatment trends[21]. Table A.11 reports an increase of 0.018 SD in combined test scores; however, this effect is not statistically significant at conventional levels. For Math, we observe a statistically significant increase of 0.027 SD, while there is no effect detected in Reading. The corresponding event studies are presented in Figure A.34.

Regarding the 6th-grade results, the literature examining the effect of food provision on educational outcomes generally finds smaller impacts in higher grades compared to lower ones. For instance, Howard 2011 investigates the relationship between food insecurity and non-cognitive skills in school-aged children, concluding that the effects are more pronounced during early childhood. Similarly, Kurtz et al. 2020 evaluates a weekend feeding program in Northwest North Carolina and finds significant improvements in Reading test scores for 3rd graders, with weaker effects observed for 4th and 5th graders. Ruffini 2021, studying the Community Eligibility Provision program in the United States, finds that the largest gains are concentrated among students in elementary grades (3 to 5), with more muted impacts for those in middle school (grades 6 to 8). Likewise, Gordanier et al. 2020, using data from South Carolina on the same program, reports improvements in Math scores for elementary students but not for middle school students. Taken together, the results for 4th and 6th graders suggest that the effectiveness of this healthy food policy is greater when implemented earlier in children's educational trajectories.

[21] This limitation arises because 6th graders do not take the test every year, with the exception of the four-year period between 2013 and 2016.

5 Suggestive Mechanisms

The adoption of the healthier SMP increases test scores by 0.039 SD in treated schools. To understand the potential mechanisms behind this result, we draw on the existing literature. The primary channel proposed is increased exposure to nutritious food: improved nutrition fosters cognitive development, which in turn enhances educational outcomes, as reviewed in Section 2.1. However, due to data limitations, we are unable to test directly biological mechanisms such as changes in micronutrient intake, energy balance, or biomarkers of cognitive development.

To address this limitation, we pursue several complementary strategies. First, in Section 5.1 we show that the students most likely to receive SMP meals (rural students, students attending full-day schools, and eligible students) are those whose test scores increase the most (between 0.043 and 0.071 SD). Second, in Section 5.2 we explore the two distinct roles of income, as a proxy for baseline dietary quality and as a determinant of SMP eligibility. Since eligibility is not recorded in our data but coverage is likely universal in the first and second terciles, we argue that the difference in effects (0.066 and 0.036), although not statistically significant, is informative about baseline diet. In Section 5.3 we examine possible spillovers of whether exposure to the SMP changes what food students buy or consume at school and what food they eat at home. We find significant effects only for the consumption of healthier snacks at school and healthier foods at home. This allows us to rule out a crowding-out mechanism, whereby healthier meals at school induce parents to substitute cheaper, less nutritious alternatives for healthier meals at home.

In Section 5.4 we examine whether the healthier meals reduced BMI, so that this channel, rather than exposure to nutritious food alone, drives the results. We find a null effect across three different datasets and empirical strategies. That is expected, since in Section 5.5 we remind that lunch, the main meal students receive at school, did not change in caloric composition, ruling out a change in calories as a mechanism. In Section 5.6 we look at attendance. Attendance can be thought of as a proxy for health: healthier meals improve health, and better health increases attendance. We find a precisely estimated zero effect on attendance, ruling out the health-through-attendance mechanism. Finally, in Section 5.7 we test several behavioral mechanisms: parental investments and expectations, and classroom behavior and student perceptions, measured through questions asked of students and teachers regarding students' self-efficacy, classroom climate, and victimization. We find no effects. Overall, among the mechanisms tested, exposure, baseline nutrition, and dietary habits are the ones consistent with the observed results.

5.1 Exposure to Nutritious Food

In Section 4.1 we find that students exposed to the reformulated SMP increase their test scores by 0.039 SD. Since we cannot administratively distinguish eligible from non-eligible students, this coefficient corresponds to an intention to treat effect. We suggest that this effect is driven by

exposure to healthier meals, and in this section we assess that interpretation by estimating effects across subgroups that face different exposure probabilities: rural vs. urban students, students attending full vs half school day schools, and eligible vs. ineligible students. Rural students and students in full school day schools are more likely to participate in the SMP, and eligible students are more likely to receive meals by definition. We find that these groups drive the results, with effects ranging between 0.043 and 0.071 SD.

A student's rurality status, enrollment in a full school day school, and program eligibility are structurally predetermined and highly costly for households to manipulate. Because it is practically impossible for families to systematically relocate geographically, change school infrastructure regimes, or alter their socioeconomic vulnerability status solely in response to a dietary upgrade, these characteristics serve as plausibly exogenous sources of variation in treatment exposure. If the nutritional channel is the primary mechanism, the impact on total test scores should be larger in groups facing weaker constraints on meal take-up, unaffected by endogenous sorting.

Table 6 presents estimates across these three dimensions of varying treatment intensity. Throughout, we interpret the ordering of the point estimates across columns rather than the statistical significance of the differences between them: equality of the estimates across subgroups cannot be rejected, but their ranking aligns monotonically with the institutional exposure margins, which is consistent with the proposed mechanism.

We first examine program eligibility with a proxy created with available information[22], which governs the baseline probability of receiving the meals. As expected, the cognitive improvements are entirely concentrated among the eligible population, who experience a highly significant 0.045 SD increase in Column 1. In contrast, the estimate for non-eligible students is a null in Column 2 (0.002 non-significant). This divergence acts as an implicit falsification test, confirming that the benefits occur only to those explicitly targeted by the nutritional intervention. The event studies are in the figure A.15 who support the validity of the estimates.

We next exploit institutional variation in time spent at school, which determines the number of meals a student can consume. Although both full and half day students are entitled to school lunch, actual take-up differs sharply: ELPI data indicate that full day students consume school lunches (where the core of the food modifications was concentrated) at a rate of approximately 78%, compared to approximately 40% among half day students. The empirical estimates track

[22]The SIMCE dataset lacks a direct administrative flag for PAE eligibility. Because of that, we construct a proxy that mirrors the program's official allocation rules. A student is classified as eligible if: (1) they attend a rural school (which confers automatic eligibility), (2) their mother lacks a complete high school education, or (3) their household income falls below the bottom 60% of the public school distribution (capturing the national 40% vulnerability threshold and associated welfare programs). This rule yields an eligible sample share of 67% in urban areas and 100% in rural areas, aligning closely with actual administrative take-up rates (∼68% and ∼95%, respectively). See Section A.1 for more detail respect the program's official allocation rules.

Table 6: The Impact of Healthier Meals on Total Test Scores by Exposure

	Eligibility Status		School Day Type		Rurality Status	
	(1) Eligible	(2) Non-Eligible	(3) Full-Day	(4) Half-Day	(5) Rural	(6) Urban
Healthier	0.045***	0.002	0.043**	0.016	0.071**	0.032*
	(0.015)	(0.035)	(0.017)	(0.034)	(0.029)	(0.019)
Observations	373,109	67,581	373,730	66,488	92,109	348,581

Notes: Table displays the overall average treatment effect on the treated (ATT) for total scores extracted from Callaway & Sant'Anna (2021) difference-in-differences models across various subsamples. Standard errors clustered at the municipality level are reported in parentheses. All models cover the 2010–2016 period and include individual and school covariates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

this exposure gap, as students enrolled in the full school day exhibit a significant gain of 0.043 SD in Column 3, while the point estimate for half day students falls to 0.016 SD and is statistically indistinguishable from zero. The event studies are in the Figure A.16 and support the validity of the estimates.

Finally, we analyze differences by geographic rurality. Students in rural schools are 100% eligible, and ELPI shows a take-up rate of roughly 95%, which likely reflects limited alternative food environments, whereas urban students display a comparatively lower take-up rate of approximately 68%. Consistent with this intensity gradient, rural students experience the largest cognitive gains in the sample, at 0.071 SD in Column 5. Urban students also benefit from the intervention, although their estimated gain is correspondingly smaller, at 0.032 SD[23]. The event studies are in the Figure A.17 and support the validity of the estimates.

As a final alternative measure of exposure, in Section A.2 of the Appendix we replicate the analysis of Krämer et al. 2021 using attendance as a proxy for treatment intensity, and find the same gradient: the estimated effect is null in the lowest attendance tercile and rises to 0.040 and 0.080 SD in the middle and top terciles (Table A.4)[24].

[23] Analysis by math and reading separately are shown in Table A.15 in the appendix. Results are similar to the total math+reading.

[24] We report these estimates in Section A.2, and not here, because attendance is potentially endogenous to the program and is therefore better treated with caution. However, Table 13 shows that the reformulation did not raise attendance (we find a precise zero), and Tables A.2 and A.3 show no heterogeneous or distributional effects, so the concern is unlikely to be quantitatively important here. Still, these results should be read with caution and as complementary to the predetermined exposure margins reported above.

5.2 Heterogeneity by Household Income: Biological Returns and Institutional Exposure

We next examine treatment heterogeneity by household income. In public schools, income plays two distinct roles, since it is correlated with baseline dietary quality (Mosdøl et al. 2008; Mark et al. 2012) and it also determines SMP eligibility. We do not observe nutritional status directly, so income should be read as a proxy for baseline diet rather than a measure of it. To separate the two roles, we compare the first and second terciles, which face the same eligibility rates, so that under comparable take-up any difference between them is informative about baseline diet rather than about differences in exposure.

Section 2.2.1 describes baseline dietary patterns. As reported in Table 2, home diets of primary school children in public establishments fall short of national clinical guidelines for most food groups. The shortfall is present in all three income terciles and is most pronounced in the lowest tercile, where intake is dominated by calorie-dense staples (bread consumption exceeds the guideline by more than four times) rather than by micronutrient-dense foods. Two facts follow: public school students face a baseline dietary shortfall relative to guidelines, and the size of that shortfall declines with household income.

A natural concern is that the income gradient reflects differences in the length of exposure rather than differences in baseline diet. If schools serving poorer students had adopted the reform earlier, those students would have accumulated more years of exposure, and the larger effects in the lower terciles would follow from timing rather than from initial dietary conditions. Mean baseline (2014) household income is somewhat lower in the earliest adoption cohort (see Table 3 in Section 3.1), but this difference is driven by the upper tail rather than by the bulk of the distribution, since baseline income percentiles coincide across the 2015, 2016 and 2017 cohorts up to the 95th percentile (Figure A.2 reports these distributions), and the cohorts differ only among a small number of high-income households[25].

Table 7 shows the effects on test scores[26] when partitioning the sample into income terciles. We find a clear monotonic decline in treatment effects: students in the first tercile experience a gain of 0.066 SD, while the second and third terciles exhibit gains of 0.033 and 0.023 SD, respectively. These results align with findings of the effect of healthy meals on disadvantaged populations in the United States (Anderson et al. 2018), while contrasting with evidence from higher-income contexts where gains concentrated among socioeconomically advantaged peers (Belot and James 2011).

Eligibility is likely to be effectively universal in the first and second income terciles, so students

[25]These households lie above the eligibility threshold and therefore never receive program meals, so they cannot account for the estimated income gradient. Comparing the distributions rather than the means, adoption timing is unrelated to household income among the students the program actually serves.

[26]Table A.16 shows the effects for Math and Reading separately. Figures A.18 and A.19 show the event studies

Table 7: The Impact of Healthier Meals by Income Tercile

	T1	T2	T3				
			All	Non-Elig.	Eligible	Half Day	Full Day
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Healthier	0.066***	0.036*	0.023	0.003	0.036	-0.014	0.032
	(0.023)	(0.021)	(0.027)	(0.035)	(0.027)	(0.046)	(0.027)
Observations	193,253	112,760	134,677	67,581	67,096	27,712	106,891

Notes: Table displays the overall average treatment effect on the treated (ATT) for total scores extracted from Callaway & Sant'Anna (2021) difference-in-differences models across various subsamples. T1, T2, and T3 means Tercile 1, 2 and 3 respectively. Standard errors clustered at the municipality level are reported in parentheses. All models cover the 2010–2016 period and include individual and school covariates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

in both groups have access to the same meals. The estimated effect nevertheless falls from 0.066 SD in the first tercile to 0.033 SD in the second. The event studies validate these estimates in figure A.18. Since access is similar across the two groups, this pattern is more consistent with diminishing returns to the same meal, where the gain is larger for students starting from a larger dietary shortfall, than with differences in exposure to the program. The same pattern appears for other proxies of socioeconomic vulnerability, with 0.045 SD for students whose mother has no college education against 0.013 SDs for those whose mother has some college[27], and 0.071 SD in rural areas against 0.032 SD in urban areas.

The estimate falls further in the third tercile, to 0.023 SD. Figure A.19 shows the event study that in general support the parallel trend assumption with the exception of five periods before adoption. Eligibility rules restrict program meals to roughly 30% of third-tercile students, so this unconditional estimate combines a minority of students who receive meals with a majority who do not. To separate exposure from baseline diet within this group, we use variation in school schedules. Students in half-day schools tend to, either go home for lunch or arrive after lunch, so many do not eat the program meal even when their school serves it[28]. Among third-tercile students in full school day schools, the estimated gain is 0.032 SD, close in magnitude to the second-tercile estimate, although it is not statistically significant. Among third-tercile students in half-day schools, the point estimate is -0.014 SD and is statistically indistinguishable from zero. The administrative eligibility proxy shows the same pattern, with a gain of 0.036 SD for eligible third-tercile students against 0.003 SD for their non-eligible peers, the two latter indistinguishable from zero. We read these contrasts in terms of magnitude rather than statistical significance, since the subgroup estimates are imprecise. The event studies associated with these estimates are in figures A.20 and A.21 in the

[27]These results are available in table A.17 and figure A.22 in the appendix. The table also shows effect by gender and the event study in the figure A.23.

[28]In Section 5.1 we show that students in schools operating a half-day schedule are less likely to have lunch at school despite being beneficiaries, with a take-up rate of 78% among full school day students compared to 40% among half-day students. This difference is not driven by eligibility.

appendix, being a little more noisier for full school day five periods before the adoption for total scores.

Taken together, these results point to two distinct margins. Within the first and second terciles, where eligibility and access are effectively universal, the estimated effect declines as household income rises, which is consistent with larger returns for students starting from a larger dietary shortfall. In the third tercile, where most students do not receive meals, the unconditional effect is small but recovers once we restrict attention to students more likely to eat at school. Within treated schools, students who are less likely to eat the program meal show effects close to zero, which suggest that the gains may operate through the consumption of the meal rather than through school-level factors. We also acknowledge that the comparisons are underpowered, but the magnitudes and gradient patterns are informative.

5.3 Spillovers on dietary habits

This subsection examines whether exposure to healthier school meals changed students' dietary patterns, focusing on the foods children ate or purchased at school, and at home. Whether institutional food provision reinforces or displaces healthy eating outside school is a central question in the economics of education literature (Afridi 2010; Adamson et al. 2013; Valizadeh and Ng 2020; Marcus and Yewell 2022), and the sign of the effect is theoretically ambiguous.

Three responses are possible. First, healthier school meals may shift students' preferences toward healthier food, so that children request such items at home (a positive spillover, or complementarity). Second, students may dislike the new menus and seek unhealthy substitutes outside school, or parents, aware that their children now eat healthier meals at school, may reallocate the household food budget toward cheaper and more processed items (a compensatory, or crowding-out, response). Third, the reform may leave dietary behavior outside school unchanged, either because home food choices are unresponsive to school menus or because complementarity and crowding out offset each other.

Sections 5.3.1 and 5.3.2 examine each of these possibilities. The first considers the consumption of healthy and unhealthy snacks at school by questions asked to students, and the second the consumption of healthier and less healthy foods at home, asked to parents. We find a positive effect on healthy snacks at school and on the Healthy Dietary Index at home, with no effect on soda, fast food, or candy at school, or on the Unhealthy Dietary Index at home. These findings are consistent with a preference formation channel (Wardle et al. 2003), in which repeated exposure to healthier meals at school gradually shifts children's food preferences, and this shift carries over into food choices at home. This has a direct implication for interpreting our main results: the effect on test scores operates through two channels, exposure to nutritious foods at school and a change in students' preferences that alters what they consume at home.

5.3.1 In school dietary behavior

Using the SIMCE questionnaire of students, we track student-level consumption patterns from 2014 to 2016 across four categories: healthy snacks, sugary drinks, fast food, and candy[29]. It is important to acknowledge a data limitation inherent to our institutional tracking window: because our panel offers a relatively limited number of pre-treatment periods (2) prior to the policy rollout, our evaluation of the parallel trends assumption should be interpreted with this constraint in mind. Nonetheless, as illustrated in Figure A.24 in the appendix, the available pre-treatment event indicators remain stable and statistically indistinguishable from zero across all specifications.

Following the implementation of the reform, we find a statistically significant 1.69 percentage points increase in the probability of children bringing healthy snacks to school[30]. Crucially, this positive behavioral shift is accompanied by a statistically insignificant effects on junk food categories: soda, fast food and candy.

These results suggest a positive behavioral adjustment rather than unintended compensatory substitution. The consumption of healthy snacks increased, while the null effects on soda, fast food, and candy indicate an absence of a compensatory rebound effect. These patterns align with recent evidence that targeted school-level nutritional interventions can shape home-packing habits without inducing adverse outside-of-school dietary substitutions.

5.3.2 Household dietary spillovers

To complement the findings on snacks choices at school, we evaluate whether the institutional nutritional improvements generated broader behavioral spillovers within the household. A key concern with public provision programs is the potential for behavioral crowd-out, where families might compensate for healthier school meals by increasing the consumption of unhealthier alternatives at home. To test this, we evaluate aggregate dietary habits using the ELPI data, leveraging the 2012 (pre-reform) and 2017 (post-reform) waves.

We construct a Healthy Dietary Index (capturing the consumption frequency of fruits, vegetables and diary[31] per week) and an Unhealthy Dietary Index (capturing processed snacks, sugary beverages,

[29]The specific questions are Healthy snacks: "I have eaten/bought healthy snacks at school" (*He consumido / comprado comida saludable dentro de mi escuela: frutas, yogur, cereal, etc.*); Sugary drinks: "I have drunk/bought sugary drinks at school" (*He consumido/comprado bebidas gaseosas dentro de mi escuela*); Fast food: "I have eaten fast food at school" (*He consumido/comprado comida rápida dentro de mi escuela*); and Snacks: "I have eaten/bought sweets and snacks at school" (*He consumido/comprado golosinas y snacks dentro de mi escuela*).

[30]The increase in healthy snacks survives the strict Holm-Bonferroni step-down correction for multiple hypothesis testing ($p < 0.0125$), supporting the conclusion that the finding is not driven by false positives across multiple outcomes.

[31]We left only 2012-2107 comparable categories. For example, Fish or Meats was not included in the index because,

Table 8: Effect of School Meal Reform on Food Consumption in School

	(1) Healthy Snacks	(2) Soda	(3) Fast-Food	(4) Candy
Healthier	0.0169***	-0.0056	-0.0002	0.0048
	(0.0059)	(0.0081)	(0.0095)	(0.0087)
Average	0.82	0.60	0.49	0.74
Observations	175,418	175,418	175,418	175,418

Notes: Estimates are group-time average treatment effects (ATT) calculated using the Callaway and Sant'Anna (2021) estimator for the 2014-2016 SIMCE panel. All dependent variables are dummy variables. The unadjusted p-value for Column 1 ($p = 0.004$) is statistically significant under the strict Holm-Bonferroni step-down correction for multiple hypothesis testing ($p < 0.0125$), confirming the result is not driven by the simultaneous evaluation of four outcomes. The specific questions are Healthy snacks: "I have eaten/bought healthy snacks at school" (*He consumido / comprado comida saludable dentro de mi escuela: frutas, yogur, cereal, etc.*); Sugary drinks: "I have drunk/bought sugary drinks at school" (*He consumido/comprado bebidas gaseosas dentro de mi escuela*); Fast food: "I have eaten fast food at school" (*He consumido/comprado comida rápida dentro de mi escuela*); and Candy: "I have eaten/bought sweets and snacks at school" (*He consumido/comprado golosinas y snacks dentro de mi escuela*). Standard errors are reported in parentheses and are clustered at the municipality level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and sweets per week) consumed at home, as reported by the main caregiver[32].

To isolate the causal effect of the meal quality reform on home consumption, we estimate a child fixed-effects difference-in-differences model. We rely on the baseline (2012) take-up of school meals to define our treatment group[33], estimating the following equation:

$$Y_{it} = \alpha_i + \beta(Take - up_i \times Post_{2017}) + \mathbf{X}'_{i,2012}\boldsymbol{\gamma} \times Post_{2017} + \lambda_t + \varepsilon_{it} \tag{6}$$

In Equation 6, Y_{it} denotes the dietary index outcome for child i in wave t; α_i are child fixed effects; λ_t are survey wave fixed effects; and $\mathbf{X}_{i,2012}$ is a vector of baseline household controls.

Our identification strategy incorporates three key elements to isolate the causal mechanism. First, it estimates an effect on the treated students since treatment status (Take-up$_i$) is fixed strictly at baseline, three years prior to the 2015 reform. This eliminates endogenous sorting, opt-in behavior, or differential program exit driven by the meal quality improvements themselves. Second, the inclusion of child fixed effects, α_i, absorbs all time-invariant individual and family unobservables, such as baseline cognitive endowment, permanent income, and stable parenting preferences. Third, to account for flexible socioeconomic trends or differential developmental trajectories over the five-year window, we interact the post-reform indicator with baseline household income, maternal education, and rural status ($\mathbf{X}_{i,2012} \times \text{Post}_{2017}$). Identification thus relies on conditional parallel

while being asked as a single category in 2012, in 2017 the survey only asked for Fish, and did not ask for Meats.

[32]The underlying unit is consumption frequency per week. The variables for each food type are standardized and then averaged to construct the respective composite healthy and unhealthy indices.

[33]In Chile's public school system, PAE beneficiary status exhibits high administrative persistence: once a child is enrolled as a meal recipient, they typically retain program access throughout primary school, even if administrative prioritization rules or household vulnerability indexes shift in subsequent years.

trends among socioeconomically similar children.

Finally, the identification strategy assumes the absence of dietary spillovers to untreated peers. This is institutionally plausible: non-participating students consume distinct meals not sharing the place where they eat, neutralizing the direct experiential exposure required for peer emulation or shifts in child-driven household food demand. Furthermore, we can empirically rule out informational spillovers to parents; SIMCE data confirms exposure to food advertising remained constant and teachers did not increase their encouragement of healthy snacks (See Section 3.4.5 and Figure A.25). Absent direct child dietary exposure or broader parental informational shocks, the reform is highly unlikely to have altered the home consumption of untreated households.

Moreover, to explore whether behavioral adjustments compound over time, we leverage the staggered rollout of the policy across municipalities. We classify the years of exposure to the policy by 2017 based on the school the child attended, allowing us to estimate an intensity (dose-response) model:

$$Y_{it} = \beta_1(\text{Take-up}_i \times \text{Post}_t) + \sum_{k\in\{2,3\}} \delta_k(\text{Take-up}_i \times \text{Exposure}_{ik} \times \text{Post}_t) + \mathbf{X}'_{i,2012}\boldsymbol{\gamma} \times \text{Post}_t + \alpha_i + \lambda_t + \varepsilon_{it} \quad (7)$$

In Equation 7, Exposure_{ik} are dummy variables for two or three years of policy exposure, leaving one year of exposure as the omitted base category. This specification compares students who receive the meals to those who do not, while simultaneously evaluating the marginal impact of prolonged exposure to the new nutritional standards.

Finally, to align the sample with the relevant school-aged population eligible for the reform, we restrict our estimation to households with children aged 7 or older. All regressions apply household longitudinal sampling weights to ensure national representativeness, and we cluster standard errors at the municipality level to account for arbitrary correlation of shocks within the administrative units where the policy was rolled out.

To further validate specification in Equation 6, Table A.14 presents a falsification test demonstrating that baseline take-up does not predict differential post-reform changes in household observable characteristics, such as income or maternal labor supply. This confirms that the nutritional reform did not act as an income transfer that inadvertently triggered broader socioeconomic shifts, thereby isolating the dietary channel.

Columns 1 to 3 in Table 9 demonstrate a positive, statistically significant increase in the Healthy Dietary Index for eligible households following the reform. Specifically, the baseline DiD estimate in Column 1 shows a significant effect of 0.151 SD, which remains stable at 0.154 SD when introducing

Table 9: Impact of Healthy Meals on Household Dietary Indices

	Healthy Dietary Index			Unhealthy Dietary Index		
	(1)	(2)	(3)	(4)	(5)	(6)
Take-up × Post	0.1514***	0.1544***	0.1434*	0.0352	0.0659	-0.0281
	(0.0403)	(0.0439)	(0.0866)	(0.0534)	(0.0575)	(0.0886)
Take-up × Exposure 2y × Post			0.0184			0.2123*
			(0.1005)			(0.1258)
Take-up × Exposure 3y × Post			0.0175			0.0893
			(0.1038)			(0.1093)
Total Effects (Dynamic Interaction)						
Exposure 1 year (Baseline)			0.1434*			-0.0281
			(0.0866)			(0.0886)
Exposure 2 years (Total Effect)			0.1618***			0.1842
			(0.0584)			(0.1123)
Exposure 3 years (Total Effect)			0.1609***			0.0613
			(0.0610)			(0.0667)
Controls × Post	No	Yes	Yes	No	Yes	Yes
Observations	9,990	9,990	9,990	9,990	9,990	9,990

Notes: Regressions use household sampling weights and restrict the sample to households with children aged 7 or older. All specifications absorb household and survey wave fixed effects. Controls interacted with the post-reform indicator include baseline maternal education, baseline rurality, and baseline household income. Take-up is baseline take-up. Columns 3 and 6 incorporate differential treatment intensity by exposure duration, omitting the base category, one year of exposure, where exposure 2 means 2 years exposed. Total effects for exposure levels 2 and 3 are calculated using linear combinations (Take-up × Post + Take-up × Exposure × Post). Number of clusters: 175. Robust standard errors adjusted for clustering at the municipality level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

time-varying baseline controls (Column 2). When allowing for heterogeneous treatment effects by exposure duration (Column 3), the total dynamic effects remain consistent and significant at approximately 0.161 SD for both exposure groups providing no evidence of dynamic effects.

Conversely, Columns 4 through 6 examine the Unhealthy Dietary Index. While the baseline interaction is positive, it is substantially smaller in magnitude (0.035 to 0.066) and statistically indistinguishable from zero across the primary static specifications. Although the dynamic specification in Column 6 hints at minor positive coefficients but non-significant in Exposure 2 years, the overall narrative reinforces our earlier findings: the reform successfully promoted healthier home dietary habits without triggering a broad, unintended surge in negative or compensatory consumption patterns.

Our findings in these two subsections matches the broader literature on how children's food preferences form. Preference formation is not confined to the home: children develop a taste for food through repeated exposure wherever they are fed, and repeated exposure, not mere availability, is the mechanism. Evidence on younger children shows that new foods typically require ten to sixteen exposures before acceptance, and that tasting, not just offering, drives acceptance. Parents, however, often abandon a food much earlier: roughly a quarter of parents of infants and toddlers concluded their child disliked a food after two or fewer tries (Savage et al. 2007). A school lunch program serving the same reformed menu five days a week for one to three years supplies exactly this

repeated, tasting-based exposure, and experimental evidence confirms that such exposure-based interventions raise both liking and consumption of target foods among schoolchildren (Wardle et al. 2003).

For a shift in a child's own preferences to manifest in the snacks brought to school and meals eaten at home, it must also translate into what is purchased at home, that is, the channel must run from child to parent, not the reverse. This is plausible: children develop a taste for the reformed menu at school, and that taste, rather than a shift in parental beliefs, is what shapes the food parents buy at home. Mothers report that children exert substantial influence over food purchasing decisions, that children know which requests mothers will grant, and that this influence operates largely irrespective of a food's healthfulness (Monalisa et al. 2023).

This bottom-up channel plausibly matters most for lower-income households, where the cost of a refused food is higher and parents are less willing to reintroduce it repeatedly, the very repeated-exposure process the reform performs on their behalf at school (Daniel 2016). Together with the shift in the child's own preferences, this easing of the household budget constraint explains why the effect on the positive index appears after one year of exposure and persists after two to three years.

5.4 Anthropometrics

An improvement in the nutritional composition of school meals may lead to a reduction in obesity rates. Consequently, finding evidence of health status improvements could help explain, at least partially, the observed increases in test scores[34]. In general, the medical literature has established a negative relationship between obesity and cognitive functioning (Liang et al. 2014).

A direct test of whether the program reduced body mass would require data on obesity rates for fourth graders. However, the data for all cohorts evaluated in the main estimation is not available. Because of that, to show evidence on effects of the healthy meals on body weight we use three complementary strategies. The first strategy uses an auxiliary Junaeb dataset, the Mapa Nutricional, as an individual panel that follows the same children across 2012, 2013, 2014 and 2018, and estimates

[34]On the one hand, there is evidence suggesting that school meal programs either reduce or have no significant impact on obesity. For instance, Holford and Rabe 2024 find that transitioning from means-tested to universally provided nutritious school meals in England led to reductions in obesity prevalence and BMI, with stronger effects observed among younger children compared to older ones. Similarly, studies conducted in the United States and Sweden report that participation in school meal programs does not lead to increased body weight (see Mirtcheva and Powell 2013, Dunifon and Kowaleski-Jones 2004 for the US and Lundborg et al. 2022 for Sweden). On the other hand, some research suggests that school meals may contribute to weight gain. Notably, Schanzenbach 2009 and Millimet et al. 2010 provide evidence from the U.S. indicating that school meal participation is associated with increased body mass. However, such concerns are less applicable in our context for several reasons. First, our analysis focuses on a change in the nutritional composition of school meals (an intensive margin adjustment), rather than changes in overall participation in the School Meals Program (an extensive margin effect). Second, the Chilean school meals under study are designed to resemble homemade dishes, with limited provision of ultra-processed or "junk" foods. Third, Caro 2020, using a regression discontinuity design based on eligibility thresholds for first-grade students in Chile, finds no evidence of increased obesity rates as a result of school meal provision.

a dynamic student fixed effects model anchored at 2014, so treatment intensity comes from whether the school adopted in 2015 or 2016 (3 to 4 versus 2 to 3 cumulative years of exposure). The second strategy uses the ELPI, with child fixed effects on the 2012 and 2017 waves, where identification comes from individual take-up interacted with post, plus a triple interaction giving a dose-response by years of exposure. Finally, we again use the Mapa Nutricional for Kindergarten and 1st grade students, applying the CS staggered DiD over 2010 to 2016, the same period and strategy as the main analysis but with younger cohorts. The first two strategies measure children close to the age of our main sample, in 5th grade in 2018 and at age 7 or older in ELPI, while the third measures Kindergarten and 1st graders.

5.4.1 Mapa Nutricional: BMI and categorical nutritional indicators

A direct evaluation of anthropometrics during our primary test score sample period (2010–2016) is constrained by historical data availability. Chile's national nutritional census (*Mapa Nutricional*, Junaeb) was historically restricted to early childhood cohorts (Pre-K, Kindergarten, and 1st Grade), and only in 2018 started collecting information for 5th graders. The survey collects precise individual-level measurements of height (cm) and weight (kg), alongside demographic identifiers (age and sex). From these raw anthropometrics, Junaeb constructs Body Mass Index (BMI) z-scores[35] and categorical nutritional indicators (underweight, normal weight, overweight, and obesity) following World Health Organization (WHO) growth standards.

To bridge this data constraint and evaluate children at the exact same developmental stage as our primary cognitive sample (9 to 10 years old), we construct a longitudinal panel linking individual student records from Mapa Nutricional across four survey waves: 2012 (Pre-K), 2013 (Kindergarten), 2014 (1st Grade), and 2018 (5th Grade). This empirical setup allows us to establish two crucial features: (i) testing for parallel pre-trends prior to policy implementation, and (ii) exploiting differential cumulative exposure duration (2 to 4 years) across adoption cohorts.

Because body mass is a stock that accumulates from daily energy balance rather than a flow responding to immediate dietary change (Grossman 1972, Cawley 2004, Currie and Almond 2011), and because the reform altered the nutritional quality of the meals without imposing a caloric deficit, any effect on body composition should emerge only after sustained exposure. This does not preclude short-run cognitive gains operating through channels unrelated to body mass, which we discuss in Section 6 and Table A.12. We therefore build the panel around cumulative exposure rather than a single post-reform year following Hoynes et al. 2016: by 2018, children in Cohort 2015 schools had received the upgraded meals for 3 to 4 years and those in Cohort 2016 schools for 2 to 3 years. This design ensures that a null result reflects the absence of an effect on body composition rather than insufficient time for physical adjustment to occur.

[35] BMI z-scores measures how much a child body mass index deviates from the average for their exact age and sex.

We estimate a dynamic two-way fixed effects specification anchored at the pre-policy baseline year of 2014 (1st Grade):

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \in \{2015,2016\}} \sum_{\tau \neq 2014} \beta_\tau^k \left(\text{Year}_\tau \times \text{Cohort}_k\right) + \varepsilon_{it} \tag{8}$$

where Y_{it} represents the anthropometric outcome for student i in year $t \in \{2012, 2013, 2014, 2018\}$. Following WHO standards, Y_{it} includes: (i) continuous BMI z-score, (ii) an indicator for combined overweight or obese (BMI $z > +1$ SD), and (iii) an indicator for obesity alone (BMI $z > +2$ SD). The parameter α_i denotes student fixed effects absorbing time-invariant individual heterogeneity, λ_t represents year fixed effects, and Cohort_k identifies schools adopting the upgraded meal program in wave $k \in \{2015, 2016\}$. Standard errors are clustered at the municipality level.

The results are presented in Table 10. Panel A reports the pre-treatment interaction terms (Year 2012 and Year 2013). Across all three health outcomes, the pre-trend coefficients are small in magnitude and statistically indistinguishable from zero. This confirms that treatment and comparison cohorts followed identical physical development trajectories prior to the intervention, validating our identification strategy.

Table 10: Dynamic Student Fixed Effects Estimates on Anthropometric Outcomes

	(1) BMI z-score	(2) Overweight or Obese (=1)	(3) Obese (=1)
Panel A: Pre-Trends (Relative to 2014 Baseline)			
Year 2012 × Cohort 2015	0.009	0.000	0.006
	(0.027)	(0.009)	(0.006)
Year 2013 × Cohort 2015	0.039	0.005	0.007
	(0.027)	(0.009)	(0.006)
Year 2012 × Cohort 2016	0.001	0.002	0.004
	(0.029)	(0.009)	(0.006)
Year 2013 × Cohort 2016	0.031	0.002	0.005
	(0.032)	(0.010)	(0.006)
Panel B: Cumulative Exposure Effects (Post-Policy 2018)			
Year 2018 × Cohort 2015 (3–4 Yrs Exposure)	0.041*	0.007	0.005
	(0.025)	(0.007)	(0.007)
Year 2018 × Cohort 2016 (2–3 Yrs Exposure)	0.039	0.011	0.007
	(0.028)	(0.009)	(0.008)
Observations	405,409	405,409	405,409
Unique Students	126,248	126,248	126,248
Mean of Dep. Var. (2014)	1.053	0.510	0.228

Notes: The sample tracks an individual panel of primary school students in *Mapa Nutricional* across 2012 (Pre-K), 2013 (Kindergarten), 2014 (1st Grade; reference category), and 2018 (5th Grade). Cohort 2015 and Cohort 2016 indicate the year the school adopted the upgraded meal program. All estimations include student and year fixed effects. Standard errors clustered at the municipality level in parentheses. Number of clusters: 342 *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Panel B reports the post-treatment estimates in 2018 (5th Grade), where students experienced differential exposure intensity depending on their school's rollout wave: 3 to 4 cumulative years of

healthier meals for Cohort 2015, and 2 to 3 years for Cohort 2016. Even after multiple years of sustained exposure, we observe no statistically significant or economically meaningful impacts on student obesity rates or overall overweight/obesity prevalence[36]. The absence of dynamic physical changes across differential exposure durations demonstrates that the academic gains documented in Section 4 are not mediated by medium-term reductions in body mass or obesity, pointing instead to direct nutritional and cognitive benefits.

5.4.2 Early Childhood Longitudinal Survey - ELPI: BMI Z-score

As an additional check, we use ELPI data that contains weight and height of children in 2012 and 2017 and the same strategy presented in Section 5.3.2 in Equations 6 and 7 using take-up variation to estimate changes in body mass. Across all specifications, we find non-significant effects. In the baseline child fixed-effects specification (Column 1), the estimated impact is near zero, with a coefficient of 0.009, which is statistically insignificant. Flexibly controlling for differential baseline growth trajectories via baseline controls interacted with the post-reform indicator (Column 2) yields a similarly negligible point estimate of -0.021, which is also statistically insignificant and allows us to rule out positive changes larger than 0.115 SD with 95% confidence.

In Column 3, we evaluate dose-response effects similar to Equation 8. The baseline estimate for the one-year exposure cohort remains small and statistically insignificant at 0.061. The differential interactions for two and three years of exposure show additional negative point shifts of -0.108 and -0.128, respectively, neither of which is statistically significant. The resulting total net effects yield statistically insignificant estimates of -0.047 (standard error 0.128) for two years of exposure and -0.067 (standard error 0.106) for three years. Taken together, these results provide suggestive evidence that the school meal quality improvement successfully enhanced dietary quality and micronutrient intake without distorting physical body composition or inducing caloric imbalances. Also, these results align with the findings in the Section 5.4.1.

5.4.3 Mapa Nutricional: BMI, overweight and obesity indicators for alternative cohorts.

Finally, we indirectly assess the reform's health impacts using Mapa Nutricional data for Kindergarten and First-grade students and the main strategy presented in the section 3.2.[37] Younger children are generally more responsive to nutritional interventions (Just and Price 2013, Belot et al. 2016; Belot et al. 2018), making them an informative test group: if no anthropometric changes

[36]The medical literature establishes that a decrease in BMI SD, on the order of 0.2 to 0.25, is the threshold for clinical significance in the pediatric obesity literature (Ford et al. 2010; Kolsgaard et al. 2011). However one paper, Reinehr et al. 2004 find that even a very small reduction in BMI z-score is associated with significant improvements in several cardiometabolic risk factors. More importantly, this entire literature on minimum thresholds is defined in relation to health and cardiometabolic outcomes among obese children, not overweight children. To the best of our knowledge, no study has established a minimum significant reduction in BMI required to affect cognitive outcomes.

[37]While Kindergarten enrollment is not compulsory in Chile, approximately 96% of children are enrolled. Estimates remain consistent across early grades, mitigating selection concerns.

Table 11: Effect of School Meal Quality Reform on Child BMI z-Score (ELPI Panel)

	Dependent Variable: BMI z-Score		
	(1)	(2)	(3)
Panel A: Main Effects			
Take-up × Post	0.009	-0.021	
	(0.064)	(0.069)	
Panel B: Differential Dose-Response (Relative to 1 Year Exposure)			
Take-up × Post			0.061
			(0.095)
Take-up × Post × 2 Years Exposure			-0.108
			(0.153)
Take-up × Post × 3 Years Exposure			-0.128
			(0.139)
Panel C: Total Net Effects by Exposure Cohort			
Total Effect: 1 Year Exposure (Baseline Coefficient)			0.061
			(0.095)
Total Effect: 2 Years Exposure			-0.047
			(0.128)
Total Effect: 3 Years Exposure			-0.067
			(0.106)
Baseline Controls × Post	No	Yes	Yes
Observations	7,498	7,498	7,498
R-squared	0.736	0.737	0.738

Notes: All regressions include Child Fixed Effects and Wave Fixed Effects, weighted by sampling weights. Baseline controls interacted with the post-reform indicator ($\mathbf{X}_{i,2012} \times$ Post) include baseline household income, maternal education, and rural status. In Column (3), $Take-up_{2012} \times Post$ captures the baseline 1-year exposure effect, while the three-way interactions measure the differential effect for 2 and 3 years of exposure. The regression excludes BMI z-Scores below -5 and above 5 since these values correspond to wrong values. All children are at least 7 years old in 2017. Standard errors clustered at the municipality level are reported in parentheses. Number of clusters: 163. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

occur at this age, significant effects in later grades are highly unlikely.

Table 12 presents the estimated effects of the meal improvements on BMI Z-scores (Panel A), the combined probability of being overweight or obese (Panel B), and obesity alone (Panel C). Across the full sample and when disaggregated by gender, the point estimates are small and statistically indistinguishable from zero. For the full sample, the reform did not alter BMI Z-scores or obesity rates. Event-study plots in Appendix Figures A.37 to A.39 broadly confirm parallel pre-trends. Apart from a marginally significant lead coefficient six years prior to adoption ($t = -6$), the pre-treatment coefficients over the five years leading into the reform show no systematic differential trends.

Across all three empirical approaches, we find no meaningful impacts on child BMI z-scores, overweight, or obesity rates. Because body composition is unaffected even after up to four years of exposure, weight loss or obesity reductions can be ruled out as the primary mechanism behind our test score findings. Instead, the academic gains point to improvements in meal quality and micronutrient intake rather than changes in overall caloric balance. This aligns with the reform's core design, which aimed to upgrade food quality rather than restrict calories. Any broader physiological

or health benefits from healthier habits may simply take longer to materialize, remaining beyond what medium-term evaluations can capture.

Table 12: The Effect of Healthier Meals on Anthropometric Outcomes for Kinder and First Graders

	Full Sample (1)	Boys (2)	Girls (3)
Panel A: BMI Z-Score			
Healthier	0.049	0.059	0.039
	(0.031)	(0.030)	(0.036)
Observations	700,660	342,114	358,546
Panel B: Overweight or Obesity (=1)			
Healthier	0.006	0.006	0.005
	(0.007)	(0.007)	(0.008)
Observations	844,471	412,463	432,008
Panel C: Obesity (=1)			
Healthier	0.006	0.008	0.004
	(0.005)	(0.005)	(0.006)
Observations	844,471	412,463	432,008

Notes: Table displays the overall average treatment effect on the treated (ATT) extracted from Callaway & Sant'Anna difference-in-differences models for Kindergarden and 1st grade. Overweight or obesity and obesity is the probability that the students falls under tis category. Standard errors clustered at the municipality level are reported in parentheses below the point estimates. All models cover the 2010 to 2016 period. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.5 Calories

A change in nutritional requirements may lead to changes in caloric intake, and the calorie count could be driving the results. As presented in Section 2.1, some evidence asserts that augmenting the number of calories increases short-term memory and attention. Figlio and Winicki 2005 found that the increase of calories in the school meals on testing days allowed the students to obtain better scores. However, McEwan 2013 found no effect in the context of an increase in calories in Chilean schools in 2006.

More importantly, as mentioned in the Section 2.3 and shown in Table 1, there was no change in the number of calories in breakfast and lunch, and there was a reduction from 250 to 200 calories in the evening snack. Despite the small reduction in calories in that last meal, it is unlikely that this change in calories could have had any meaningful impact since, mainly, the students have breakfast and lunch, and only a small number of students eat the evening snack at school (see Section 2.2). Therefore, we rule out that this channel is responsible for the results.

5.6 Attendance

The mechanism through which healthier meals improve these scores through an increased attendance operates as follows. Better nutrition leads to improved overall health status and, consequently, students are less likely to miss school due to illness[38] (Anderson et al. 2018). As a result, students are exposed to instructional time more consistently, which translates into improved academic performance. To empirically assess this mechanism, we employ two strategies. First, we estimate differences in attendance rates across repeated cross-sectional cohorts using our main specification, equations 1 and 2. Second, we leverage on the availability of individual-level attendance data for all students across years to estimate the impact of healthier meals on attendance[39].

Table 13 presents the results of the attendance analysis. Columns 1 and 2 report estimates using the CS method and the TWFE specification, respectively, based on repeated cross-sectional samples. Columns 3 and 4 provide estimates using individual-level panel data. Column 1 shows that the adoption of the healthier SMP is associated with a negligible reduction in attendance of 0.05 percent, while column 2 reports a slightly larger, though still minor, decline of 0.25 percent. Although the latter estimate is statistically significant, it is not economically meaningful, corresponding to a reduction of just 0.5 school days, or approximately 4 hours of instruction per year. The results from the individual-level models in columns 3 and 4 are consistent, confirming the absence of a meaningful effect[40]. It is worth emphasizing that these are precise zero estimates, as indicated by the small standard errors. The event studies are shown in figure A.36 and validates the estimates for Callaway and Sant'Anna 2021 estimates.

These findings suggest that the implementation of healthier school meals did not have a significant impact on student attendance. This conclusion is consistent with the broader literature on school meal programs. Studies evaluating both the provision[41] and nutritional quality of school meals at scale generally report null effects on attendance. In terms of meal quality and compared to the other papers, our results align with Anderson et al. 2018, who find no significant effect on attendance, and deviates from Belot and James 2011, who document a modest reduction of 0.7 days in authorized absenteeism, an effect too small to account for their observed academic improvements.

[38] Another commonly hypothesized channel is that students may attend school more frequently to access healthier meals. While this may hold in contexts where the treatment involves the provision of meals versus no meals at all, it is unlikely that students would increase attendance merely to consume more vegetables or salads. Thus, we argue that if an attendance-related channel is operative, it is more plausibly linked to improved health status.

[39] For the case of TWFE, we estimate an individual fixed effects model. The model to estimate in this case is:

$$y_{ist} = \theta + \delta H_{ist} + \beta X_{it} + \alpha Z_{st} + \lambda_s + \gamma_t + \omega_i + \varepsilon_{ist} \tag{9}$$

[40] Figure A.36 shows the event studies associated to the estimates from table 13.

[41] In terms of meal provision, as discussed in Section 5, most evaluations of large-scale programs in medium and high-income countries also find no significant impact on attendance (McEwan 2013, Leos-Urbel et al. 2013, Dotter 2013, Imberman and Kugler 2014, Frisvold 2015, Corcoran et al. 2016, Kurtz et al. 2020, Cuadros-Meñaca et al. 2022, Schwartz and Rothbart 2020).

Table 13: The effect of the healthier SMP on attendance

	Cross-Sections		Individual level panel	
	(1) CS	(2) TWFE	(3) CS	(4) TWFE
Healthier	-0.0468	-0.2480**	-0.0709	-0.2409**
	(0.1201)	(0.1024)	(0.1010)	(0.1037)
Mean Y	92.77	92.77	92.48	92.48
% change	-0.05%	-0.25%	-0.07%	-0.25%
Observations	445,768	445,768	2,032,909	2,032,909
$Adj - R^2$	-	0.152	-	0.481

Notes: Each column represents a separate regression. The dependent variable is the percentage of attendance from 0 to 100 unit (0 to 100%). Observations are at the individual year level. Standard errors are clustered at the municipal level, and they appear in parentheses, number of clusters: 340. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Columns 1 and 2 estimate using cross repeated sections and columns 3 and 4 use an individual panel of students. Column 4 uses a FE model from equation 9. Columns 2 and 4 include year and school fixed effects. Column 4 includes year, school and individual fixed effects. Columns 1 and 3 use Callaway and Sant'Anna 2021 approach (CS) and columns 2 and 4 TWFE. * p < 0.1, ** p < 0.05, *** p < 0.01.

5.7 Behavior

5.7.1 Parental investment and expectations

A central concern when evaluating school meal quality reforms is whether the estimated cognitive gains reflect direct biological channels rather than behavioral adjustments within the household. Specifically, better school food could theoretically motivate parents: visible meal upgrades signal institutional quality, improved child alertness raises the perceived returns to home investments, and alleviated household stress frees up parental bandwidth. If these factors increased parental motivation, parents might boost expectations and home investments, confounding the direct treatment effect.

Tables 14 and 15 test this potential channel by examining whether the reform altered parental engagement[42] and educational expectations regarding college attainment. Across all specifications, we find no evidence of these behavioral spillovers. As reported in Table 14, applying the empirical strategy outlined in Section 5.3.2 (Equations 6 and 7) yields a small and statistically insignificant effect of 0.00614 on the Parental Engagement Index. Similarly, Table 15 shows no effects on parental educational expectations across both administrative SIMCE data (-0.0029) and ELPI survey models (0.0122). Figures A.26 shows that the pre-trend for the SIMCE version are noisy, however, the

[42]The parental engagement index is constructed as a standardized z-score index, taking the simple average of individual z-scored of the time per week that the parents engage in the following activities with the child: reading books, telling stories, singing songs, visiting parks, taking cultural trips, talking or drawing together, running errands together, and visiting friends or relatives. Parental expectations is measured as a binary or standardized indicator reflecting whether the parent expects the child to attain tertiary (college) education.

coefficients are small and not economically meaningful providing suggestive evidence of the absence of the impact.

Table 14: Impact of Healthy Meals on Parental Engagement Index

	Parental Engagement Index		
	(1)	(2)	(3)
Take-up × Post	0.0614	0.0686	0.0331
	(0.0569)	(0.0649)	(0.0825)
Take-up × Exposure 2 × Post			0.1161
			(0.1431)
Take-up × Exposure 3 × Post			-0.0114
			(0.1107)
Total Effects (Dynamic Interaction)			
Exposure 1 year (Baseline)			0.0331
			(0.0825)
Exposure 2 years (Total Effect)			0.1492
			(0.1297)
Exposure 3 years (Total Effect)			0.0217
			(0.0746)
Controls × Post	No	Yes	Yes
Observations	9,900	9,900	9,900

Notes: All specifications absorb household and survey wave fixed effects for children older than 7 years old in 2017. Controls interacted with the post-reform indicator include baseline maternal education, baseline rurality, and baseline household income. Column 3 incorporates differential treatment intensity by exposure duration, omitting the base category. Total effects for exposure levels 2 and 3 are calculated using linear combinations (Take-up × Post + Take-up × Exposure × Post). It uses analytical sampling weights. Number of clusters: 176. Robust standard errors adjusted for clustering at the municipality level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Because neither parental expectations nor home-based engagement respond significantly to the intervention, we can rule out motivation-driven home investments or intra-household reallocation as primary drivers. The documented cognitive gains thus operate independently of parental behavioral adjustments, reinforcing the interpretation of a direct school-level nutritional mechanism.

5.7.2 Classroom behavior and student perception

Beyond household-level adjustments, a secondary spillover of the improved meals is that can alter the aggregate learning environment. Diets lower in ultra-processed foods and refined sugars could theoretically reduce student hyperactivity and behavioral issues. If the nutritional reform led to fewer classroom disruptions, improved peer relations, or increased student motivation, the observed cognitive gains could be the byproduct of a calmer, more productive school climate in addition to the direct physiological benefits of the food itself.

Table 15: Impact of Healthy Meals on Parental Educational Expectations

	Parent expects child to attain tertiary education			
	SIMCE	ELPI		
	(1)	(2)	(3)	(4)
Healthier	-0.0029			
	(0.0031)			
Take-up × Post		0.0218	0.0122	0.0089
		(0.0174)	(0.0170)	(0.0308)
Take-up × Exposure 2 × Post				-0.0293
				(0.0389)
Take-up × Exposure 3 × Post				0.0358
				(0.0443)
Total Effects (Dynamic Interaction)				
Exposure 1 year (Baseline)				0.0089
				(0.0308)
Exposure 2 years				-0.0204
				(0.0232)
Exposure 3 years				0.0446
				(0.0315)
Controls × Post	–	No	Yes	Yes
Observations	465,228	9,990	9,990	9,990

Notes: Column (1) presents the overall average treatment effect on the treated (ITT) from administrative SIMCE data questionnaire estimated using the Callaway and Sant'Anna difference-in-differences estimator. Number of clusters: 342. Columns (2)-(4) report survey-based parental educational expectations from the ELPI dataset. These specifications absorb household and survey wave fixed effects for children older than 7 years old in 2017. Controls interacted with the post-reform indicator include baseline maternal education, baseline rurality, and baseline household income. Column 3 incorporates differential treatment intensity by exposure duration, omitting the base category. Total effects for exposure levels 2 and 3 are calculated using linear combinations (Take-up × Post + Take-up × Exposure × Post). It uses analytical sampling weights. Number of clusters: 176. Robust standard errors adjusted for clustering at the municipality level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To evaluate this psychosocial and behavioral channel, we analyze survey data from students, parents, and teachers regarding self-efficacy, peer dynamics, and classroom order. Tables 16 and 17 present the estimates for these mechanisms.

The results consistently demonstrate that the nutritional intervention did not alter the psychosocial or disciplinary environment of the schools. As shown in Table 16, student-reported measures of self-efficacy (e.g., enjoying studying, doing homework despite difficulty) and classroom climate (e.g., following order, peer respect) yield point estimates that are practically zero and statistically insignificant. Furthermore, neither students nor parents report any shifts in peer victimization, aggression, or school anti-bullying measures.

Crucially, Table 17 corroborates these findings from the teachers' perspective. If the nutritional improvements had smoothed out behavioral volatility, we would expect a reduction in classroom disruptions. Instead, we observe a no effect on the likelihood of a teacher interrupting class due

Table 16: Overall ATT Estimates for Student Outcomes and School Climate

Outcome Variable	Respondent	Estimate	Outcome Variable	Respondent	Estimate
Panel A: Student Self-Efficacy			**Panel B: Classroom Climate**		
Learns as easily as peers	Student	-0.005	Peer respect in classroom	Student	0.005
		(0.010)			(0.016)
Does homework despite difficulty	Student	-0.013	Student respects teachers	Student	-0.002
		(0.009)			(0.010)
Enjoys studying	Student	0.005	Follows classroom order	Student	0.000
		(0.012)			(0.014)
			Keeps classroom clean	Student	0.002
					(0.018)
Panel C: Peer Victimization			**Panel D: Bullying Policies and Measures**		
Peer victimization: scares/coerces	Student	0.002	Bullying policies communicated	Parent	0.004
		(0.017)			(0.007)
Peer victimization: threatens to hit	Student	-0.006	Bullying incidents reported	Parent	0.012
		(0.018)			(0.010)
			Bullying measures taken	Parent	0.006
					(0.008)
			Parents informed on bullying prevention	Parent	0.011
					(0.007)

Notes: Table displays the overall average treatment effect on the treated (ATT) extracted from Callaway & Sant'Anna difference-in-differences models. Data comes from the SIMCE questionnaire of students and parents. Standard errors clustered at the municipality level are reported in parentheses. Estimates in Panels A through C cover the 2011–2016 period (or 2012–2016 for victimization outcomes), while estimates in Panel D are calculated specifically for the 2015 cross-section using data from 2012–2015. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

to disorder. Teacher reports on student-to-student respect (0.003) and respect toward teachers (-0.004) similarly confirm the absence of behavioral shifts.

Figures A.27 to A.32 in the appendix show the event studies that in general validate the parallel trend assumption and the causal interpretation of these possible mechanisms.

6 Discussion

The estimates from this paper suggest a 0.039 SD increase in test scores, a magnitude consistent with the findings of Anderson et al. 2018, who find an effect of 0.031 SD evaluating the impact of healthier school meals in California. Our estimates are smaller than those reported in Belot and James 2011, who document effects ranging from 0.11 to 0.2 SD, although the authors caution that their estimates may be unusually large and are best read as evidence of sign rather than magnitude. Both studies, like ours, evaluate the intensive margin of school meal provision, that is, improvements in nutritional quality rather than access to meals.

Table 17: Overall ATT Estimates for Student Interactions and Classroom Dynamics (Teacher Questionnaire)

Outcome Variable	Estimate	Observations
Panel A: Student Interactions		
Students listen respectfully to classmates	0.003	45,210
	(0.011)	
Students help each other, even if not friends	0.001	44,890
	(0.009)	
Panel B: Classroom Respect and Environment		
Students treat teachers with respect	-0.004	45,120
	(0.012)	
Students keep classroom clean	0.002	44,950
	(0.015)	
Teacher interrupts class due to disorder	-0.008	45,300
	(0.010)	

Notes: Table displays the overall average treatment effect on the treated (ATT) extracted from Callaway & Sant'Anna difference-in-differences models. Data comes from the teacher questionnaire. All outcome variables are dummy variables. Standard errors clustered at the municipality level are reported in parentheses. All models cover the 2011–2016 period. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Studies of the extensive margin (the provision of meals versus none) report a much wider range, from 0.027 to 0.032 SD (Ruffini 2021, Schwartz and Rothbart 2020), through 0.050 to 0.100 SD (Gordanier et al. 2020, Y. Kim 2021, Imberman and Kugler 2014), up to 0.110 to 0.300 SD (Dotter 2013, Chakraborty and Jayaraman 2019, Aurino et al. 2020). These programs bundle several mechanisms and none isolates nutritional quality, so our estimate should sit, as it does, at the lower end of that distribution. Under the benchmarks in Cohen 1969 an effect of this size would be classified as small, but the sample-adjusted guidelines in Duncan and Magnuson 2013 and Kraft 2020 place it in line with other studies of more than 2,000 participants.

All values estimated in this paper are intention-to-treat effects, estimated over all test-takers in treated schools, while only students meeting the program's income eligibility criteria actually receive the meals. With no evidence of spillovers to non-eligible peers (Section 5.1), the effect on participating students is approximately the ITT scaled by coverage, that is 0.051 to 0.053 SD, close to the 0.044 SD obtained when the model is estimated directly on the proxy-eligible subsample, and 0.071 SD among rural students, whose take-up is almost universal. These figures indicate that the ITT understates the effect on the students who actually receive the meals, since roughly one quarter of students in treated schools do not receive them.

The size of the effect is also informative about which mechanisms are operating. Section 5 rules out the channels through which school meals are usually thought to raise achievement in the extensive-margin literature: caloric intake, body mass, school attendance, parental investment and

expectations, and classroom behavior. What remains is exposure to the meal itself, with effects concentrated in the groups most likely to eat it and largest where the baseline diet is furthest from the guidelines (Sections 5.1 and 5.2). In that light, 0.039 SD is the return to nutritional quality lunch plus at home spillovers, isolated from the other margins that school meal programs usually move.

For policy design, one relevant question is what happens outside the school. A standard concern with in-kind provision is crowding out, whereby households reallocate the food budget toward cheaper and more processed items once the child is fed at school, which would offset part of the nutritional gain the program pays for. We find the opposite sign: healthier snacks at school increase, junk food categories do not move, and the positive dietary index at home rises by roughly 0.15 SD with no corresponding increase in the negative index (Section 5.3). The mechanism consistent with this pattern is preference formation through repeated exposure, since a school lunch serving the same reformed menu daily for one to three years performs a repetition that households, particularly lower-income ones facing a higher cost of a refused food, are unlikely to sustain on their own. The effect on test scores therefore combines two channels, exposure to nutritious foods at school and a change in students' preferences that alters what they consume at home, and the case for reformulating meal quality is strongest in the schools where targeting already concentrates the program.

A related question is whether the two-year window over which we estimate the ITT is long enough for improvements in diet quality to generate the estimated cognitive gains. Our own estimates speak to this directly: the effect of the healthier menu is larger in the second year of exposure than in the first (Figure 4), so the response builds over the observation window rather than being truncated by it. This is consistent with the physiological time lags documented in the nutrition and medical literature (Table A.12), where behavioral and attentional gains appear within weeks to a few months of dietary improvement and gains in processing speed and executive function within roughly six to twenty four months. These channels operate independently of body weight, consistent with our finding of significant test-score effects in the absence of any detectable change in overweight, obesity, or BMI (Section 5.4).

Conducting a comprehensive cost-benefit analysis is difficult, as SMPs affect outcomes across multiple domains, including education, health, and social protection (Gelli et al. 2014). Nevertheless, a basic calculation is informative. The average cost of a healthy vendor meal contract is $258 (2026 USD) per test-taker per academic year[43]. With 180 school days in a year, this corresponds to approximately $1.43 per student per day. The improved nutritional standards increased the cost of the program by 18.2 percent, which translates into an additional $47 per test-taker annually. Using this cost difference and our estimated effect size of 0.039 SD, we calculate that achieving a 0.1 SD

[43]Calculations made by the authors based on Resolución Exenta N 44 (2015), Resolución Exenta N 125 (2016), and Resolución Exenta N 4 (2017) of Junaeb.

improvement in test scores through this intervention would cost approximately $121.9 per student per year. This is comparable to the estimate reported in Anderson et al. 2018 for California.

By contrast, other educational interventions are often considerably more expensive. For instance, raising test scores by 0.1 SD in the Tennessee STAR class size reduction experiment cost an estimated $2,031 per student per year[44]. Likewise, Kurtz et al. 2020 report that Florida's extended school day program achieved a 0.05 SD increase in reading scores at a cost of $1,069 per student per year (Figlio et al. 2018), while North Carolina's weekend feeding program generated a 0.09 SD gain at a cost of $297[45]. Relative to these interventions, improving meal quality appears cost-effective, and its benefits may extend into adulthood: Lundborg et al. 2022 find that improved school nutrition in Sweden led to increased educational attainment and a 3 percent rise in lifetime earnings[46].

From a policy perspective, it is also important to consider factors beyond cost and effect size, such as scalability. As emphasized by Aurino et al. 2020, small-scale programs often depend on complex or unsustainable supply chains, raising concerns about their viability at scale. This is not a concern for Chile's SMP, which is a nationally-mandated, government-run program with virtually universal school coverage. First, there are no additional coordination costs across subnational jurisdictions with divergent policies. Second, the vendor assignment system was explicitly designed to minimize costs and reduce market concentration, implying that the program may scale efficiently even as the number of beneficiaries increases.

7 Conclusion

This paper exploits variation in the nutritional quality of school meals generated by the staggered adoption of a healthier menu in 2015 to estimate its impact on the academic performance of fourth-grade students in Chilean public schools. The sequential rollout was determined by a late 1990s policy aimed at promoting vendor competition and minimizing procurement costs, with no direct connection to student outcomes. Leveraging this plausibly exogenous rollout and employing the method of Callaway and Sant'Anna 2021 to implement a difference-in-differences strategy under staggered adoption and dynamic heterogeneous treatment effects, we find that the adoption of healthier school meals increases combined Math and Reading test scores by 0.039 SD. Disaggregated effects are 0.033 SD for Reading and 0.038 SD for Math. The effects are more pronounced and remain statistically significant among students from the poorest households and those attending rural schools.

We perform a series of robustness checks that support the identifying assumption that the timing of the healthier menu rollout is unrelated to other time-varying factors influencing test scores.

[44] Original cost estimates from Krueger 1999, inflation-adjusted by Anderson et al. 2018, and then updated by the authors.

[45] Inflation-adjusted to 2026 USD.

[46] They argue that their benefit-to-cost ratio is twice as high as that of the Head Start program, underscoring the long-term returns to high-quality school nutrition.

Additionally, we explore potential mechanisms. While the most likely channel is the biological link between nutrient intake and cognitive development, our data do not allow for direct testing of this mechanism. Nevertheless, we plausibly rule out alternative channels such as caloric intake, body mass, school attendance, parental investment and expectations, and classroom behavior. These findings offer suggestive evidence consistent with the nutritional quality of meals being a key driver of improved academic performance.

Two additional patterns support this reading. The gains line up with exposure to the meal rather than with attending a treated school, since they are concentrated among proxy eligible students, students in full-day schools, and rural students, and are close to zero among students in treated schools who do not receive the meal. Beyond the school, the reform raised the healthy dietary index at home without any increase in the unhealthy index, so the improvement in school food was not offset by household substitution toward cheaper and more processed items. Repeated daily exposure to the reformed menu, and the shift in children's food preferences it generates, is the channel most consistent with both patterns, so the effect on test scores operates through two routes: the nutritious foods students eat at school, and the change in preferences that carries over to what they consume at home.

Although the estimated effects may appear modest, they align with updated empirical benchmarks that account for heterogeneity in educational interventions, as discussed in Kraft 2020. From a cost-effectiveness perspective, the intervention is highly favorable: it would cost approximately $121.9 to raise test scores by 0.1 SD. In contrast, comparable educational interventions are significantly more expensive, including the Tennessee STAR experiment ($2,031 per 0.1 SD), Florida's extended school day program ($1,069 per 0.05 SD in Reading), and North Carolina's weekend feeding program ($267 per 0.09 SD in Reading).

Finally, from a policy standpoint, improving the nutritional quality of school meals represents a cost-effective, scalable, and administratively feasible intervention. The existing centralized procurement and vendor assignment process allows for efficient expansion of the program's reach, suggesting that healthier school meals can be a powerful and practical policy tool for improving educational outcomes, particularly among disadvantaged populations.

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A Appendix

Figure A.1: Examples of meals in the SMP



A.1 Eligibility

The program targets all students enrolled in public or charter schools (i.e., private subsidized schools), which together serve over 90 percent of Chilean students. Eligibility is based on socioeconomic status. Under the program's rules, students in rural schools are automatically eligible. For urban students, a vulnerability index with explicit administrative weights determines priority. Specifically, having a mother without a complete high school education accounts for 25% of the priority score. Belonging to the 40% most vulnerable households (according to the Registro Social de Hogares, RSH) accounts for 6%, alongside participation in other poverty-targeted programs like Chile Solidario (17%) and Fonasa A (12%). The remaining 40% is for children in the foster care system or under the custody of Child Protective Services (CPS). Less than 5% of children under 18 years old fall in any of those three categories[47]. Because a narrow percentage of students fall under the last three categories, maternal education and household vulnerability serve as the most relevant determinants of eligibility.

[47] Currently, 5% of children under 19 are in Fonasa A, 2% of children under 18 are part of Chile Solidario, and 0.1% of children under 18 are under the custody of CPS (Protección 2026; Dipres 2023, and authors' calculations based on Fonasa beneficiaries).

Table A.1: Food spending, by Household Income Tercile EPF 2012

	Monthly spending				Monthly spending		
	T1	T2	T3		T1	T2	T3
Meats and eggs	7,192	7,676	10,084	Dairy	2,594	3,059	4,508
	(5,399)	(6,042)	(8,764)		(2,716)	(3,030)	(4,887)
Fish	678	712	1,411	Bread	4,975	5,220	5,444
	(1,560)	(1,253)	(2,272)		(3,107)	(3,075)	(3,362)
Fruits	1,075	1,346	1,818	Snacks	271	389	544
	(1,548)	(1,676)	(3,275)		(575)	(821)	(1,034)
Vegetables	3,209	3,240	4,483	Sugary drinks	2,495	3,007	3,821
	(2,394)	(2,843)	(3,853)		(2,346)	(2,772)	(3,289)
Legumes	272	285	260	Treats	2,654	2,937	4,369
	(513)	(560)	(633)		(2,945)	(2,769)	(4,547)

Notes: Author's calculations using the VII Household Budget Survey *Encuesta de Presupuestos Familiares* (INE, 2011-2012). Sample: children enrolled in public primary education (one observation per child). Cells report monthly household spending on each food group divided by household size, in May 2017 pesos, adjusted with the INE consumer price index. Households that purchased nothing in a category during the survey reference period enter with a value of zero. Means are weighted by the survey's expansion factor, standard deviations (in parentheses) are weighted the same way. Households are split into terciles of per capita disposable household income (excluding imputed rent, computed using the survey's expansion factor) among households with a primary-school child attending public schools. T1 is the lowest tercile and T3 the highest. Spending is the product of quantity and unit value, so differences across terciles reflect both the amount of food households buy and the prices they pay for it, and cannot be read as differences in quantities alone.

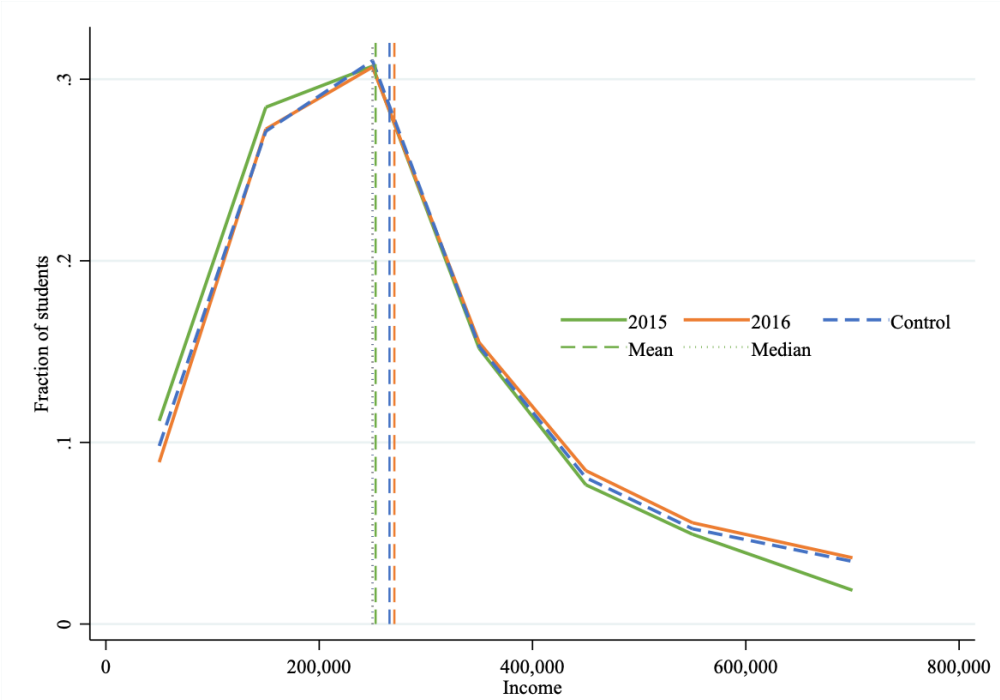
Since there is no direct administrative flag for SMP student eligibility, and our data lacks individual-level information to identify participants in Chile Solidario, Fonasa A, or the foster care system, we construct a comprehensive eligibility proxy. This proxy mirrors the administrative formula using observable data on maternal education and household income. A student classifies as eligible if they meet any of the following criteria: (1) they attend a rural school, (2) their mother has less than a high school education, or (3) their self-reported household income falls below the 60% threshold of the public school distribution. This threshold calibrates to the national 40% RSH vulnerability criteria and captures the associated Fonasa A and Chile Solidario populations.

Our eligibility assignment rule categorizes 67% of urban students as eligible and, by construction, sets rural eligibility at 100%. To assess its accuracy as a proxy for participation, we benchmark this constructed measure against actual PAE take-up data from the Longitudinal Early Childhood Survey (ELPI). The ELPI data reveal an actual take-up rate of 68% in urban settings and 95% in rural areas. The tight correspondence between our eligibility assignment and actual take-up provides confidence that our measure accurately captures the treated population.

A.2 Exposure to nutritious food: attendance

As an additional robustness check following Krämer et al. 2021, we explore whether the impact of the school meal reform varies by student attendance, which we use as a proxy for treatment intensity. Because attendance is potentially endogenous, we discuss potential identification caveats and present suggestive descriptive checks below; consequently, these heterogeneous effects should be interpreted as suggestive patterns rather than causal estimates.

Figure A.2: Groups 2015, 2016 and 2017 (control)
Income distribution



Note: Income distribution using income information from the SIMCE survey year 2014 (baseline). The sample includes students in grade 4 aged 8 to 11 with valid test scores and trimmed at 95th percentiles of the income distribution within each group. Vertical dashed lines denote group means, and vertical dotted lines denote group medians.

Healthier meals do not affect attendance. We acknowledge the potential for endogeneity concerns, and therefore, these results should be interpreted with caution. One possible behavioral response is that parents might increase their children's school attendance due to the availability of healthier meals. However, according to Junaeb, parents were not informed about the nutritional improvements, which limits the likelihood of such anticipatory behavior. Another plausible mechanism is that healthier meals could reduce illness among children, thereby indirectly increasing attendance. To address the latter concern, we conduct several robustness checks to evaluate whether the treatment had any effect on student attendance. First, Table 13 shows that the introduction of healthier meals had no statistically significant effect on attendance. The estimates are precise, with small standard deviations, indicating a negligible average treatment effect. Second, Table A.2 reveals no evidence of heterogeneous effects across different student characteristics, including income terciles, on attendance and Table A.3 presents the results over the distribution. It confirms that both the average and quantile treatment effects on attendance are statistically insignificant.

To further address concerns regarding potential selection bias, we conduct additional robustness checks. Specifically, we consider the possibility that the introduction of healthier meals may have influenced student attendance patterns, thereby altering the composition of the treatment and control groups in terms of observable characteristics. To evaluate this, we replicate our main regression analysis, conditioning on current attendance levels (categorized as low, medium, and

Table A.2: The effect of healthier SMP on Attendance: Heterogeneity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Gender		Mother Education		Household Income			Zone	
	Boy	Girl	Less High School	More High School	Low	Medium	High	Rural	Urban
Healthier	-0.001	-0.001	-0.001	-0.001	0.000	0.000	-0.001	0.001	-0.001
	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0011)	(0.0011)	(0.0011)	(0.0011)
Observations	915,031	931,349	1,128,130	718,275	787,378	492,868	566,029	387,818	1,447,929

Notes: Each column represents a separate regression using Callaway and Sant'Anna 2021 approach. The dependent variable is the attendance rate at individual level. Observations are at the individual year level. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Standard errors clustered at the municipality level appear in parentheses, number of clusters: 340. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table A.3: The effect of healthier SMP on quartile of attendance

	(1)	(2)	(3)	(4)	(5)
		Quartile			
	Average	First	Second	Third	Fourth
Healthier	-0.0016	0.0017	-0.0001	0.0006*	0.000
	(0.0010)	(0.0041)	(0.0003)	(0.0003)	(0.0004)
Observations	1,846,512	1,846,512	1,846,512	1,846,512	1,846,512

Notes: Each column represents a separate regression using TWFE approach and quantile regression method. The dependent variable is the attendance rate at individual level. Observations are at the individual year level. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. All regressions include school and year fixed effects. Standard errors clustered at the municipality level appear in parentheses, number of clusters: 340. * p < 0.1, ** p < 0.05, *** p < 0.01.

high), separately for treated and untreated students. The results indicate no significant changes in observable characteristics across attendance levels, suggesting minimal selection bias.

Furthermore, we assess the potential differential impact of the intervention by stratifying the sample according to pre-intervention attendance levels. We then estimate the treatment effects of healthy meals on current attendance, conditional on historical attendance categories. Across all groups, we find no statistically significant effects of the intervention on current attendance. Full results from the last two supplementary analyses are available upon request.

Exposure effects. Taken together, the preceding analyses help reduce concerns regarding attendance endogeneity. Continuing with the analysis in terms of exposure, if healthier meals meaningfully impact cognitive outcomes, students who are more frequently exposed to them should experience greater gains, as they consume these meals more regularly. In this context, students with higher attendance rates are more likely to be exposed to the nutritional benefits of the program. Table A.4 presents the program's effect on test scores across terciles of attendance[48]. In Panel A, we examine the effect on combined test scores. Columns 1 to 3 indicate that the estimated impact increases with attendance. Students in the lowest attendance tercile show no measurable improvement,

[48]Figure A.4 presents the event studies associated to Table A.4 for Math+Reading.

whereas those in the middle and top terciles experience statistically significant increases of 0.04 and 0.08 SD, respectively. A similar pattern is observed when Math and Reading test scores are analyzed separately. These findings suggest that students who attend school more regularly benefit more from healthier meals, likely due to sustained exposure to improved nutrients. This result is consistent with Krämer et al. 2021, who find that the provision of double-fortified salt in school meals improves Math and Reading outcomes only for students in the highest attendance tercile.

Table A.4: The effect of a healthier SMP by rate of attendance.

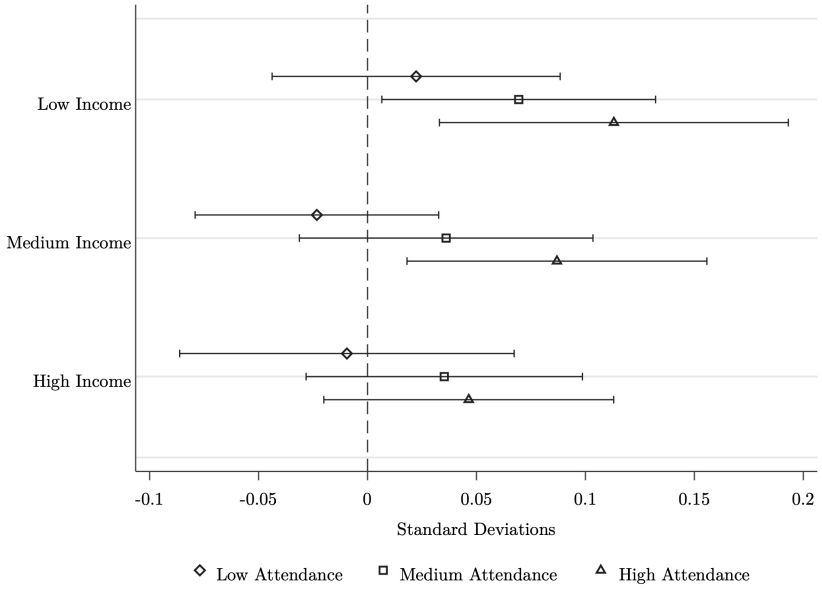
	(1)	(2)	(3)
		Attendance	
	Low	Medium	High
Panel A: Total (SD)			
Healthier	-0.0003	0.0402*	0.0802***
	(0.0217)	(0.0222)	(0.0231)
Observations	124,373	130,463	127,486
Panel B: Math (SD)			
Healthier	0.0102	0.0302	0.0736***
	(0.0215)	(0.0223)	(0.0239)
Observations	126,038	131,561	128,202
Panel C: Reading (SD)			
Healthier	-0.0090	0.0408*	0.0743***
	(0.0211)	(0.0219)	(0.0218)
Observations	125,164	130,972	127,905
Mean Attendance	0.865	0.946	0.983

Notes: Each column represents a separate regression using the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Observations are at the individual year level. Individual covariates include years of mother education, household income, gender and age of the student. School covariates include a dummy for the rurality of school and students to teacher ratio. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 340. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

We next investigate whether the pattern identified in Table A.4 is more pronounced among groups that are more likely to be nutritionally deprived, namely children from lower-income households. Figure A.3 illustrates an increasing treatment effect with attendance level among students from low-income households, and to a lesser extent among those from middle-income households[49]. One potential explanation is that students from low-income families are both more likely to suffer from nutritional deficiencies (as shown in table 2 and section in the appendix A.1 and more likely to participate in the free meals program (section A.1). In contrast, among students from higher-income households, we do not observe a similar pattern, and the estimates are statistically insignificant. This is consistent with the idea that wealthier students are both less likely to be nutritionally

[49]From Figures A.5 to A.7 show the event studies for Math+Reading associated to Figure A.3. Additionally, Figures A.8 and A.9 shows the estimates by level of attendance and income for Math and Reading alone.

Figure A.3: The effect of a healthier SMP on Total score by level of attendance and income

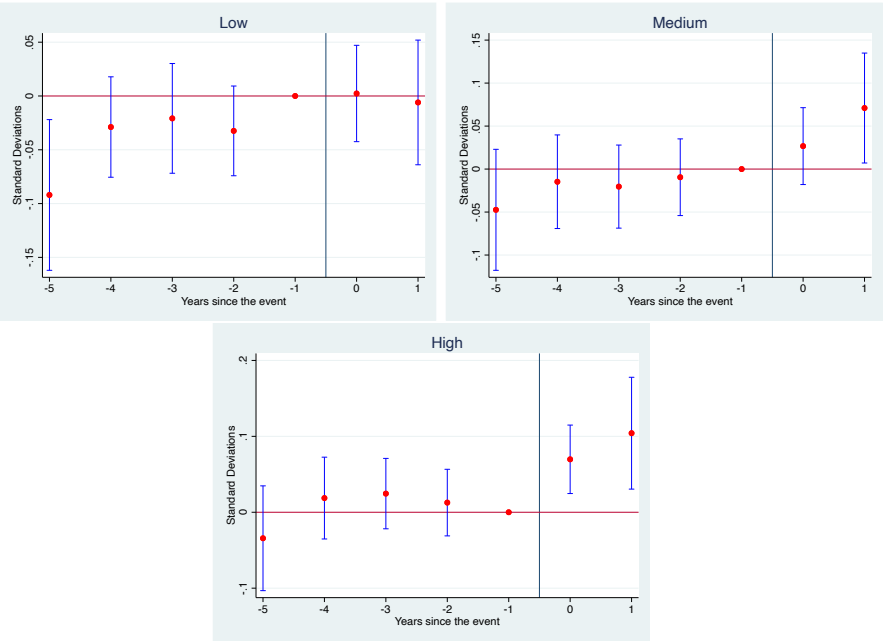


Notes: The figure presents the estimates using the Callaway and Sant'Anna 2021 approach for income/attendance groups. The dependent variable is the test scores measured in standard deviations. Observations are at the individual year level. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teachers ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

deprived and less likely to consume the school-provided meals.

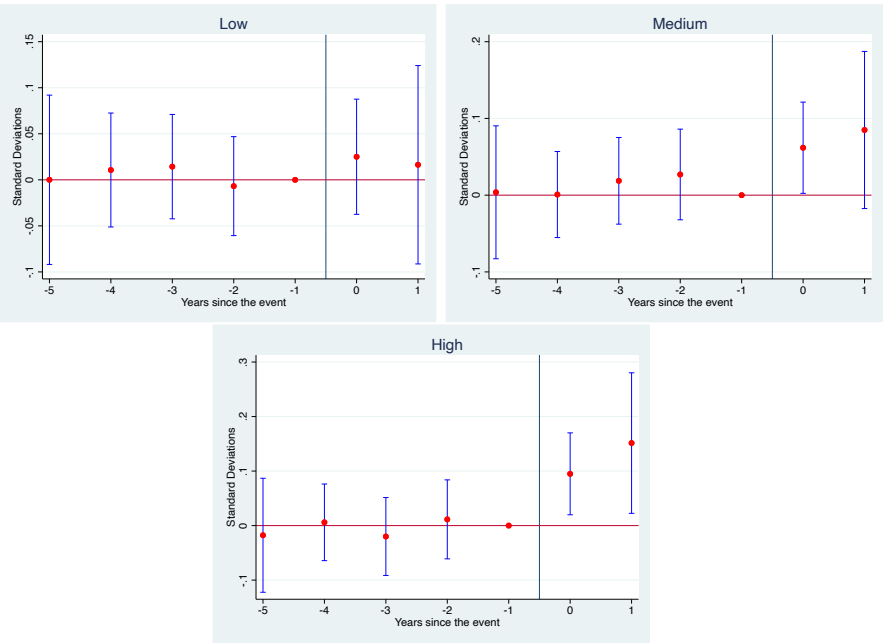
Overall, the results demonstrate that students who are more frequently exposed to healthier meals experience the largest improvements in academic performance. Moreover, students who are more likely to be both nutritionally deprived and eligible for free school meals also exhibit the most substantial gains. While this does not constitute direct evidence that improved nutrition is the causal channel, the results provide indirect support for the hypothesis that nutrient quality plays a key role in explaining the observed effects on top of the results presented in section 5.1.

Figure A.4: Effect of the SMP on Math+Reading by level of attendance



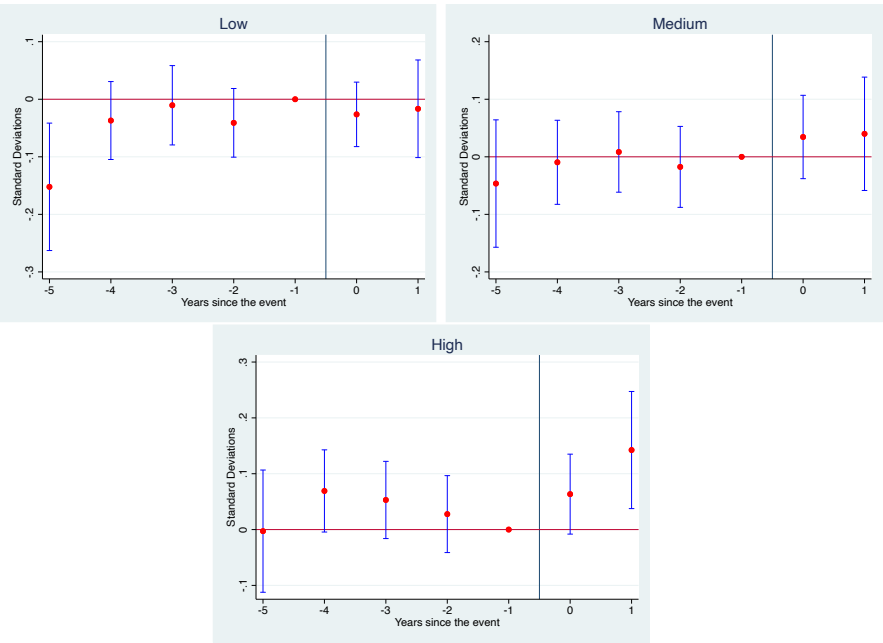
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach for different attendance groups. The dependent variable is the test scores measured in standard deviations. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.5: Effect of the SMP on Math+Reading and Low Income students by level of attendance



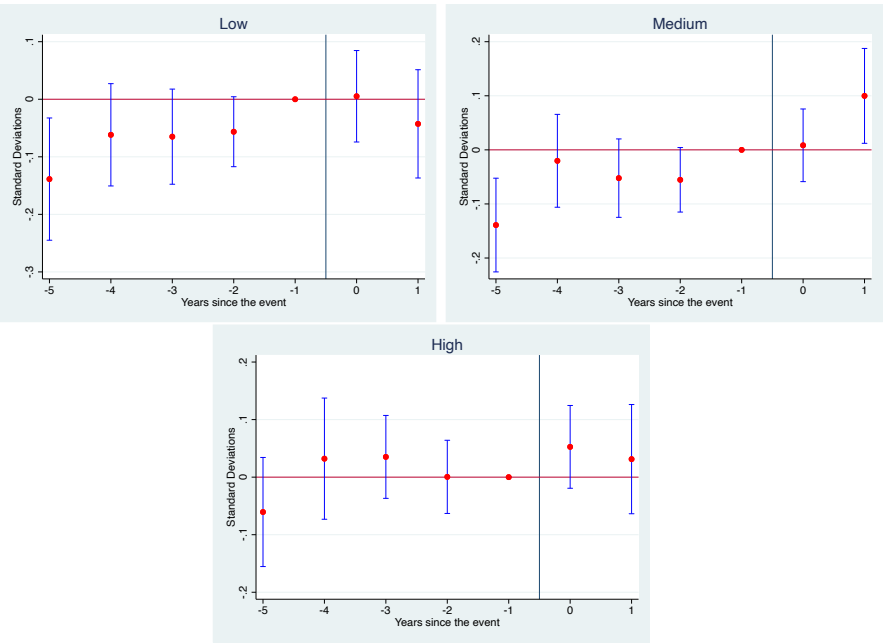
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach for different attendance groups. The dependent variable is the test scores measured in standard deviations. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.6: Effect of the SMP on Math+Reading and Medium Income students by level of attendance



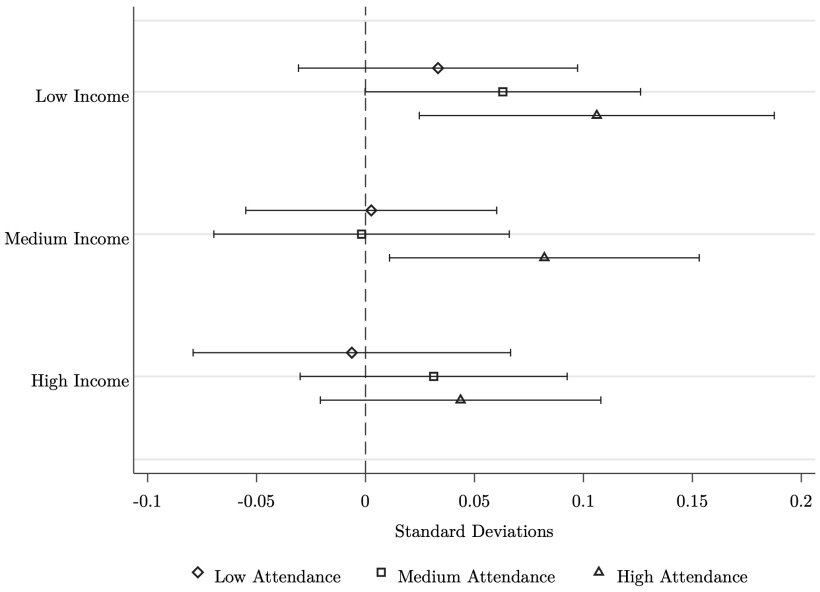
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach for different attendance groups. The dependent variable is the test scores measured in standard deviations. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.7: Effect of the SMP on Math+Reading and High Income students by level of attendance



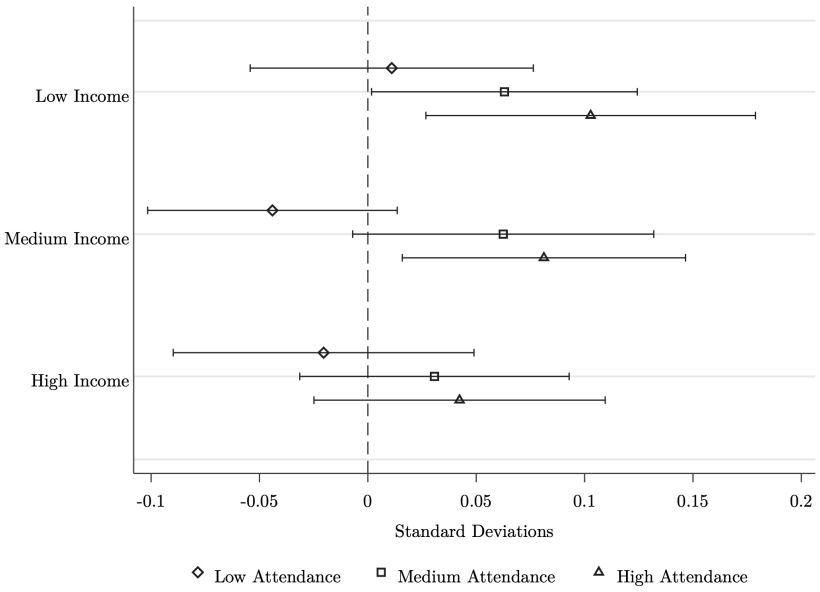
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach for different attendance groups. The dependent variable is the test scores measured in standard deviations. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.8: The effect of a healthier SMP on Math by level of attendance and income (CS).



Note: The figure presents the estimates using the Callaway and Sant'Anna 2021 approach for different groups of interest for different income/attendance groups. The dependent variable is the test scores measured in standard deviations. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.9: The effect of a healthier SMP on Reading by level of attendance and income (CS).



Note: The figure presents the estimates using the Callaway and Sant'Anna 2021 approach for different income/attendance groups. The dependent variable is the test scores measured in standard deviations. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school, full-school day, and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Table A.5: Food or Compound Effects on Cognition

Food or Compound	Effect	Channel	Magnitude	Age Group	Literature Review
Antioxidants & Polyphenols	Positive	Flavonoid and anthocyanin intake; proposed via reduction of oxidative stress and increased cerebral blood flow.	Working memory, processing speed: Cohen's d = 0.16-0.98. Verbal memory: 33.25 vs. 30.75 words (p=0.035); delayed recall 6.49 vs. 6.15 words (p=0.04). No effect on reading.	Children and Adolescents (7-18).	L. Bell et al. 2015; Barfoot et al. 2019; Zhou et al. 2024
Vitamins & Micronutrients	Positive	Micronutrient adequacy supports neurotransmitter synthesis, myelination, and gene expression regulation relevant to learning and memory.	Verbal learning/memory: +0.23 SD. Associative memory: β=0.76. Verbal recognition memory: β=1.00. Working memory (Digit Span): 32% vs. 13% change from baseline (p=0.027).	Children and Adolescents (5-16).	Kurpad et al. 2013; Osendarp et al. 2007; Eilander et al. 2010; Lam and Lawlis 2017; Taljaard et al. 2013
Iron	Positive	Iron deficiency impairs dopaminergic neurotransmission, myelination, and hippocampal function relevant to attention and memory; supplementation is proposed to reverse these deficits, particularly in children who are anemic at baseline.	Intelligence: SMD 0.46 (0.79 if anemic). Attention: SMD 0.45. Memory: SMD 0.45 (0.47 if anemic). School achievement: SMD 0.00. Separate meta-analysis: cognitive performance SMD 0.50 (0.29 anemic, 0.01 non-anemic). Large fortification trial: no significant effect on any test.	Children (6-12).	Gutema et al. 2023; Krämer et al. 2021; Low et al. 2013
Omega-3 Fatty Acids	Positive	Maintain neuronal membrane function, reduce neuroinflammation.	Reading age: +0.8 months (for bottom 20th percentile); +1.9 months (for bottom 10th percentile). Parent-rated behavior improved; teacher-rated: no effect. No effect in full, unselected sample.	Children (7-9 years).	Stonehouse 2014; Richardson et al. 2012

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Table A.5 – *continued from previous page*

Food or Compound	Effect	Channel	Magnitude	Age Group	Literature Review
Water	Positive	Adequate hydration is associated with improved visual attention, visual search, and short-term memory in children; proposed mechanisms include maintenance of cerebral blood flow and hydration-sensitive neuronal function.	Reaction time: faster with higher intake; interference r=-0.27 to -0.31. Visual attention/search: improved. Short-term memory: worse when dehydrated (p=0.024). No effect on accuracy.	Children (6-12 years).	N.A. Khan et al. 2015; Bar-David et al. 2005; Edmonds and Burford 2009; Edmonds and Jeffes 2009
Animal-Source Foods (Meat/Milk)	Positive	Provides bioavailable iron, vitamin B12, folate, zinc, and riboflavin; in the primary trial evidence, score gains were attributed to this micronutrient content rather than to protein or energy intake.	Total test scores: +57.5 vs. control ($p<0.05$); +44.8 vs. plant snack ($p<0.05$); milk +39.4 vs. control ($p<0.05$). 6/7 subjects improved ($p<0.05$ each). Digit span: r=0.129 (forward), r=0.140 (backward).	Children and adolescents (6-14 years).	Hulett et al. 2014; Neumann et al. 2007; J. Y. Kim and S. W. Kang 2017
Complex Carbohydrates (Glycemic Index)	Mixed	Postprandial blood glucose response is proposed to affect arousal, cortisol, and glucose availability to the brain, with effects varying by cognitive domain rather than uniformly favoring low or high GI.	Verbal memory: low-GI 15.2-16.0 vs. high-GI 14.1-15.2 words (p=0.03). Processing speed/vigilance: high-GI better (p=0.01-0.03). Secondary memory: low-GI cereal -30.7 vs. high-GI -47.2 pts (p=0.02). Largest RCT (n=188): no significant effect.	Children and adolescents (6-14 years).	Micha et al. 2011; Ingwersen et al. 2007; Jansen et al. 2020
Dietary Fiber	Mixed	Proposed via short-chain fatty acid production and gut-brain axis signaling from fiber fermentation, and via the lower glycemic response of higher-fiber foods.	Flanker test - Task accuracy: β=0.32 and β=0.35 for total fiber and insoluble respectively. Spatial memory: 17.0 vs. 14.0; 15.9 vs. 13.0 correct items. Auditory attention: 36.6 vs. 26.8 hits. Brown rice trial: null once age controlled. Inulin pilot: delayed recall 9.6 vs. 4.2 words.	Children and adolescents (6-14 years).	Naiman A Khan et al. 2015; Mahoney et al. 2005; Gruneck et al. 2022; Horasan Sagbasan et al. 2024

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Table A.5 – *continued from previous page*

Food or Compound	Effect	Channel	Magnitude	Age Group	Literature Review
Refined Sugars	Negative	Proposed via chronic glycemic variability and insulin signaling effects on hippocampal-dependent memory and executive-control circuits; human mechanistic evidence is limited, most proposed pathways are drawn from rodent studies.	Academic performance: −4.9 points. Executive-function reaction time: dose-response, reaction time got progressively slower as the group's SSB frequency increased. 2-back task: average reaction time was 960ms in the no-SSB group, 1075ms in the 1x/week group, 1226ms in the 2+x/week group. Block-design errors: dose-response, the proportion classified as abnormal on this test increased step by step as the SSB category increased; digit span OR=4.76 in overweight adolescents; moderate intake showed lower odds on 3 other tests.	Children and adolescents (12-17 years).	Yan et al. 2022; Ren et al. 2022; Zhang et al. 2022

Notes: RCT = randomized controlled trial. SD = SD. Cohen's d = standardized effect size expressing the difference between two group means in pooled SD units (conventionally, 0.2, 0.5, and 0.8 denote small, medium, and large effects). SMD = standardized mean difference, the effect size metric used in meta-analyses to pool results across studies that measured the same construct on different scales; interpreted on the same scale as Cohen's d. β = unstandardized regression coefficient, reporting the estimated change in the outcome, in its original scale units, associated with the intervention. HVLT = Hopkins Verbal Learning Test. SSB= sugar-sweetened beverage. ms=milliseconds

Table A.6: 2015 Changes in Nutritional Requirements in the SMP.

Food or Compound	Cognition	Food (change in # servings)
Antioxidants	Positive	Salads (↑ 70%), Vegetables Side Dish (↑ 20%), Fruit (↑ 25%).
Vitamins, Omega-3 Fatty Acid	Positive	Avocado (↑ 33%), Salads (↑ 70%), Vegetables Side Dish (↑ 20%), Fruit (↑ 25%) and Fish (↑ 15%)
Water	Positive	Mandatory on trait
Protein	Positive	Beef, chicken, pork (↑ 25%)
Fiber	Positive	Granola (added), Oats (↑ 50%) and Whole wheat bread (↑ 50%).
Refined Carbohydrates	Negative	White Bread (↓ 30%), Cereal (eliminated), Rice, Pasta (↓ 12%).
Refined Sugars	Negative	Jam, Honey, Caramel (↓ 50%), Cereal and Oat Cookies (eliminated) Jelly with canned fruits (↓ 50%)

Note: Table summarizes the changes introduced in nutritional requirements of the school meal program. The total number of calories was unchanged, however, the number of servings of unprocessed foods increased, and the processed foods decreased. Unprocessed foods positively affect cognition, unlike processed foods which often have the opposite effect.

Table A.7: The effect of healthier SMP on Math+Reading: Chaisemartin and D'Haultfoeuille.

	CS	Chaisemartin and D'Haultfoeuille				
	(1)	(2)	(3)	(4)	(5)	(6)
Healthier	0.0389^{**}	0.0400^{***}	0.0401^{***}	0.0395^{***}	0.0397^{***}	0.0498^{**}
	(0.0163)	(0.0150)	(0.0142)	(0.0150)	(0.0142)	(0.0202)
Individual Controls	Yes	Yes	No	Yes	No	Yes
School Controls	Yes	Yes	Yes	No	No	Yes
Municipality Trends	No	No	No	No	No	Yes
Observations	440,690	440,690	440,690	440,690	440,690	440,690

Notes: Each column represents a separate regression using the De Chaisemartin and d'Haultfoeuille 2020 approach. The dependent variable is the test scores measured in standard deviations. Observations are at the individual year level. Standard errors are clustered and appear in parentheses. Individual covariates include the mother's education, household income, gender and age of the student. School covariates include a dummy for the rurality of school and students to teacher ratio. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Effect of Healthier SMP by Subject (Chaisemartin and D'Haultfœuille)

	CS	Chaisemartin and D'Haultfoeuille				
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Math						
Healthier	0.038**	0.0388**	0.0360**	0.0384**	0.0356**	0.0409*
	(0.0164)	(0.0152)	(0.0143)	(0.0152)	(0.0143)	(0.0217)
Observations	444,404	444,404	444,404	444,404	444,404	444,404
Panel B: Reading						
Healthier	0.033**	0.0337**	0.0362**	0.0332**	0.0357**	0.0476**
	(0.0158)	(0.0151)	(0.0145)	(0.0151)	(0.0145)	(0.0186)
Observations	442,538	442,538	442,538	442,538	442,538	442,538
Individual Controls	Yes	Yes	No	Yes	No	Yes
School Controls	Yes	Yes	Yes	No	No	Yes
Municipality Trends	No	No	No	No	No	Yes

Notes: Each column represents a separate regression using the De Chaisemartin and d'Haultfoeuille 2020 approach. The dependent variable is the test scores measured in standard deviations. Observations are at the individual year level. Standard errors are clustered and appear in parentheses. Individual covariates include education of the mother, household income, gender and age of the student. School covariates include a dummy for the rurality of school, and students to teacher ratio. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9: The effect of healthier SMP by different clusters.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
		School			Province			Region	
	Total	Reading	Math	Total	Reading	Math	Total	Reading	Math
Healthier	0.0389**	0.0334**	0.0380**	0.0389**	0.0334**	0.0380**	0.0389**	0.0334**	0.0380**
	(0.0153)	(0.0162)	(0.0140)	(0.0172)	(0.0171)	(0.0154)	(0.0178)	(0.0192)	(0.0145)
# Clusters	4,245	4,245	4,245	55	55	55	18	18	18
Observations	440,690	444,404	442,538	440,690	444,404	442,538	440,690	444,404	442,538

Notes: Each column represents a separate regression using different clusters and the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Observations are at the individual year level. Standard errors are clustered and appear in parentheses. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for the rurality of school and students to teacher ratio. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.10: Estimates using Different Geographic Linear Trends

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	School trends			Municipality trends			Province trends			Region trends		
	Total (SD)	Math (SD)	Reading (SD)	Total	Math	Reading	Total	Math	Reading	Total	Math	Reading
Healthier	0.0389**	0.0350**	0.0337**	0.0394**	0.0344**	0.0350**	0.0277*	0.0227	0.0263*	0.0260*	0.0192	0.0252*
	(0.0156)	(0.0160)	(0.0150)	(0.0159)	(0.0164)	(0.0151)	(0.0162)	(0.0168)	(0.0154)	(0.0159)	(0.0167)	(0.0151)
Observations	440,690	444,404	442,538	440,690	444,404	442,538	440,690	444,404	442,538	440,690	444,404	442,538
$Adj - R^2$	0.152	0.161	0.115	0.150	0.157	0.117	0.148	0.155	0.116	0.148	0.154	0.116

Notes: Each column represents a separate regression. The dependent variable is the test scores measured in standard deviations. Each regression controls for different geographic linear trends using TWFE. Observations are at the individual year level. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Also, all regressions include school and year fixed effects. Standard errors are clustered at the municipality level and appear in parentheses. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table A.11: The effect of healthier SMP on 6th grade

	(1)	(2)	(3)	(4)	(5)	(6)
	Total		Math		Reading	
	CS	TWFE	CS	TWFE	CS	TWFE
Healthier	0.0183	0.00182	0.0277*	0.0126	0.0056	-0.00871
	(0.0150)	(0.0124)	(0.0152)	(0.0128)	(0.0147)	(0.0121)
Individual Controls	Yes	Yes	Yes	Yes	Yes	Yes
School Controls	Yes	Yes	Yes	Yes	Yes	Yes
School FE	No	Yes	No	Yes	No	Yes
Year FE	No	Yes	No	Yes	No	Yes
Observations	241,769	241,769	244,698	244,698	243,342	243,342
Adjusted R2	-	0.170	-	0.178	-	0.132
Pretrend test = 0 (p-value)	0.9589		0.705		0.9844	

Notes: Each column represents a separate regression using Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Observations are at the individual year level. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for the rurality of school and students to teacher ratio. Standard errors are clustered and appear in parentheses. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table A.12: Physiological Time Lags Between Improved Meal Quality and Cognitive Outcomes

Time Horizon	Physiological Mechanisms	Supporting Evidence	References
Short-Term (Up to 6 Months)	Within roughly six months, improvements in diet quality are associated with more stable glycemic control and improved insulin sensitivity, which attenuates the neural insulin resistance linked to diets high in refined sugars and saturated fats. Greater intake of essential micronutrients and omega-3 fatty acids also supports neuronal membrane function and neurotransmitter synthesis.	Stabilizing blood glucose is associated with short-run improvements in working memory and behavioral regulation, defined as the capacity to maintain stable mood and to limit impulsive or hyperactive behavior. Intervention trials report measurable cognitive gains, such as faster processing speed, within a five-to-six-month window.	Mahoney et al. 2005; Osendarp et al. 2007; Petrova et al. 2019
Medium-Term (6 to 24 Months)	Between six and twenty four months, sustained shifts toward less processed, nutrient-dense diets are associated with changes in the gut microbiome and the resolution of micronutrient deficiencies. Soluble fibers act as prebiotic substrates for short-chain fatty acid production, which is linked to lower chronic, low-grade systemic inflammation, while consistent intake of iron, iodine, and B vitamins supports myelination and neurogenesis.	Pediatric studies associate this reduction in inflammatory markers with improved synaptic plasticity, and trials correcting micronutrient deficits over roughly an academic year report gains in executive function and attentional inhibition.	Gómez-Pinilla 2008; Taljaard et al. 2013; Naiman A Khan et al. 2015; Sadler et al. 2023

Notes: The channels above operate independently of body weight, consistent with the absence of significant effects on overweight, obesity, and BMI reported in Section 5.4). Structural changes documented over longer horizons (for example, white matter integrity) lie beyond this paper's two-year observation window and are therefore were not included.

Table A.13: Law 20.606: Nutrient Thresholds Triggering the "High In" Warning Label, by Stage

	Stage 1 (June 2016)	Stage 2 (June 2018)	Stage 3 (June 2019)
Solid foods (per 100 g)			
Energy (kcal)	350	300	275
Sodium (mg)	800	500	400
Total sugars (g)	22.5	15	10
Saturated fat (g)	6	5	4
Liquids (per 100 ml)			
Energy (kcal)	100	80	70
Sodium (mg)	100	100	100
Total sugars (g)	6	5	5
Saturated fat (g)	3	3	3

Note: A product must carry the "high in" warning label, and becomes subject to the advertising ban and school-sales prohibition, once it exceeds the threshold for any one of these four nutrients. Between Stage 1 and Stage 3, the solid-food threshold falls by 50 percent for sodium and by 55.5 percent for total sugars; for liquids, only the energy and sugar thresholds tighten over the same period (sugars fall once, between Stage 1 and Stage 2), while the sodium and saturated fat thresholds are unchanged across all three stages. Source: Decreto 13 (Minsal 2015).

Table A.14: Impact of Program Eligibility on Household Characteristics

	(1) Household Income	(2) Maternal Labor Force	(3) Father Present	(4) Number of Siblings	(5) Household Size
Take-up × Post	-243.75	-0.0045	-0.0237	-0.0084	-0.0524
	(8,322.56)	(0.0248)	(0.0242)	(0.0351)	(0.0776)
Observations	9,990	9,990	9,990	9,990	9,990
R^2	0.7467	0.6267	0.8104	0.8676	0.7752

Notes: Regressions use sampling weights and restrict the sample to public schools and children older than 7 years old in 2017. All specifications include child fixed effects, survey wave fixed effects, and baseline income, mother labor force participation and rural status interacted with a post-reform indicator. Number of clusters: 175. Cluster-robust standard errors at the municipality level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.15: The Impact of Healthier Meals on Math and Reading Test Scores by Exposure

	Eligibility Status		School Day Type		Rurality Status	
	(1) Eligible	(2) Non-eligible	(3) Full-Day	(4) Half-Day	(5) Rural	(6) Urban
Panel A: Math						
Healthier	0.045***	-0.009	0.043***	0.015	0.054*	0.035*
	(0.016)	(0.035)	(0.016)	(0.043)	(0.031)	(0.019)
Observations	376,312	68,092	376,865	67,064	92,771	351,633
Panel B: Reading						
Healthier	0.037**	0.012	0.035**	0.015	0.076***	0.024
	(0.015)	(0.033)	(0.018)	(0.025)	(0.027)	(0.018)
Observations	374,680	67,858	375,281	66,783	92,406	350,132

Notes: Table displays the overall average treatment effect on the treated (ATT) for math and verbal scores extracted from Callaway & Sant'Anna (2021) difference-in-differences models across various subsamples. Standard errors clustered at the municipality level are reported in parentheses. All models cover the 2010-2016 period and include individual and school covariates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.16: The Impact of Healthier Meals by Income

	T1 Low	T2 Medium	T3 High				
			All	Non-Elig.	Eligible	Half Day	Full Day
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Math							
Healthier	0.067***	0.034	0.018	-0.009	0.033	-0.013	0.024
	(0.023)	(0.021)	(0.026)	(0.035)	(0.027)	(0.057)	(0.023)
Observations	194,888	113,760	135,756	68,092	67,664	27,954	107,728
Panel B: Reading							
Healthier	0.056**	0.032	0.025	0.014	0.034	-0.012	0.034
	(0.022)	(0.020)	(0.026)	(0.033)	(0.027)	(0.035)	(0.029)
Observations	194,066	113,227	135,245	67,858	67,387	27,854	107,317

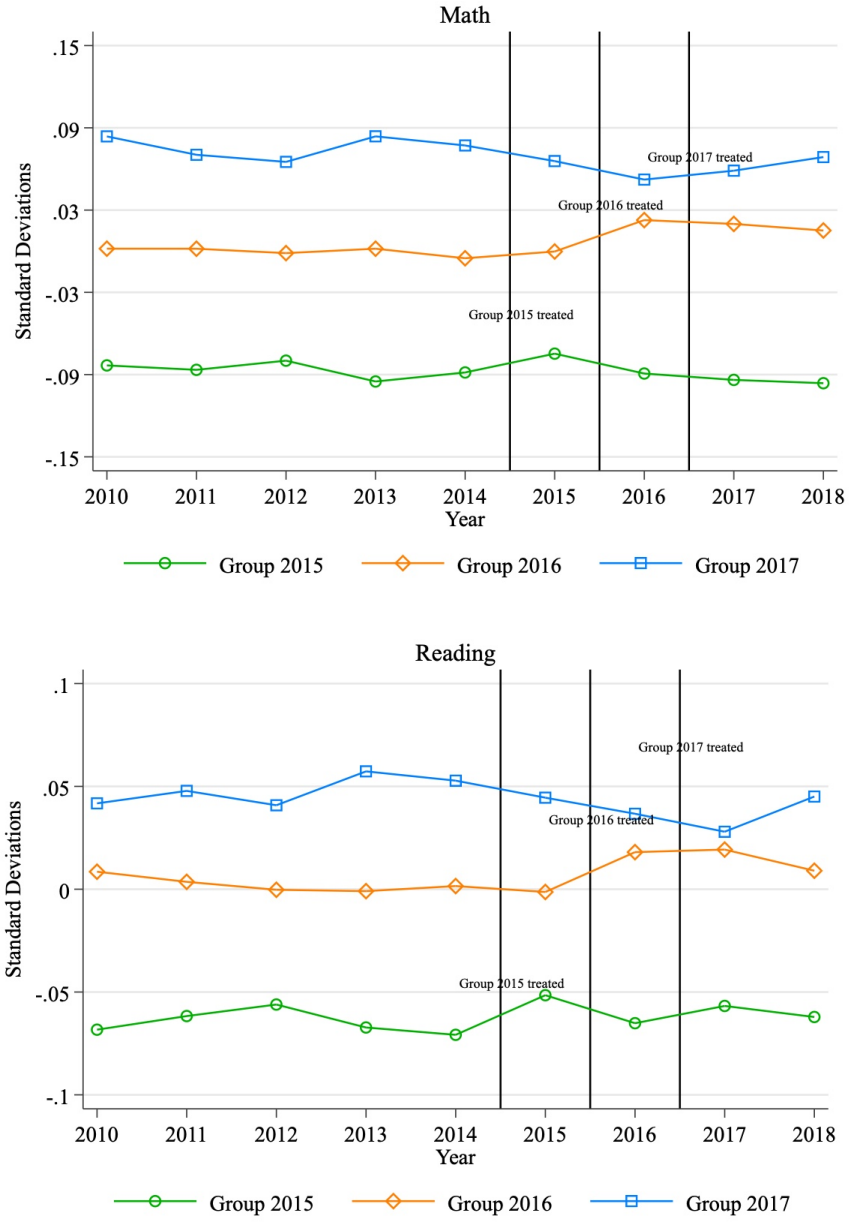
Notes: Table displays the overall average treatment effect on the treated (ATT) for total scores extracted from Callaway & Sant'Anna (2021) difference-in-differences models across various subsamples. Standard errors clustered at the municipality level are reported in parentheses. All models cover the 2010–2016 period and include individual and school covariates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.17: Heterogeneous Treatment Effects by Mother's Education and Student Gender

	Mother's Education		Student Gender	
	(1) Non-College	(2) Some College	(3) Boys	(4) Girls
Panel A: Total				
Healthier Meals	0.045^{***}	0.013	0.037^{*}	0.031^{*}
	(0.016)	(0.033)	(0.021)	(0.018)
Observations	357,350	83,340	224,492	227,883
Panel B: Math				
Healthier Meals	0.043^{***}	0.010	0.033	0.032^{*}
	(0.016)	(0.033)	(0.021)	(0.018)
Observations	360,424	83,980	226,577	229,625
Panel C: Reading				
Healthier Meals	0.040^{**}	0.011	0.036^{*}	0.024
	(0.016)	(0.030)	(0.020)	(0.019)
Observations	358,863	83,675	225,590	228,693

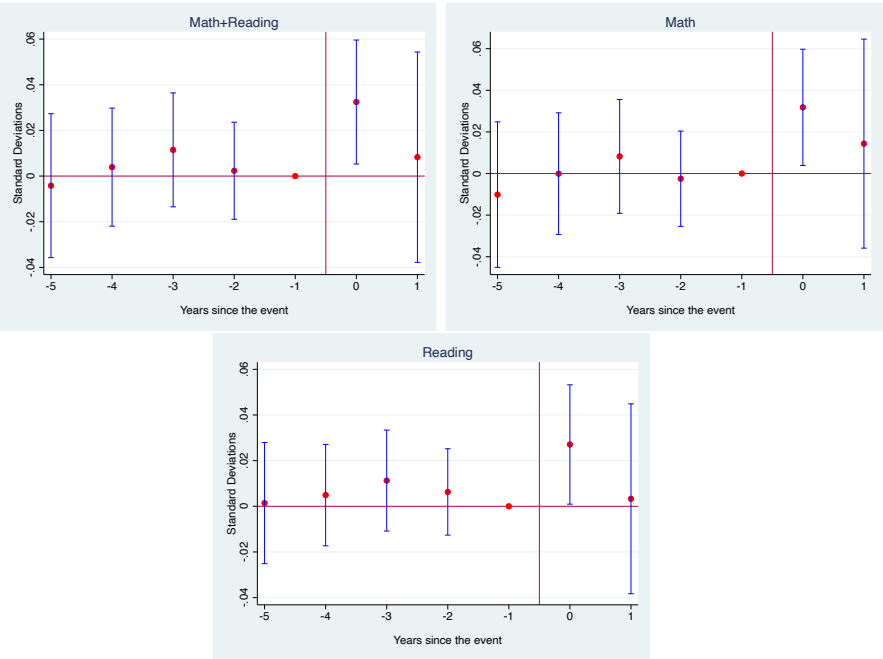
Notes: Table displays heterogeneity analyses across mother education levels and student gender using Callaway & Sant'Anna (2021) DiD framework. Standard errors clustered at municipality level in parentheses. $\star p < 0.10$, $\star\star p < 0.05$, $\star\star\star p < 0.01$.

Figure A.10: Raw Test Scores 4th: 2011-2018



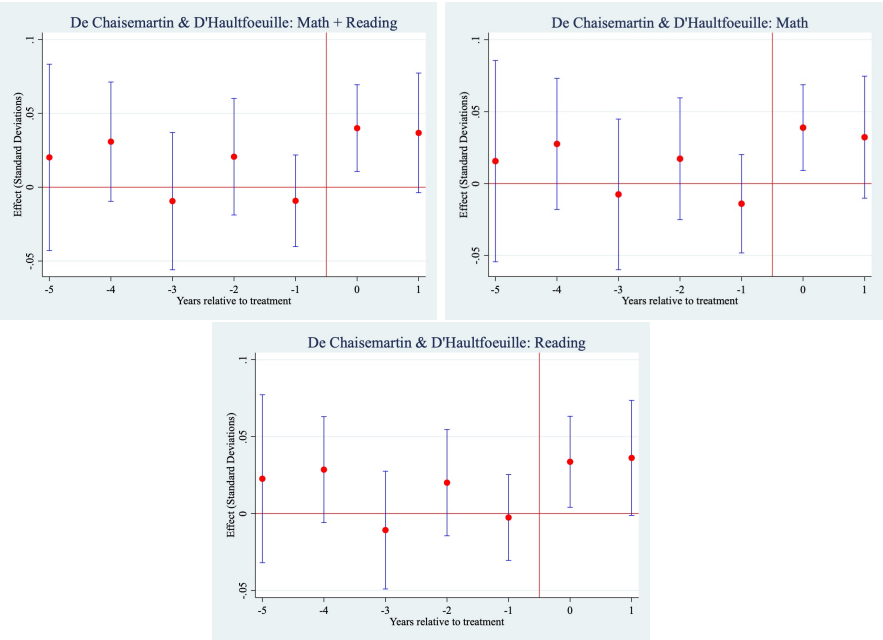
Note: These figures present the raw test scores for 4th graders in public schools in Math and Reading, respectively, between 2010 and 2018. The line with the circle marker corresponds to schools that started to receive nutritious meals in 2015. The line with a diamond marker and the line with a square marker corresponds to Schools 2016 and 2017 began to receive the healthy meals in 2016 and 2017, respectively.

Figure A.11: Event Study: TWFE



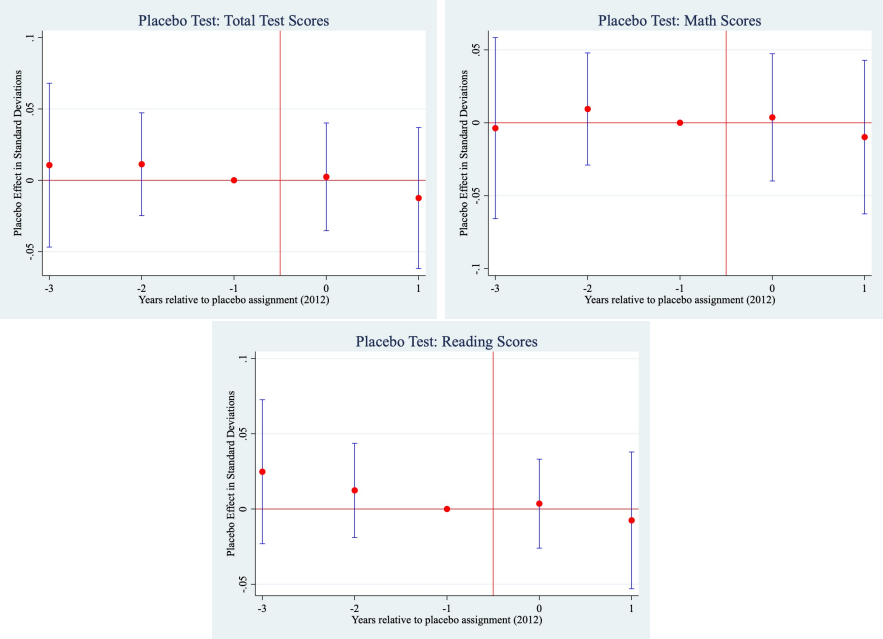
Notes: The figure presents results from the event study estimates using TWFE. The dependent variable is the test scores measured in standard deviations. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio and year and school fixed effects. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.12: Chaisemartin and D'Haultfoeuill's method



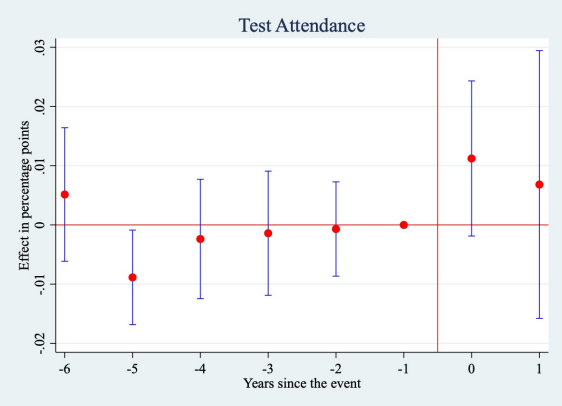
Note: The figure presents results from the event study estimates from the approach from the De Chaisemartin and d'Haultfoeuille 2020 approach. The dependent variable is the test scores measured in standard deviations. Pre-treatment coefficients are placebo estimates. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.13: Placebo Estimates for 2012-14 Meal Vendor Assignment



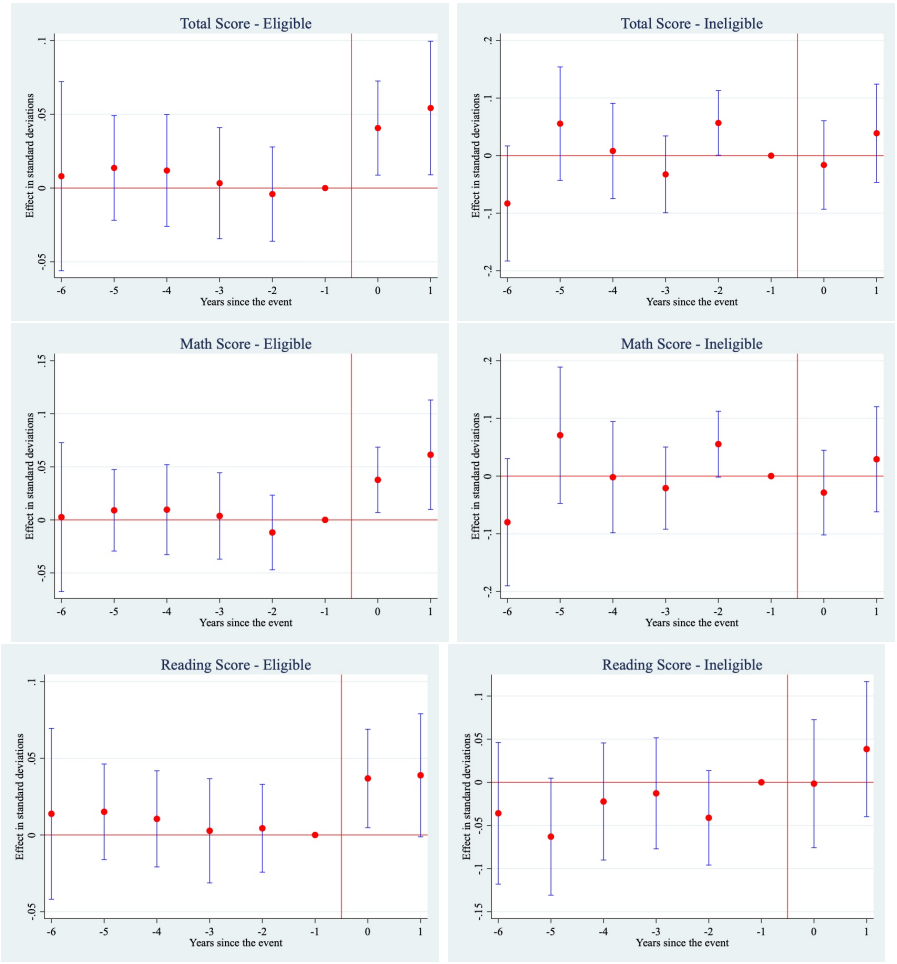
Note: The figure presents results from the event study estimates for the meal vendor assignment in 2012-14. The dependent variable is the test scores measured in standard deviations. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.14: Test Attendance



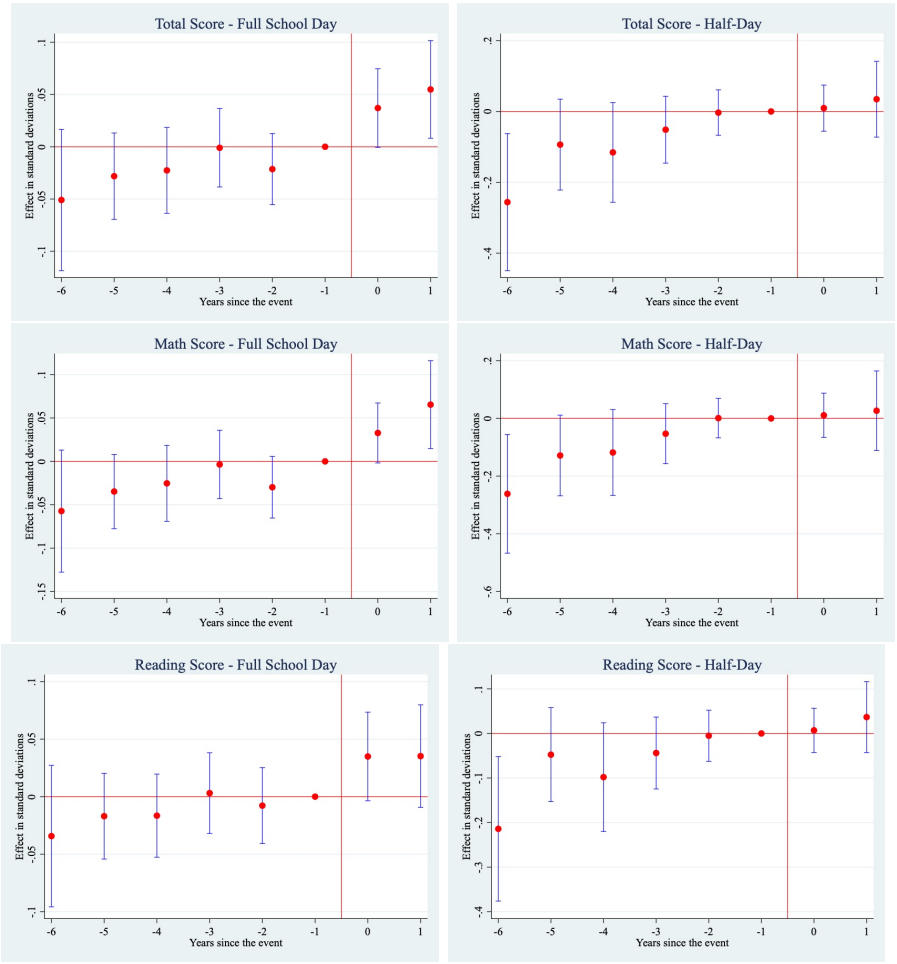
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach. The dependent variable is a dummy variable if the student attended the test. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.15: Effects Proxied by Eligibility



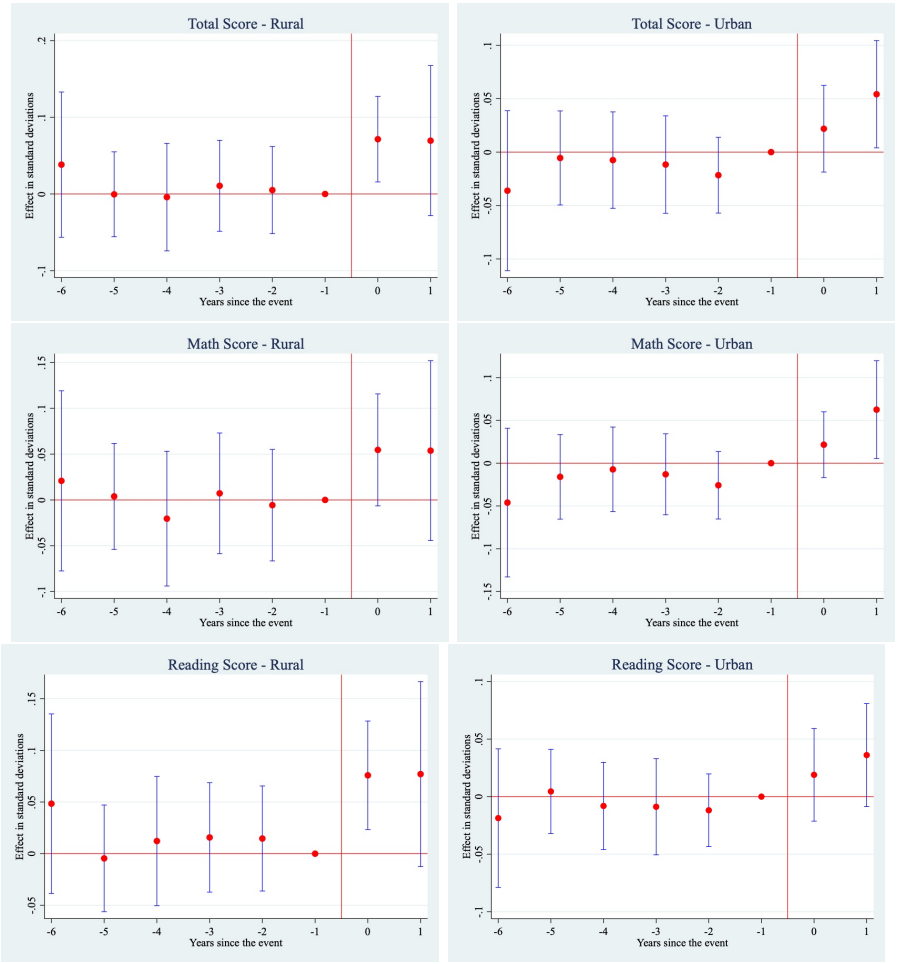
Note: The figure presents results from the event study estimates from the approach from the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Eligible students are those from rural schools, with mother with less than secondary education complete and in the 60% of the income distribution. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.16: Effects of Healthier Meals by Full or Half-Day School



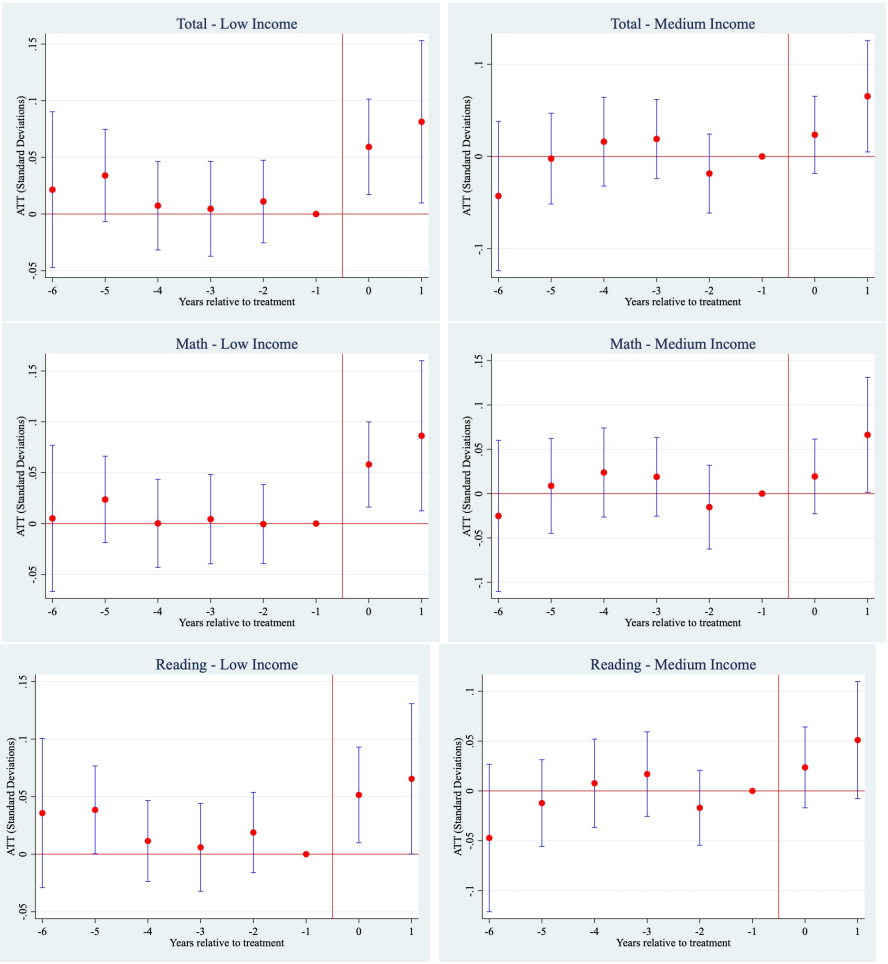
Note: The figure presents results from the event study estimates from the approach from the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Eligible students are those from rural schools, with mother with less than secondary education complete and in the 60% of the income distribution. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.17: Effects of Healthier Meals by Zone: Rural or Urban



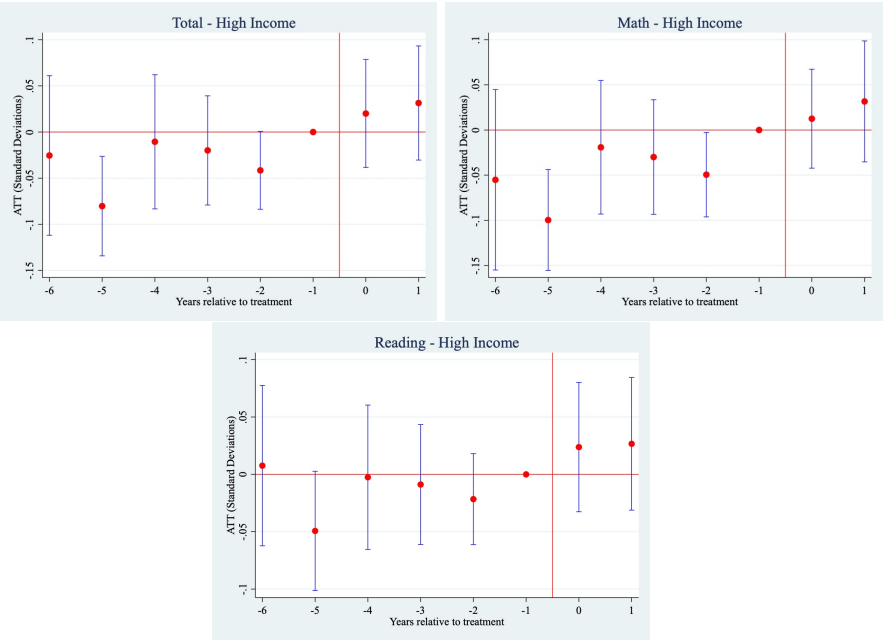
Note: The figure presents results from the event study estimates from the approach from the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Eligible students are those from rural schools, with mother with less than secondary education complete and in the 60% of the income distribution. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.18: Effects of Healthier Meals by Low (T1) and Medium (T2) Income



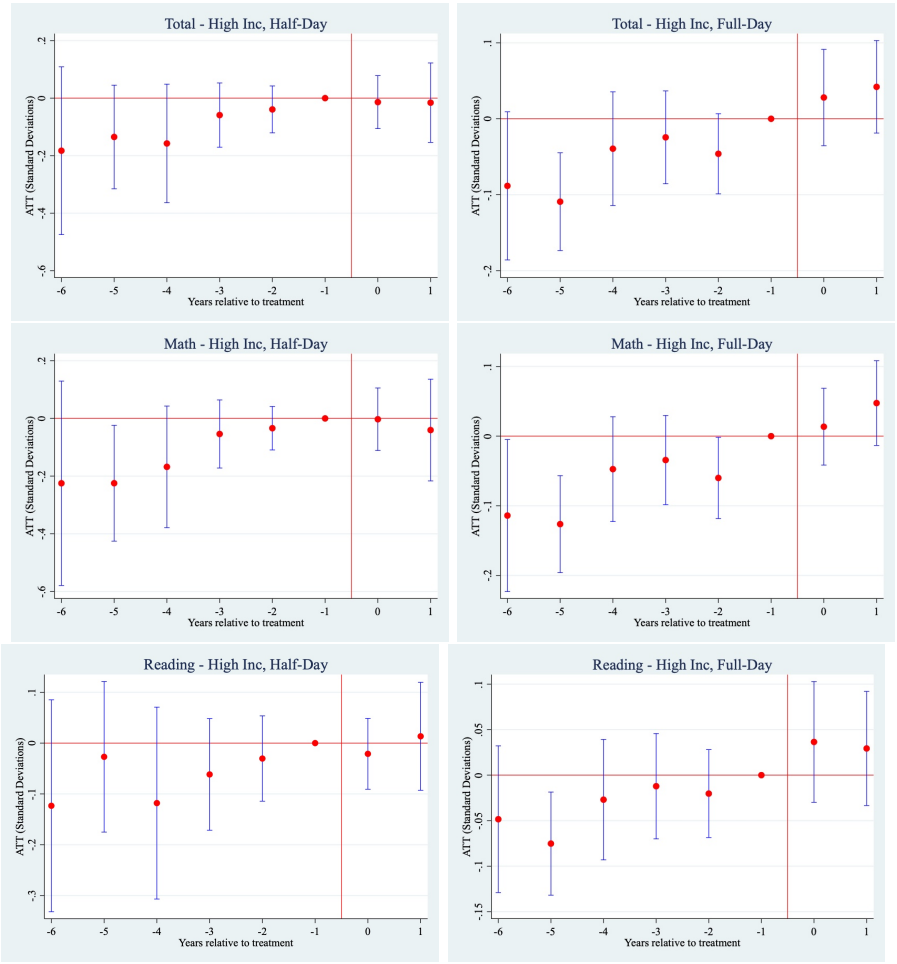
Note: The figure presents results from the event study estimates from the approach from the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Eligible students are those from rural schools, with mother with less than secondary education complete and in the 60% of the income distribution. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.19: Effects of Healthier Meals by High (T3) Income



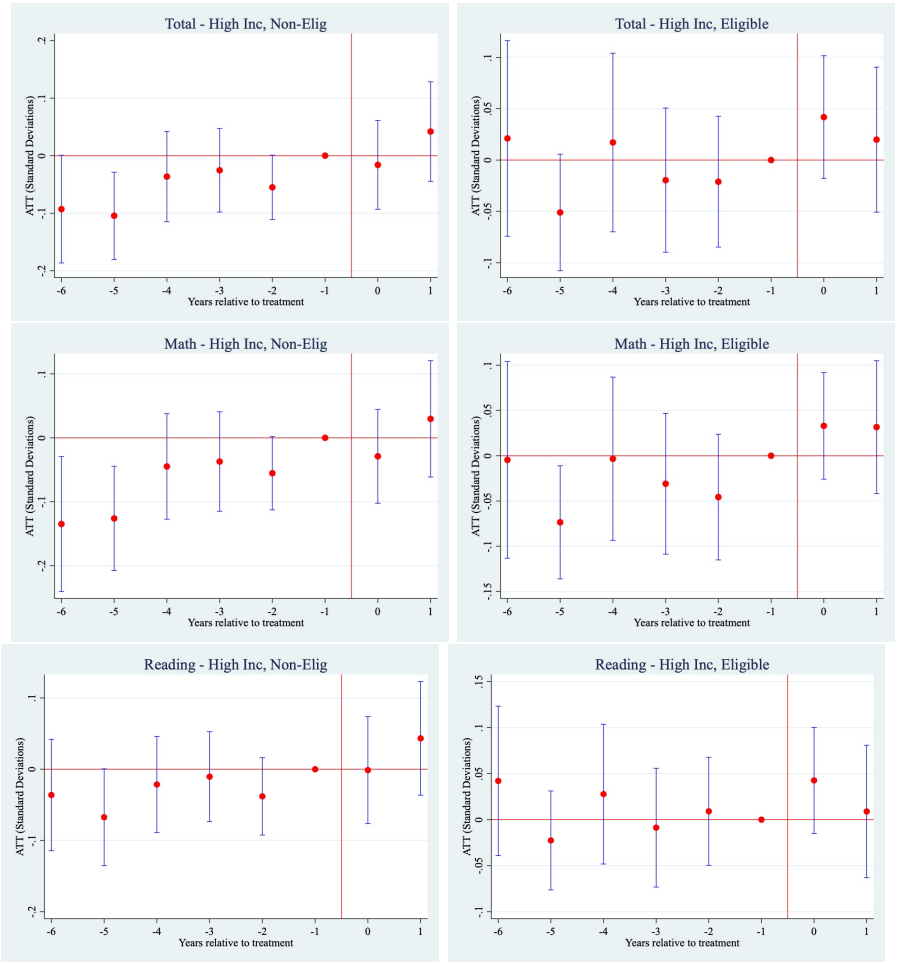
Note: The figure presents results from the event study estimates from the approach from the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Eligible students are those from rural schools, with mother with less than secondary education complete and in the 60% of the income distribution. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.20: Effects of Healthier Meals for High Income by Full and Half Day



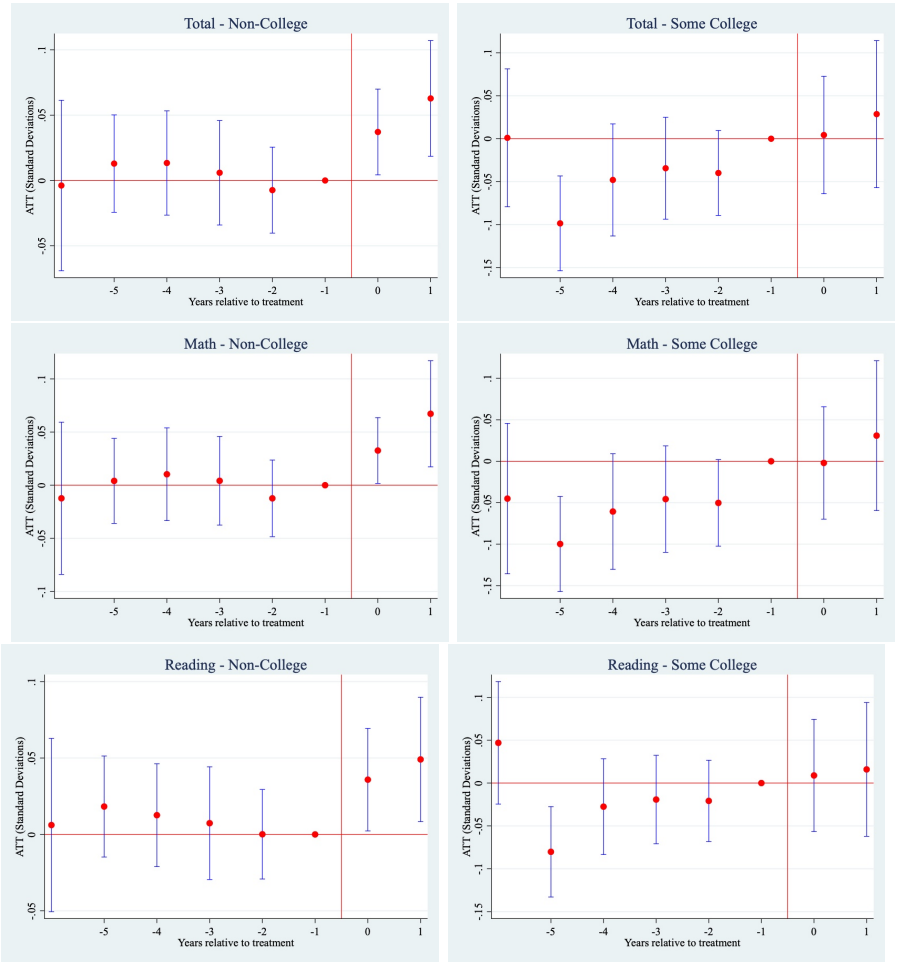
Note: The figure presents results from the event study estimates from the approach from the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Eligible students are those from rural schools, with mother with less than secondary education complete and in the 60% of the income distribution. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.21: Effects of Healthier Meals for High Income by Eligibility



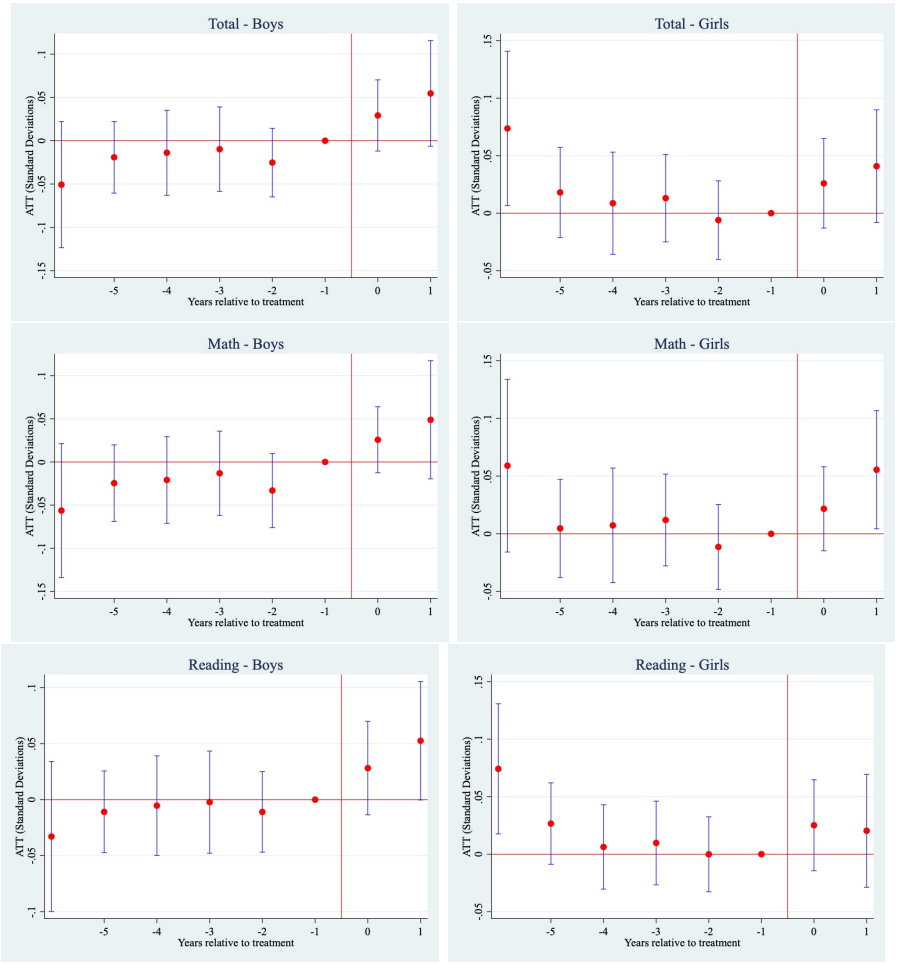
Note: The figure presents results from the event study estimates from the approach from the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Eligible students are those from rural schools, with mother with less than secondary education complete and in the 60% of the income distribution. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.22: Effects of Healthier Meals by Maternal Education



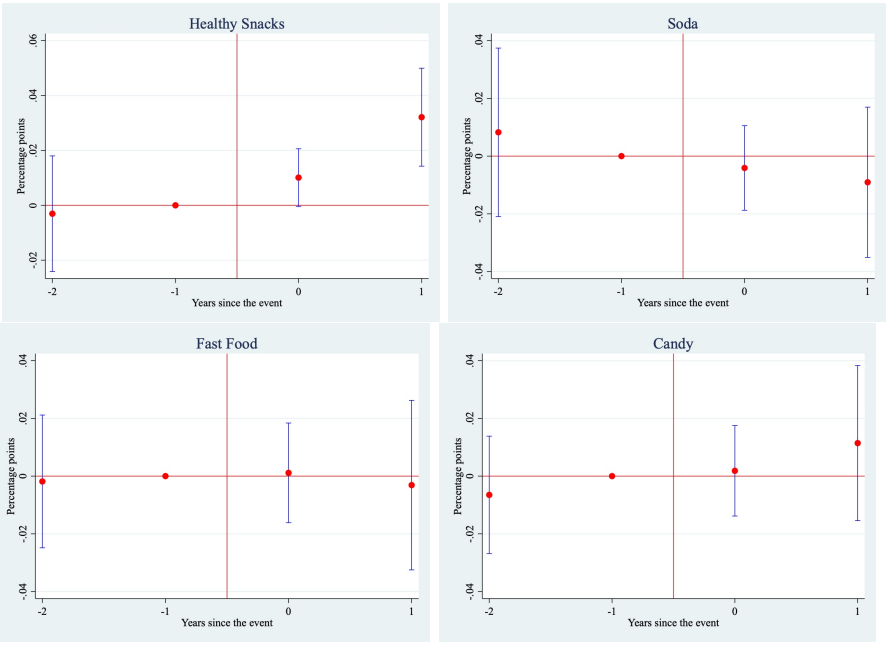
Note: The figure presents results from the event study estimates from the approach from the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Eligible students are those from rural schools, with mother with less than secondary education complete and in the 60% of the income distribution. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.23: Effects of Healthier Meals by Gender



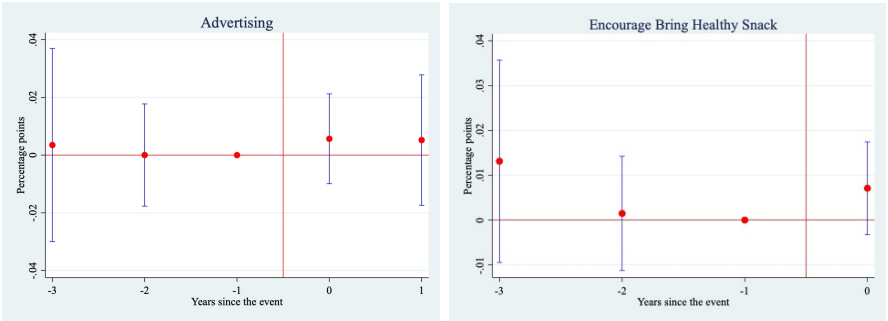
Note: The figure presents results from the event study estimates from the approach from the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Eligible students are those from rural schools, with mother with less than secondary education complete and in the 60% of the income distribution. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.24: Impact of Healthier Meals on Foods Brought to School



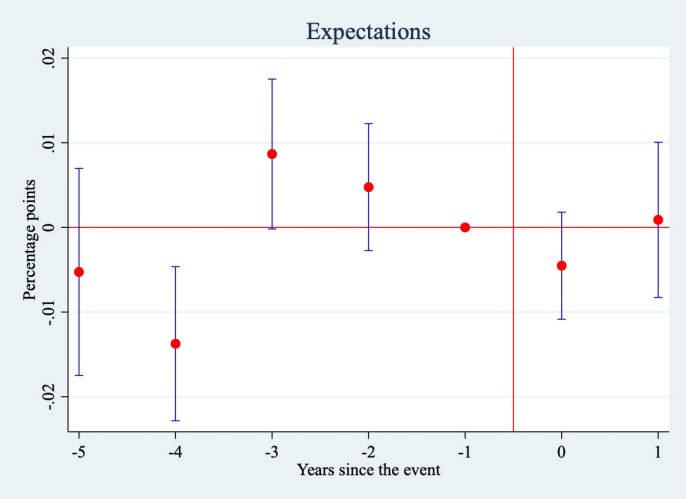
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach. The dependent variables are binary indicators for whether the student ate/bought healthy snacks, soda, fast food, or candy. Individual covariates include years of education of the mother, household income, gender, and age of the student. School covariates include a dummy for rurality of the school and the student-to-teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.25: Impact of Reform on Advertising of Unhealthy Food and Encouragement of Healthy Food



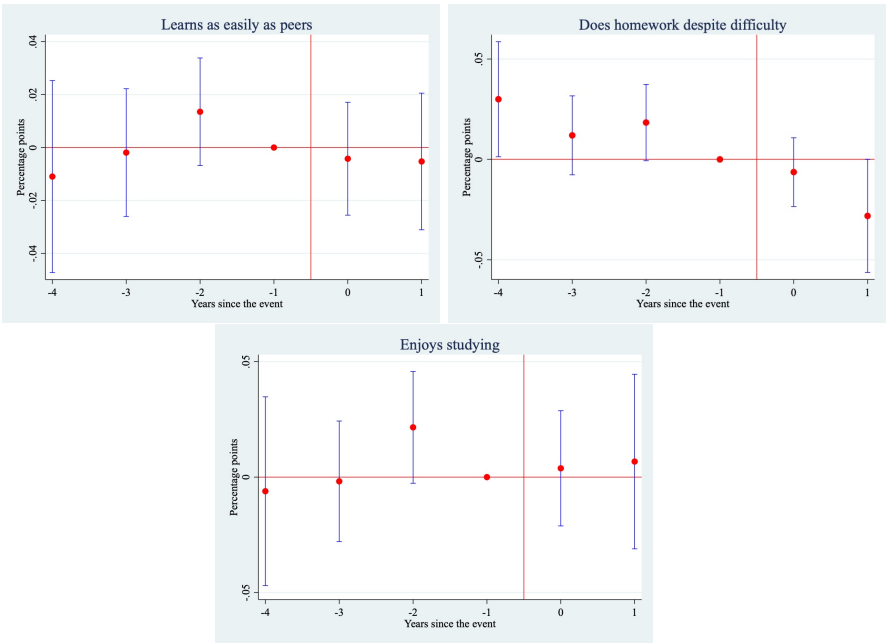
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach. The Advertising question asked students whether they have been exposed to advertising promoting sugary drinks or fast food (question available from 2012 to 2016). The Encourage bring healthy food asks students if their teachers promoted to bring healthy snacks to school (question available from 2012 to 2016). Individual covariates include years of education of the mother, household income, gender, and age of the student. School covariates include a dummy for rurality of the school and the student-to-teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.26: Impact of Healthier Meals on Parents Expectations



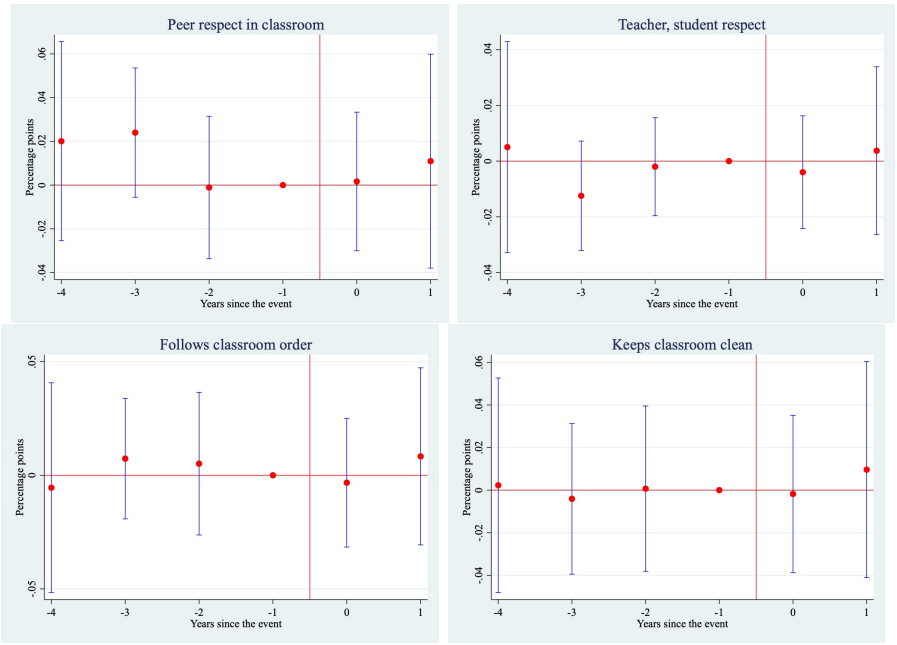
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach. The dependent variables are binary indicators for whether the student brought healthy snacks, soda, fast food, or candy from home. Individual covariates include years of education of the mother, household income, gender, and age of the student. School covariates include a dummy for rurality of the school and the student-to-teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.27: Impact of Healthier Meals on Student Self-Efficacy - Reported by Students



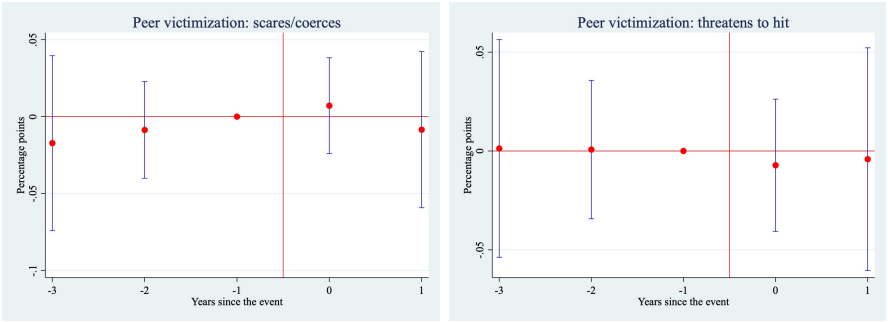
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach. The dependent variables are binary indicators. Individual covariates include years of education of the mother, household income, gender, and age of the student. School covariates include a dummy for rurality of the school and the student-to-teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.28: Impact of Healthier Meals on Classroom Climate - Reported by Students



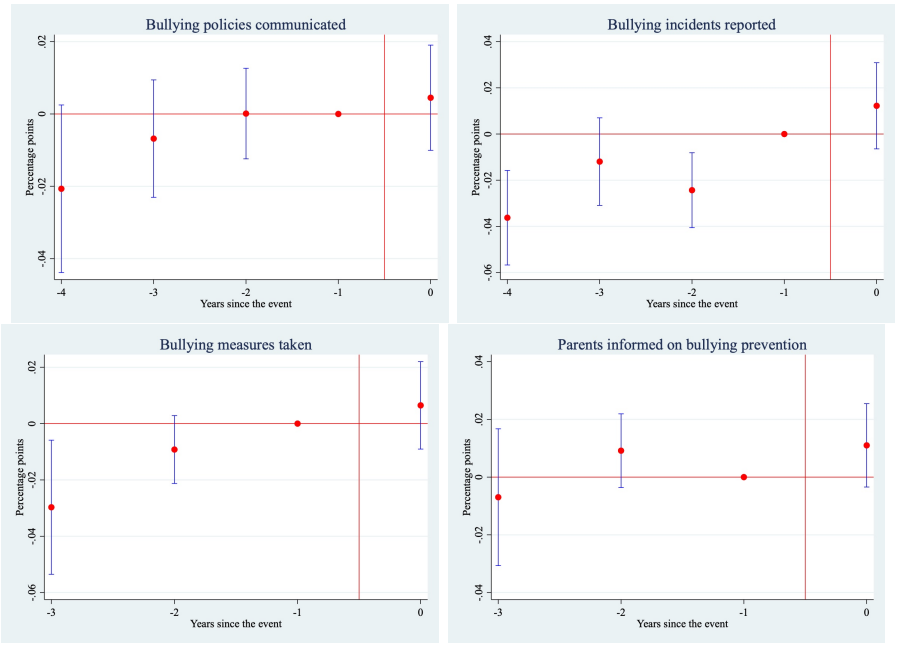
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach. The dependent variables are binary indicators. Individual covariates include years of education of the mother, household income, gender, and age of the student. School covariates include a dummy for rurality of the school and the student-to-teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.29: Impact of Healthier Meals on Peer Victimization - Reported by Students



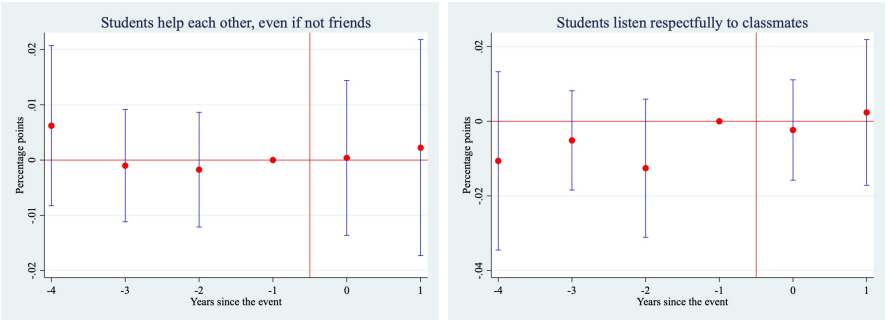
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach. The dependent variables are binary indicators. Individual covariates include years of education of the mother, household income, gender, and age of the student. School covariates include a dummy for rurality of the school and the student-to-teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.30: Impact of Healthier Meals on Bullying - Reported by Parents



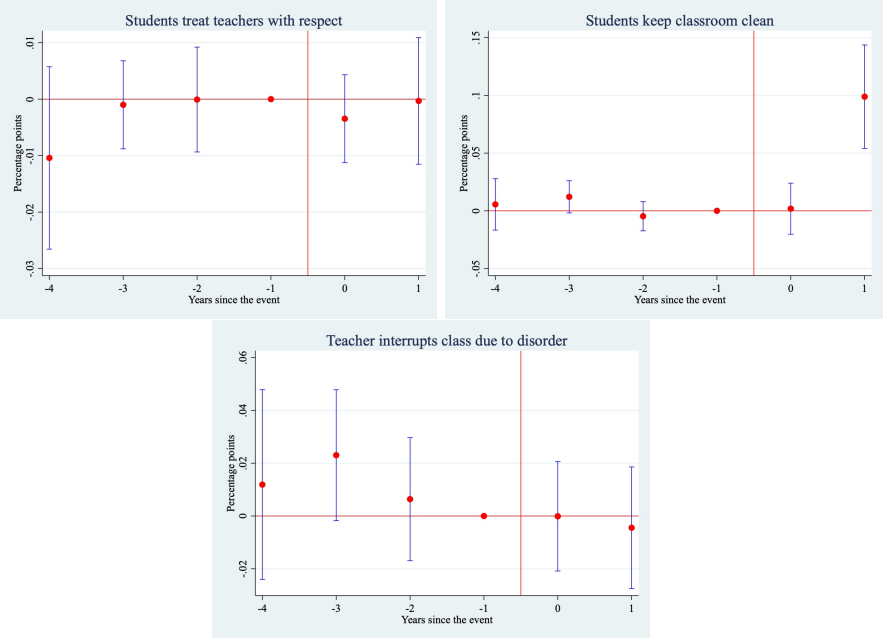
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach. The dependent variables are binary indicators. Individual covariates include years of education of the mother, household income, gender, and age of the student. School covariates include a dummy for rurality of the school and the student-to-teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.31: Impact of Healthier Meals on Student Interactions - Reported by Teachers



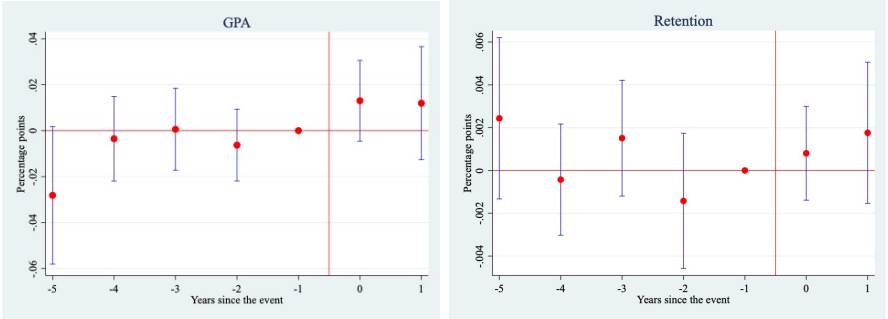
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach. The dependent variables are binary indicators. Individual covariates include years of education of the mother, household income, gender, and age of the student. School covariates include a dummy for rurality of the school and the student-to-teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.32: Impact of Healthier Meals on Class Respect and Environment - Reported by Teachers



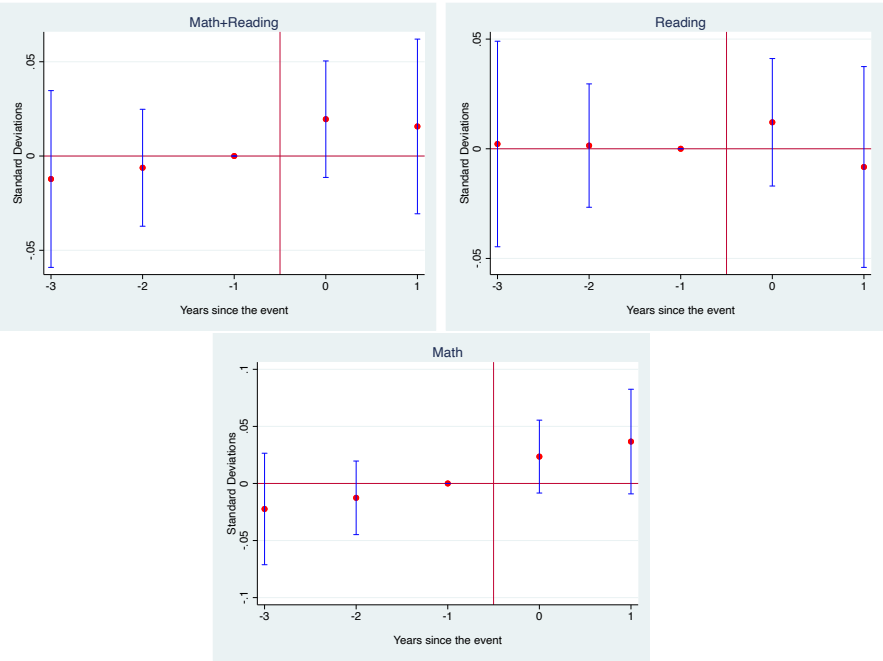
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach. The dependent variables are binary indicators. Individual covariates include years of education of the mother, household income, gender, and age of the student. School covariates include a dummy for rurality of the school and the student-to-teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.33: Impact of Healthier Meals on GPA and Retention



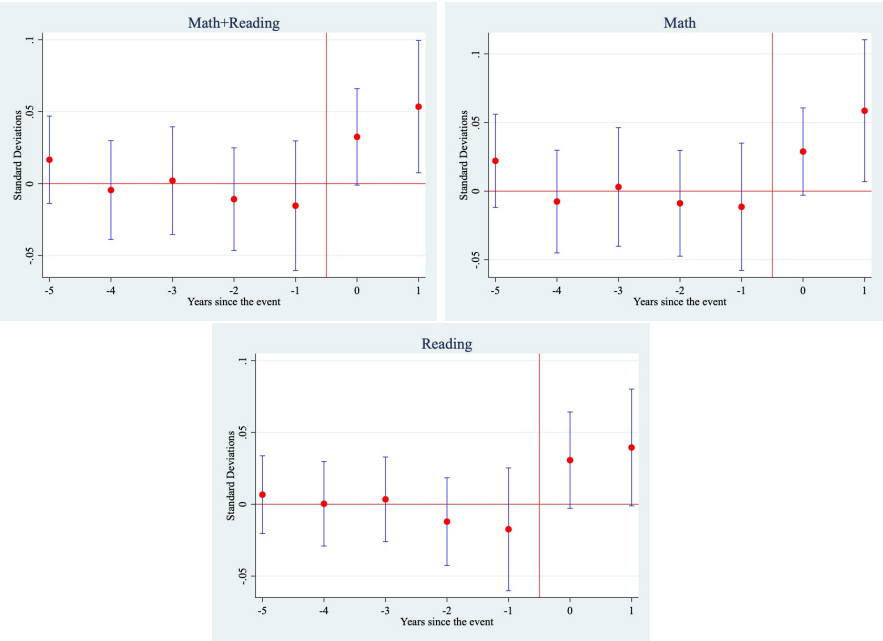
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach. The dependent variables are binary indicators. Individual covariates include years of education of the mother, household income, gender, and age of the student. School covariates include a dummy for rurality of the school and the student-to-teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.34: Event Study: Sixth Grade



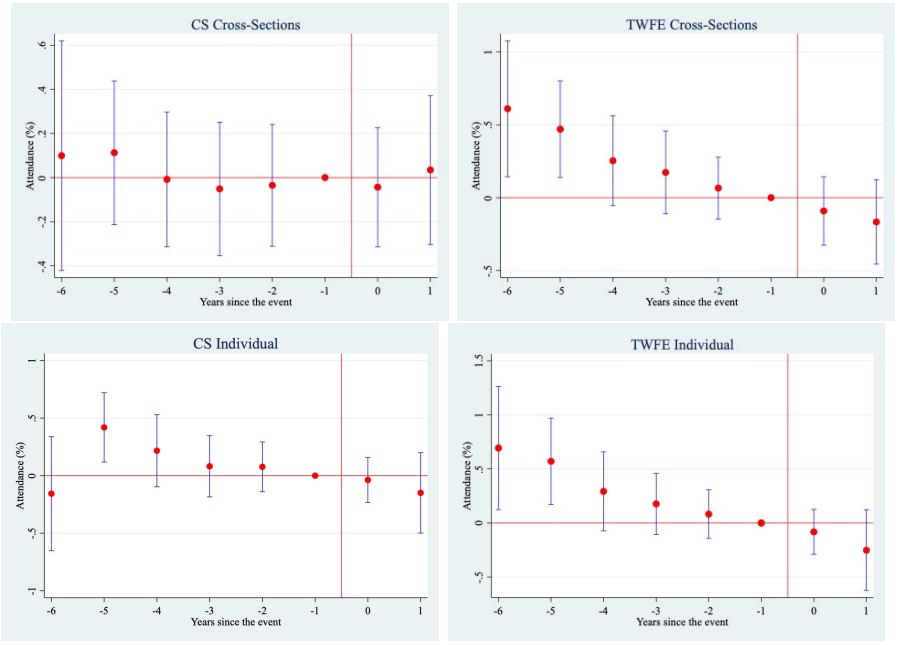
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.35: Event Study with Placebos



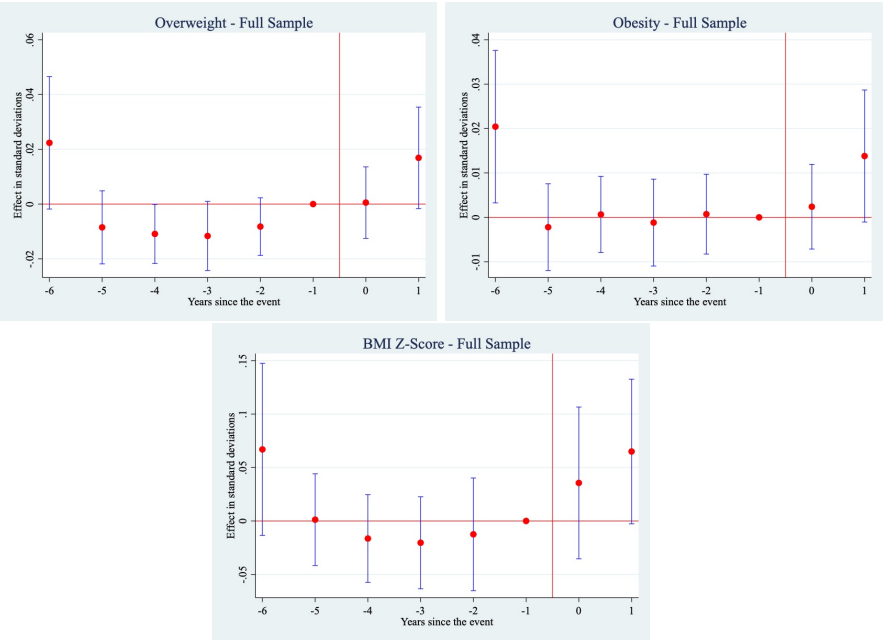
Note: The figure presents results from the event study estimates using the Callaway and Sant'Anna 2021 approach. The dependent variable is the test scores measured in standard deviations. Pre-treatment coefficients are placebo estimates. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.36: Effect of the SMP on Attendance



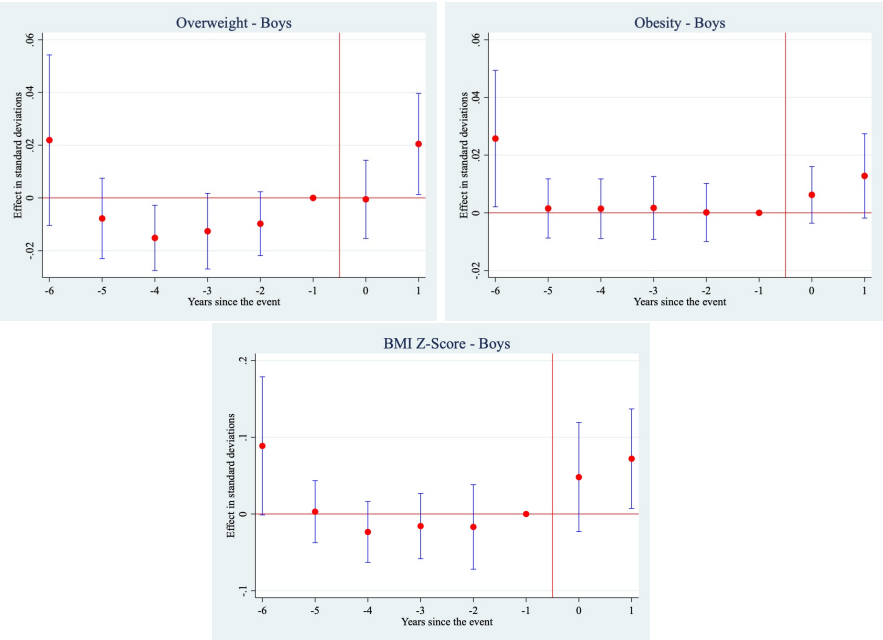
Note: The dependent variable is the percentage of attendance that takes values from 0 (0%) to 1 (100%). CS is Callaway and Sant'Anna 2021 and TWFE is Two Way Fixed Effects. Cross sections compares attendance of cross sections of students that took the national test and Individual Level is a panel of attendance at the student level of students in the main sample. Individual covariates include years of education of the mother, household income, gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. TWFE controls for year and school fixed effects and additionally for student fixed effect in the individual level estimates. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.37: Effect of the SMP on Obesity



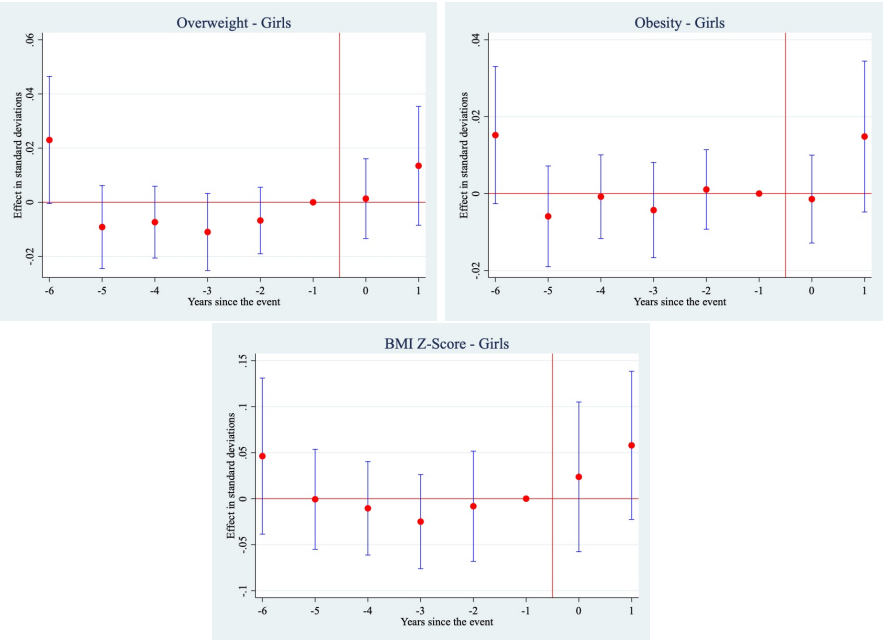
Note: The figure presents the event studies estimates using the Callaway and Sant'Anna 2021 approach on the probability of being overweight or more, obese and BMI. Individual covariates include years of gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.38: Effect of the SMP on Obesity: Boys



Note: The figure presents the event studies estimates using the Callaway and Sant'Anna 2021 approach on the probability of being overweight or more, obese and BMI. Individual covariates include years of gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.

Figure A.39: Effect of the SMP on Obesity: Girls



Note: The figure presents the event studies estimates using the Callaway and Sant'Anna 2021 approach on the probability of being overweight or more, obese and BMI. Individual covariates include years of gender and age of the student. School covariates include a dummy for rurality of school and students to teacher ratio. Bars denote 95 percent confidence intervals from robust standard errors and are clustered at the municipality level.